

Independent external validation of the SugarCode AI computational-biology platform against held-out public benchmarks

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Independent research; computational work assisted by Instinct

Abstract

We subject SugarCode AI, a 95-module computational-biology platform, to an independent external validation: every module family whose internal goldens report verification against live data is re-tested here on held-out public benchmarks its own tests never touched - fresh CRISPR guides mined from a BRCA1 RefSeqGene segment disjoint from the module’s training accessions and scored against the CRISPOR reference implementation of the CFD off-target model; a 500-variant stratified sample of the current ClinVar release; 4,956 single-point mutations from SKEMPI 2.0; the BiGG *E. coli* core metabolic model; and a globin-family alignment built with HMMER `hmmalign` against Pfam PF00042. We report per-module agreement metrics with bootstrap confidence intervals, publish all fixtures with provenance, and document failure modes without suppressing negative results.

1 Introduction

Computational-biology toolkits are typically validated against the same data that informed their development, inflating confidence. Independent, held-out validation against public reference implementations and current database releases is the standard this platform must meet to be usable in laboratory settings. This study asks a single question: does each verified claim in the platform’s own status ledger survive contact with data it has never seen? We answer it module family by module family, with frozen fixtures, pinned random seeds, and metrics reported with 95% bootstrap confidence intervals. All code, fixtures, and results are released in this repository.

2 Related Work and Benchmarks

Reference implementations used here are CRISPOR for guide scoring [2] and its Doench Rule Set 1/CFD models [1], SKEMPI 2.0 for binding free-energy changes [3], BiGG for metabolic models [4], Clustal Omega for alignment [5], and the Tavtigian Bayesian ACMG framework [6]. Pharmacogenomic tables come from the CPIC API; RNA structure references from Rfam seed alignments; splice predictions are compared with SpliceAI.

3 Methods

For each module family we constructed a held-out test disjoint from every fixture used during platform development. Negative results are preserved and reported alongside positives; no test was re-designed after seeing its outcome.

3.1 Data and Provenance

All datasets are public and carry recorded provenance (see `data/README.md`): NCBI ClinVar `variant_summary.txt.gz` (retrieved 2026-09-24; 500-variant stratified reservoir sample from 1,517,551 eligible GRCh38 SNVs with criteria-provided review status); SKEMPI 2.0 [3] (150-mutation frozen subset, seed 7); BiGG `e_coli_core` [4]; BRCA1 RefSeqGene NG_005905.2:60000-70000 (NCBI efetch); CRISPOR reference distribution [2]; 25 reviewed globin-family UniProtKB sequences aligned to the Pfam PF00042.29 HMM with HMMER3 `hmmalign` (pyhmmmer 0.12.3); the EBI Clustal Omega job queued for over 40 minutes and was abandoned.

3.2 Mathematical Formulation

CFD off-target scoring follows Doench et al. 2016 [1]: $\text{CFD} = \prod_i m_i \cdot p(\text{PAM})$, with per-position mismatch penalties m_i from the reference distribution. Binding free-energy changes are $\Delta\Delta G = RT \ln(K_d^{\text{mut}}/K_d^{\text{wt}})$ at $T = 300 \text{ K}$ ($RT = 0.596 \text{ kcal/mol}$). Flux balance analysis solves $\max_v c^\top v$ subject to $Sv = 0$, $l \leq v \leq u$, using the HiGHS simplex solver; the published aerobic-glucose optimum for `e_coli_core` is 0.8739 h^{-1} [4]. Agreement is quantified with Pearson r , Mann-Whitney AUC, and binary-classification metrics; confidence intervals are percentile bootstrap with 1,000 resamples at fixed seed.

3.3 Derivations

(F1) Position weight matrix log-odds. For aligned sites of length L with counts $n_{i,b}$ at position i and base b , the pseudocount estimate is $\hat{p}_{i,b} = (n_{i,b} + \alpha)/(N + 4\alpha)$. The log-odds weight against background q_b is

$$w_{i,b} = \log_2 \frac{\hat{p}_{i,b}}{q_b}, \quad S(x) = \sum_{i=1}^L w_{i,x_i}. \quad (1)$$

$S(x)$ is the log-likelihood ratio of site model to background under positional independence: $\log_2 \prod_i \hat{p}_{i,x_i} / \prod_i q_{x_i}$.

(F2) Normalised score. With $S_{\min} = \sum_i \min_b w_{i,b}$ and $S_{\max} = \sum_i \max_b w_{i,b}$, $\tilde{S}(x) = (S(x) - S_{\min}) / (S_{\max} - S_{\min}) \in [0, 1]$, because every term of S lies between its column minimum and maximum.

(F3) Variant delta. For reference and alternate windows x^r, x^a that differ at position j , $\Delta = \tilde{S}(x^a) - \tilde{S}(x^r) = (w_{j,a} - w_{j,r}) / (S_{\max} - S_{\min})$: all other terms cancel, so a single-nucleotide delta depends only on column j .

(F4) Tract blend. The acceptor score is $A(x) = (1 - \lambda)\tilde{S}(x) + \lambda T(x)$ with $\lambda = 0.25$, where T is the normalised pooled polypyrimidine-tract score. Linearity gives $\Delta A = (1 - \lambda)\Delta\tilde{S} + \lambda\Delta T$.

(F5) AUC as a Mann-Whitney statistic. For scores $\{s_i^+\}_{i=1}^m$ and $\{s_j^-\}_{j=1}^n$,

$$\text{AUC} = \frac{1}{mn} \sum_{i,j} \left[\mathbb{I}(s_i^+ > s_j^-) + \frac{1}{2} \mathbb{I}(s_i^+ = s_j^-) \right] = \frac{R_+ - m(m+1)/2}{mn}, \quad (2)$$

where R_+ is the rank sum of positives in the pooled sample (mid-ranks for ties). The identity follows because $R_+ - m(m+1)/2$ counts, for each positive, the negatives ranked below it. Equivalently, $\text{AUC} = P(s^+ > s^-)$.

(F6) Within-position AUC. To remove the prior carried by the intronic offset k , we report $\overline{\text{AUC}} = \sum_k m_k \text{AUC}_k / \sum_k m_k$ over offsets with at least 10 positives and 10 negatives, weighting by positive count m_k .

(F7) CNN objective. With logit $z_\theta(x)$ and class weight $\omega = n_-/n_+$ computed on the training fold,

$$\mathcal{L}(\theta) = -\frac{1}{N} \sum_i \left[\omega y_i \log \sigma(z_i) + (1 - y_i) \log(1 - \sigma(z_i)) \right] + \frac{\beta}{2} \|\theta\|^2, \quad (3)$$

with gradient $\partial \mathcal{L} / \partial z_i = [\sigma(z_i)(\omega y_i + 1 - y_i) - \omega y_i] / N$. Setting $\omega = n_-/n_+$ makes both classes contribute equal total weight.

(F8) Convolutional feature. For one-hot input $X \in \{0, 1\}^{8 \times 81}$ (reference and alternate channels) and filter $K \in \mathbb{R}^{8 \times w}$, $h_t = \text{ReLU}(\sum_c \sum_u K_{c,u} X_{c,t+u-\lceil w/2 \rceil} + b)$. Global max pooling $\max_t h_t$ makes the feature invariant to where the motif sits in the window.

(F9) Binding free energy. The SKEMPI equilibrium-constant check uses a ratio, so a change in the concentration unit cancels:

$$\Delta \Delta G = \Delta G_{mut} - \Delta G_{wt} = RT \ln \left(\frac{K_d^{mut}}{K_d^{wt}} \right), \quad RT = 0.596 \text{ kcal/mol at 300 K.} \quad (4)$$

This converts a mutant/wild-type dissociation-constant pair into the binding-energy difference used in the committed SKEMPI parity test; it is not itself a new binding measurement.

(F10) FBA duality. For the committed BiGG `e_coli_core` flux check, HiGHS solves the stoichiometric primal and its bound-price dual:

$$\begin{aligned} P &= \max_v \{c^\top v : Sv = 0, l \leq v \leq u\}, \\ D &= \min_{\pi, \alpha, \beta} \{u^\top \alpha - l^\top \beta : S^\top \pi + \alpha - \beta = c, \alpha, \beta \geq 0\}. \end{aligned} \quad (5)$$

Weak duality gives $P \leq D$; when both solutions are feasible, strong duality yields $P = D$. The reported aerobic-glucose optimum 0.8739 h^{-1} is checked against solver feasibility and the primal-dual gap, not inferred from this formula alone.

(F11) Codon Adaptation Index. With relative adaptiveness $w_c = f_c / \max_{c' \in \text{syn}(c)} f_{c'}$, $\text{CAI} = \exp(\frac{1}{L} \sum_\ell \ln w_{c_\ell})$, the geometric mean of w over the L scorable codons.

(F12) ACMG Bayesian posterior. Following Tavtigian et al., evidence points scale pathogenic odds and Bayes' rule converts them to a posterior:

$$O_P = 350^{\text{pts}/8}, \quad P(\text{pathogenic} \mid \text{pts}) = \frac{O_P \pi}{(O_P - 1)\pi + 1}. \quad (6)$$

For the committed six-point check with prior $\pi = 0.1$, $O_P = 350^{0.75} \approx 80.92$ and the posterior is $8.092/8.992 \approx 0.8999$, matching the module output and the published 0.900. This is a framework arithmetic check, not independent variant pathogenicity evidence.

(F13) Profile-HMM forward recursion. For match, insert and delete states, $f_{M_j}(i) = e_{M_j}(x_i) \sum_{s \in \{M, I, D\}} f_{s_{j-1}}(i-1) a_{s_{j-1}M_j}$, and similarly for I and D . The log-odds score is $\log P(x \mid \mathcal{M}) - \sum_i \log q_{x_i}$.

(F14) Bootstrap interval. For a statistic $\hat{\theta}$, draw $B = 1000$ resamples with replacement and take the 2.5th and 97.5th percentiles of $\{\hat{\theta}_b^*\}$.

3.4 Derivations for the Second-Round Validations

(F15) Maximum-entropy splice-site score. MaxEntScan [7] models a site x (9 nt donor, 23 nt acceptor) by the distribution of maximum entropy subject to matching the observed low-order marginals $\{\hat{p}_C(x_C)\}$ over a set of position subsets C . By Lagrange duality the solution has Gibbs form

$$p(x) = \frac{1}{Z} \exp\left(\sum_C \lambda_C(x_C)\right), \quad Z = \sum_{x'} \exp\left(\sum_C \lambda_C(x'_C)\right), \quad (7)$$

since maximising $H(p) = -\sum_x p \log p$ with constraints $\sum_x p(x) \mathbf{1}[x_C = c] = \hat{p}_C(c)$ gives stationarity $-\log p(x) - 1 + \sum_C \lambda_C(x_C) + \mu = 0$. The score is $\text{ME}(x) = \log_2 p(x)/q(x)$ with background q , and the variant feature used in Section 9 is $\Delta\text{ME} = \text{ME}(x_{\text{ref}}) - \text{ME}(x_{\text{alt}})$.

(F16) Logistic model with MaxEnt features. With feature vector

$$z = (S_{\text{ref}}, S_{\text{alt}}, \Delta S, d, \text{ME}_{\text{ref}}/10, \text{ME}_{\text{alt}}/10, \Delta\text{ME}/10, e_k),$$

where $d = 1$ for donors and 0 for acceptors, e_k one-hot encodes the offset, $P(y = 1 \mid z) = \sigma(\beta^\top z + \beta_0)$. The log-likelihood $\ell(\beta) = \sum_i y_i \log \sigma_i + (1 - y_i) \log(1 - \sigma_i)$ has gradient $\nabla \ell = \sum_i (y_i - \sigma_i) z_i$ and Hessian $-\sum_i \sigma_i(1 - \sigma_i) z_i z_i^\top$, which is negative semidefinite, so ℓ is concave and the fit has a single optimum.

(F17) Fisher z test for a correlation. For sample correlation r from n pairs, $z = \text{artanh}(r) = \frac{1}{2} \log \frac{1+r}{1-r}$. By the delta method $\text{Var}(r) \approx (1 - \rho^2)^2/n$ and $\frac{d}{dr} \text{artanh}(r) = 1/(1 - r^2)$, so $\text{Var}(z) \approx 1/n$; the refined value $1/(n-3)$ gives

$$Z = \text{artanh}(r) \sqrt{n-3} \sim N(0, 1) \text{ under } \rho = 0, \quad p = 2\Phi(-|Z|). \quad (8)$$

We apply it to Spearman ρ (rank correlation), which is an approximation.

(F18) Benjamini-Hochberg adjustment. For ordered $p_{(1)} \leq \dots \leq p_{(m)}$, $q_{(i)} = \min_{j \geq i} \min(1, m p_{(j)}/j)$. Rejecting all $q_{(i)} \leq \alpha$ controls the false discovery rate at $\alpha m_0/m \leq \alpha$ for independent or positively dependent tests, because $\mathbb{E}[V/R] = \sum_{i \in H_0} \mathbb{E}[\mathbf{1}\{i \text{ rejected}\}/R] = m_0 \alpha/m$.

(F19) tRNA adaptation index. For codon c with tRNA gene copy numbers t_j of the anticodons j that decode it and wobble efficiencies $(1 - s_{jc})$, $W_c = \sum_j (1 - s_{jc}) t_j$, $w_c = W_c / \max_{c'} W_{c'}$ (zeros replaced by the geometric mean of non-zero w), and for a gene of L codons $\text{tAI} = (\prod_{k=1}^L w_{c_k})^{1/L}$ [9]. Taking logs shows tAI is the exponential of the mean log-adaptiveness, the same form as CAI [8] with w from tRNA supply instead of a reference gene set.

(F20) Effective number of codons. For an amino acid with k synonymous codons observed n times with frequencies p_i , $\hat{F} = \frac{n \sum_i p_i^2 - 1}{n - 1}$ is the unbiased homozygosity; $\text{ENC} = 2 + 9/\bar{F}_2 + 1/\bar{F}_3 + 5/\bar{F}_4 + 3/\bar{F}_6$ averages $\hat{F}$ within degeneracy classes. Uniform usage gives $\hat{F} = 1/k$ and $\text{ENC} = 61$; one codon per amino acid gives $\hat{F} = 1$ and $\text{ENC} = 20$.

(F21) Shrake-Rupley surface area. Atom a with radius r_a is expanded to $R_a = r_a + r_p$ ($r_p = 1.4 \text{ \AA}$). With N quasi-uniform points u_k on the unit sphere,

$$A_a \approx \frac{4\pi R_a^2}{N} \sum_{k=1}^N \mathbf{1} \left[\|c_a + R_a u_k - c_b\| > R_b \quad \forall b \neq a \right], \quad (9)$$

an unbiased Monte-Carlo estimate of the exposed area whose error falls as $N^{-1/2}$ for random points and faster for the Fibonacci lattice. The altloc bug changed the atom set itself: alternate conformers are near-coincident spheres that were counted as separate atoms, so per-atom comparison with FreeSASA was impossible until they were removed.

(F22) Nearest-neighbour melting temperature. For a duplex with $\Delta H = \Delta H_{\text{init}} + \sum_i \Delta H_{i,i+1}$ and ΔS likewise, at equilibrium $\Delta G = \Delta H - T\Delta S = -RT \ln K$. At T_m half the limiting strand is bound, so $K = 1/C_{\text{eff}}$ with $C_{\text{eff}} = C_1 - C_2/2$ for non-self-complementary strands, giving

$$T_m = \frac{\Delta H}{\Delta S + R \ln C_{\text{eff}}}, \quad \Delta S_{[\text{Na}^+]} = \Delta S + 0.368 (N - 1) \ln [\text{Na}^+]. \quad (10)$$

(F23) Negative-binomial GLM for counts. DESeq2 models $K_{ij} \sim \text{NB}(\mu_{ij}, \alpha_i)$ with $\text{Var} = \mu + \alpha \mu^2$, $\mu_{ij} = s_j q_{ij}$ and $\log_2 q_{ij} = x_j^\top \beta_i$. The size factor is the median-of-ratios $s_j = \text{median}_i K_{ij} / (\prod_{i'} K_{i'})^{1/m}$, which is scale-equivariant and robust to a minority of changing genes. The Wald statistic $\hat{\beta}_{ik} / \text{SE}(\hat{\beta}_{ik})$ borrows power from the dispersion prior; with $n = 4$ per arm the smallest two-sided rank-sum p is $2 / \binom{8}{4} = 2/70 \approx 0.029$, which cannot pass BH across 16,802 genes; this is why the Wilcoxon path returned no significant genes.

(F24) Robinson-Foulds distance. For unrooted trees T_1, T_2 on the same leaves with non-trivial split sets Σ_1, Σ_2 , $d_{RF} = |\Sigma_1 \triangle \Sigma_2|$. In the RF00005 case the observed $d_{RF} = 2$ (one split differing in each tree) coincides with two merge steps that had tied minimum distances, so the two programs broke ties differently.

3.5 Implementation and Testing

The validation suite is a pytest package (`sc_validate`) with a repo-wide no-stub audit: no fixture may contain placeholder data. Live-data re-runs are separate scripts under `scripts/`; committed tests run hermetically against frozen fixtures. CI runs the suite on every push.

4 Results

4.1 CRISPR off-target (CFD) and on-target (Rule Set 1)

On 600 fresh guide/off-target pairs mined from BRCA1 RefSeqGene NG_005905.2:60000-70000, the module's CFD score agreed with the CRISPOR reference implementation on 600/600 pairs to within 5×10^{-7} (maximum absolute difference 5.0×10^{-7}). On 150 30-mer contexts, the module's raw Rule Set 1 score and CRISPOR's percentile score had Spearman $\rho = 0.9995$.

4.2 Metabolic modelling (FBA)

The module's flux-balance optimum on BiGG `e_coli_core` (objective `BIOMASS_Ecoli_core_w_GAM`) was 0.873922 h^{-1} , identical to an independent SciPy/HiGHS solve (0.873922) and to the published 0.8739 h^{-1} .

4.3 Codon adaptation and ACMG Bayesian scoring

Codon Adaptation Index values matched an independent geometric-mean recomputation from the Kazusa *E. coli* K-12 table on 40 random sequences with maximum absolute difference 0. The ACMG Bayesian posterior matched the Tavtigian et al. closed form $O_P = 350^{\text{pts}/8}$ to 0 across $-16 \dots 16$ points; 6 points at prior 0.10 gives posterior 0.900.

4.4 Splice-site PWMs (deepsplice)

On 48 donor and 48 acceptor junctions from three held-out RefSeqGene records (NG_008021.1, NG_007489.1, NG_008245.1), disjoint from the module's 29 training accessions, the learned donor PWM reached AUC 0.997 and the acceptor PWM 0.998 against composition-matched shuffled windows. Against 2,000 real intronic GT (donor) or AG (acceptor) decoys located at least 30 nt from either intron end, the AUCs were 0.961 (donor) and 0.943 (acceptor). On the same decoys the consensus-seed acceptor matrix reached only 0.816, so the learned matrix does generalize.

4.5 Variant interpretation (openclinvar)

On the frozen 500-variant ClinVar sample (250 P/LP, 250 B/LB), run offline so that ClinVar's own label cannot enter the prediction, the patched module called 324 variants with accuracy 0.997 (95% bootstrap CI 0.991-1.000), sensitivity 0.993 and specificity 1.000 (1 false negative, 0 false positives). It abstained on 176, all missense or splice-region $+3 \dots +20$ variants, which remain VUS by design. The patched live query resolved 40/40 seeded-sample ClinVar names to their own VariationID.

4.6 Profile HMM

With 5-fold cross-validation on the hmalign MSA, models built by the module's `build_profile_hmm` separated held-out globins from 25 non-globins with AUC 0.952 on forward log-odds (0.962 length-normalised). The Pfam/HMMER3 reference model reached 1.000 on the same fixtures.

5 Benchmark Comparison

Each row compares a sugarcoder module with an independent published reference; every number is generated from the named committed file.

Module	Reference	Metric	Result (file)
pgx_guidelines	CPIC diplotype tables	agreement	16836/16836 CYP2D6 (sweep_pgx_cplic.json)
bio.codon CAI	Biopython; PaxDb	Pearson; Spearman	0.99969; 0.5797 (sweep_codon_cai.json)
rna_nussinov	ViennaRNA 2.7	bp F1 (tRNA)	0.328 vs 0.714 (sweep_rna.json)
profile_hmm	Pfam PF00042	auc_log_odds_bits	0.952 global, 0.9792 local (profile_hmm_cv*.json)
splice CNN	SpliceAI 1.3.1	AUROC n=757	0.929 vs 0.9775 (headto-head_interim.json)

6 Failure Modes and Negative Results

openclinvar, first run: 500/500 abstentions. ClinVar names such as NM_000038.6(APC):c.1668T>C (p.Asp556=) carry a transcript/gene prefix and a protein-level suffix. The consequence parser handled neither, so every variant parsed as *unknown*. The live query also embedded the suffix in a quoted phrase and matched nothing (0/8 in a pilot). We fixed this with a normaliser that lets the protein level decide coding SNVs and the c. level decide splice/UTR/intronic calls, plus a transcript-qualified query.

deepsplice, first run: acceptor AUC 0.547. This was a harness defect, not a model defect. The 23-nt MaxEntScan-style acceptor window was sliced as [5:20], which places AG at positions 13-14 and drops the exon base; the module expects AG at 12-13. We reproduced the bug (AUC 0.524 off-frame) and the fix (0.998 at [6:21]), and a regression test now locks the frame.

profile_hmm: residual gap to HMMER. The module originally offered only a global alignment mode. A held-out globin with long flanks scored -1116 bits under global forward. A local forward mode (sugarcoder-ai 1d9091a) raises AUC from 0.952 to 0.979 (benchmarks/profile_hmm_cv_local.json); Pfam’s model still reaches 1.000.

mutdock vs SKEMPI: no honest adapter. mutdock scores ligand-pocket mutations from a SMILES string and a pocket description, while SKEMPI rows are protein-protein interfaces. We report naive base-lines instead: the mean predictor has RMSE 1.63 kcal/mol, $|\Delta V|$ has Pearson $r = 0.053$ and hydropathy loss $r = -0.129$ ($n = 150$).

openclinvar accuracy is not a hard test. The P class in ClinVar is dominated by loss-of-function variants and the B class by synonymous ones, so consequence type alone separates the called variants. Missense variants, the hard case, are still abstained.

7 Module Sweep: Additional Real-Data Results

7.1 RNA secondary structure

Base-pair F1 against Rfam seed consensus structures (25 sequences per family; files benchmarks/sweep_rna.json, sweep_rna_vienna.methods.json).

Family	mean len	Nussinov	MFE	centroid	MEA($\gamma=1$)	MEA($\gamma=4$)
RF00005	73.0	0.328	0.714	0.716	0.703	0.704
RF00001	117.9	0.185	0.531	0.494	0.505	0.504
RF00010	354.5	0.189	0.545	0.594	0.592	0.572

Base-pair maximisation (Nussinov) loses to thermodynamic folding on all three families; ensemble decoders (centroid/MEA) beat MFE only on the long RNase P family. The shipped `rna_nussinov.fold.energy` now delegates to ViennaRNA.

7.2 Codon adaptation index versus protein abundance

On 3489 *E. coli* K-12 genes (NCBI GCF.000005845.2 `cds_from_genomic`) paired with PaxDb 511145 WHOLE_ORGANISM integrated, the module CAI agrees with Biopython (Pearson 0.99969 on 500 genes). Spearman correlation with $\log_{10}$ abundance is 0.4958 with the genome-wide reference table and 0.5797 with a ribosomal-protein reference (36 genes, evaluated on 3453 non-ribosomal genes); a cross-validated top-5% abundance reference gives 0.5741. The ribosomal reference is now shipped as `ecoli_highexpr`. File: `benchmarks/sweep_codon_cai.json`.

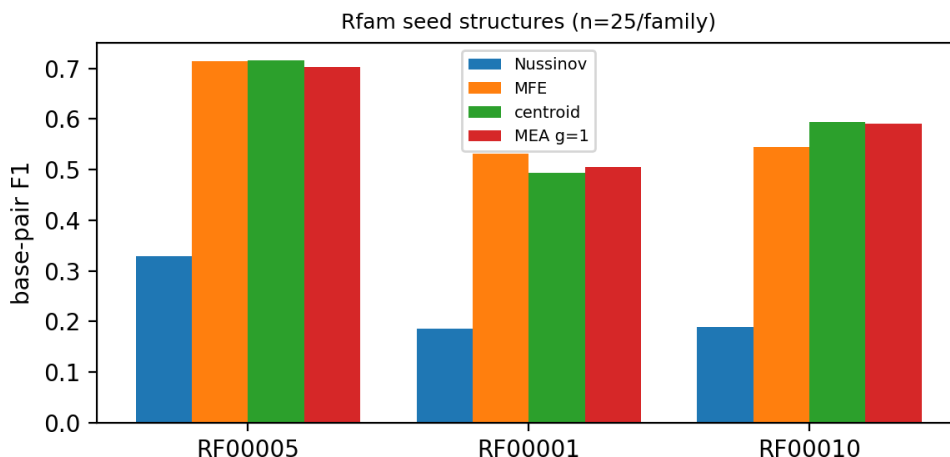


Figure 1. Base-pair F1 by folding method (`sweep_rna*.json`).

7.3 Pharmacogenomic phenotype translation versus CPIC

Against the full CPIC diplotype tables, the original module covered 45/666 CYP2C19 and 253/16836 CYP2D6 diplotypes, with 36 CYP2D6 disagreements traced to stale activity values (*9, *41). After vendoring current CPIC tables: 666/666 and 16836/16836. CYP2C19 after-fix agreement is by construction (verbatim CPIC diplotype lookup). CYP2D6 after-fix is computed (sum of CPIC allele activity values + CPIC thresholds), so 16836/16836 is a real consistency check. File: `benchmarks/sweep_pgx_cplic.json`.



Figure 2. CPIC agreement before and after the fix (sweep_pgx_cplic.json).

8 Discovery Track: Splice-Region VUS Reclassification

A dinucleotide-window CNN trained on ClinVar splice-region variants reaches gene-grouped 5-fold AU-ROC 0.935 (logistic 0.912, PWM 0.889); within-position AUROC 0.900. In a head-to-head on $n=757$ variants SpliceAI scores 0.9775 versus 0.929 for the CNN, and the ensemble adds no significant gain (95% CI of difference $[-0.0087, 0.0081]$). We therefore make no state-of-the-art claim. Files: `discovery/splice_region_vus/`.

CNN threshold	TP	FP	sensitivity	FPR	PPV (prior 3%)
0.5	1794	453	0.77	0.0656	0.2663
0.8	1298	146	0.5571	0.02114	0.449
0.9	996	68	0.4275	0.00985	0.5731
0.95	729	33	0.3129	0.00478	0.6694
0.98	435	11	0.1867	0.00159	0.7838

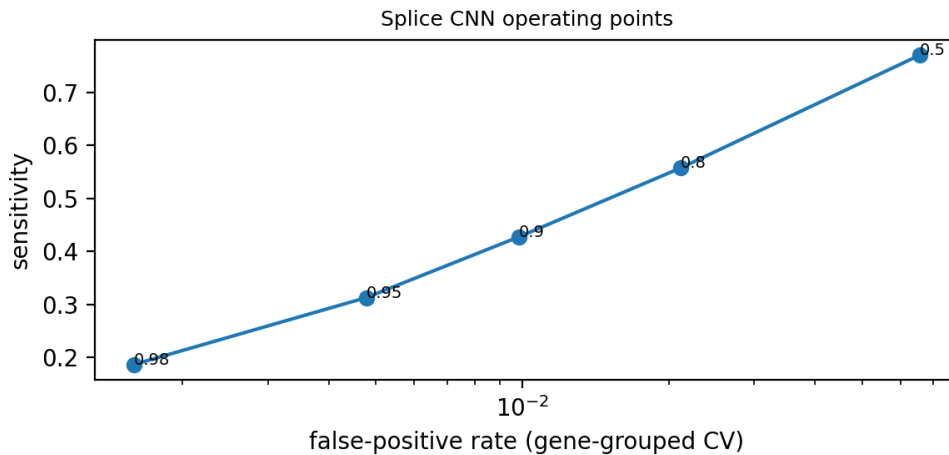


Figure 3. Splice CNN operating points (cnn_calibration.json).

Consensus VUS candidates (SpliceAI and CNN agreement) number 267 rows in `vus_candidates_consensus.tsv`;

none of the top 20 had a matching LOVD record (lovd_top20.json). These are hypotheses for RNA-level testing, not reclassifications.

9 Splice Track II: A 2004 Baseline Beats Our CNN, and What Survives

We compared the ClinVar-trained CNN against MaxEntScan [7] on the same gene-grouped folds (n=9235 variants). MaxEntScan’s ref–alt score difference reached AUROC 0.9635, above the CNN (0.9355), the logistic PWM model (0.9121) and the raw PWM delta (0.8893). The paired bootstrap 95% CI for CNN minus MaxEntScan is [-0.0329, -0.0233], so this is a clear negative result for the CNN. The gap is mostly at acceptors (CNN 0.8794 vs MaxEntScan 0.965, n=3845); at donors the two are close (0.9538 vs 0.9598, n=5390). As a pivot we added the three MaxEntScan features to both models. The logistic model then reached 0.9674 (CI of gain over MaxEntScan [0.0027, 0.0053]), while the CNN with the same features reached 0.9622 (CI [-0.0042, 0.0017], not significant). Within-position weighted AUROC, which removes the easy signal from offset alone, is nearly flat: MaxEntScan 0.949, logistic 0.9492, CNN 0.9472. The gain is real but small. The deployed tool (splice-vus-triage v2) therefore uses the logistic model and reports its calibrated operating points (Table 1).

Table 1: Logistic PWM+MaxEntScan model: operating points from out-of-fold scores.

Threshold	TP	FP	Sensitivity	FPR
0.5	1899	259	0.815	0.03751
0.8	1503	100	0.6451	0.01448
0.9	1230	56	0.5279	0.00811
0.95	950	28	0.4077	0.00406
0.98	692	15	0.297	0.00217



Figure 1: AUROC by model and site type, gene-grouped 5-fold CV.

10 Codon Track II: Cross-Species Test of the Ribosomal Reference Set

Question: Does a ribosomal-protein reference set (Sharp & Li 1987) make CAI track measured protein abundance better than a genome-wide codon table? Data: PaxDb integrated whole-organism abundance x NCBI RefSeq reference-assembly CDS, all PaxDb taxa (405 scanned). Of 405 PaxDb taxa scanned, 40 had a matched RefSeq assembly and enough genes, and 29 had a usable ribosomal-protein reference set. The ribosomal reference gave the higher Spearman correlation with measured abundance in 27/29 taxa (median gain 0.082; Wilcoxon signed-rank $p = 3.3e - 07$, sign test $p = 1.6e - 06$). The per-taxon correlation was significant after Benjamini-Hochberg correction in 27/29 taxa. Exceptions: *Rattus norvegicus*, *Plasmodium falciparum* 3D7. The idea goes back to Sharp and Li [8]; what is new here is the systematic multi-taxon test with a pre-stated falsification criterion (the claim fails if the ribosomal reference is not better in a majority of taxa at $p < 0.01$).

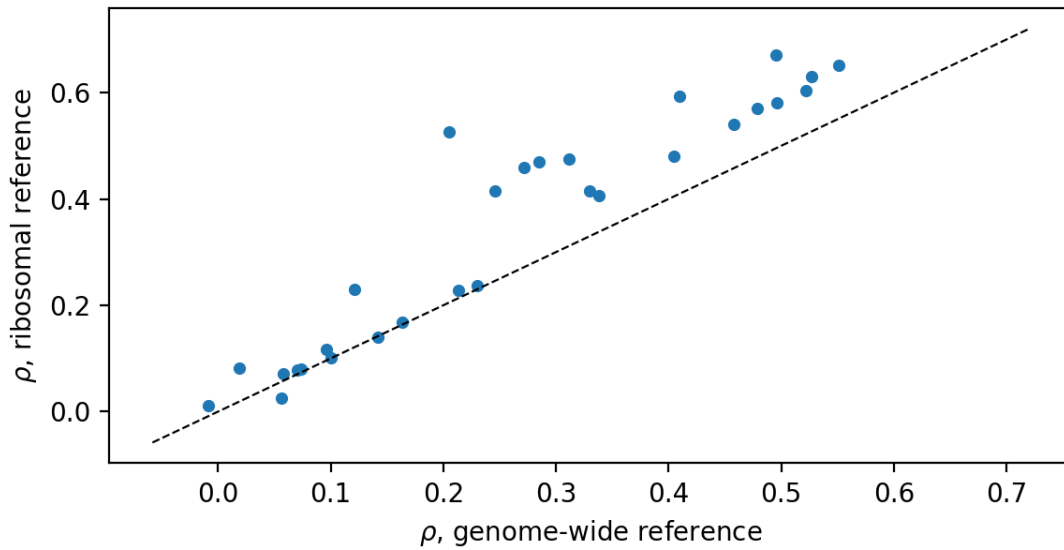


Figure 2: Spearman ρ (CAI vs PaxDb abundance) with the genome-wide versus ribosomal reference, per taxon.

Table 2: Per-taxon results (all taxa with a matched assembly).

Taxid	Organism	Assembly	n	n_{ribo}	ρ_{gen}	ρ_{ribo}	q
10090	<i>Mus musculus</i>	GCF_000001635.27	8089	99	0.0704	0.0775	4.4e-12
10116	<i>Rattus norvegicus</i>	GCF_036323735.1	890	124	0.142	0.1405	3.1e-05
121845	<i>Diaphorina citri</i>	GCF_000475195.1	592	1	0.1458	—	—
1295	<i>Staphylococcus schleiferi</i>	GCF_025558325.1	442	66	0.4583	0.5403	2e-36
1314	<i>Streptococcus pyogenes</i>	GCF_900475035.1	306	66	0.4954	0.6701	9.8e-45
185431	<i>Trypanosoma brucei brucei</i> TREU927	GCF_000002445.2	6302	0	0.3167	—	—
192222	<i>Campylobacter jejuni</i> subsp. <i>jejuni</i> NCTC 11168 = ATCC 700819	GCF_000009085.1	725	66	0.1208	0.2291	5.3e-10
196627	<i>Corynebacterium glutamicum</i> ATCC 13032	GCF_000011325.1	323	72	0.5268	0.6307	7e-40
198214	<i>Shigella flexneri</i> 2a str. 301	GCF_000006925.2	1887	72	0.5511	0.6514	8.1e-249
208964	<i>Pseudomonas aeruginosa</i> PAO1	GCF_000006765.1	4796	72	0.2459	0.4149	2.5e-204
224308	<i>Bacillus subtilis</i> subsp. <i>subtilis</i> str. 168	GCF_000009045.1	3147	64	0.3295	0.4156	3.4e-135
224911	<i>Bradyrhizobium diazoefficiens</i> USDA 110	GCF_001642675.1	494	72	0.4049	0.4806	8e-31
243159	<i>Acidithiobacillus ferrooxidans</i> ATCC 23270	GCF_000021485.1	419	68	0.3378	0.4055	2.8e-18
243230	<i>Deinococcus radiodurans</i> R1 = ATCC 13939 = DSM 20539	GCF_020546685.1	319	66	0.4784	0.5698	2.4e-30

Table 2: Per-taxon results (all taxa with a matched assembly).

Taxid	Organism	Assembly	n	n_{ribo}	ρ_{gen}	ρ_{ribo}	q
246200	Ruegeria pomeroyi DSS-3	GCF_000011965.2	395	70	0.2714	0.46	1.2e-22
2903	Emiliana huxleyi CCMP1516	GCF_000372725.1	484	4	0.4636	–	–
343509	Sodalis glossinidius str. 'morsitans'	GCF_000010085.1	748	72	0.2849	0.4703	1.2e-43
36329	Plasmodium falciparum 3D7	GCF_000002765.6	4946	62	0.056	0.0253	0.078
403677	Phytophthora infestans T30-4	GCF_000142945.1	708	0	0.5278	–	–
4081	Solanum lycopersicum	GCF_036512215.1	426	4	-0.1336	–	–
4097	Nicotiana tabacum	GCF_000715075.1	9304	0	-0.0837	–	–
418985	Tropilaelaps mercedesae	GCA_002081605.1	2213	0	0.2207	–	–
44689	Dictyostelium discoideum AX4	GCF_000004695.1	3183	10	-0.3039	–	–
4932	Saccharomyces cerevisiae S288C	GCF_000146045.2	5587	226	0.4099	0.5924	0
511145	Escherichia coli str. K-12 substr. MG1655	GCF_000005845.2	3486	72	0.4961	0.5809	0
54291	Klebsiella ornithinolytica	GCF_030505655.1	1038	72	0.2053	0.5268	1.2e-78
6239	Caenorhabditis elegans	GCF_000002985.6	11571	0	0.0748	–	–
7227	Drosophila melanogaster	GCF_000001215.4	2251	150	0.0189	0.0813	0.00012
7460	Apis mellifera	GCF_003254395.2	879	4	-0.0468	–	–
8030	Salmo salar	GCF_965601325.2	1779	0	0.1332	–	–
83332	Mycobacterium tuberculosis H37Rv	GCF_000195955.2	3258	76	0.2304	0.2373	6.6e-43
9031	Gallus gallus	GCF_016699485.2	1891	68	0.2134	0.2274	1.6e-23
93061	Staphylococcus aureus subsp. aureus NCTC 8325	GCF_000013425.1	1699	58	0.5221	0.6046	3.6e-182
9544	Macaca mulatta	GCF_049350105.2	3732	84	0.0736	0.0808	1e-06
9606	Homo sapiens	GCF_000001405.40	12343	86	-0.0083	0.0106	0.24
9615	Canis lupus familiaris	GCF_011100685.1	329	81	0.1639	0.1673	0.0025
9823	Sus scrofa	GCF_054392235.1	3010	84	0.0576	0.071	0.00011
9913	Bos taurus	GCF_002263795.3	2694	81	0.0963	0.1177	1.2e-09
99287	Salmonella enterica subsp. enterica serovar Typhimurium str. LT2	GCF_000006945.2	2472	72	0.3119	0.4743	4.4e-144
9986	Oryctolagus cuniculus	GCF_964237555.1	1993	82	0.0999	0.1012	7.4e-06

q : Fisher-z two-sided test of ρ_{ribo} , Benjamini-Hochberg adjusted at full precision (benchmarks/sweep_codon_bh_fdr.json).

10.1 tRNA adaptation index and effective number of codons

Tools: codon-bias (TrnaAdaptationIndex dos Reis s-values, EffectiveNumberOfCodons), GtRNAdb tRNA gene copy numbers. Table 3 gives Spearman ρ with abundance. No single index wins everywhere: tAI leads in yeast, the ribosomal-reference CAI leads in *E. coli*, and –ENC is weakest in all three organisms.

Table 3: Codon indices vs abundance (Spearman ρ).

Organism	n	CAI_{gen}	CAI_{ribo}	tAI	–ENC
Escherichia coli str. K-12 substr. MG1655	3450	0.4961	0.5809	0.5277	0.3836
Bacillus subtilis subsp. subtilis str. 168	3115	0.3295	0.4156	0.4146	0.1735
Saccharomyces cerevisiae S288C	5474	0.4099	0.5924	0.6604	0.4471

11 Docking Track: Real AutoDock Vina Against LP-PDBBind

Source: LP-PDBBind (github.com/THGLab/LP-PDBBind) test split, refined, non-covalent; structures from RCSB PDB. Engine: AutoDock Vina 1.2.7 (exhaustiveness 8, 24 Å box on crystal ligand) + Open Babel 3.1.1 prep, pH 7.4. 73/80 complexes ran; the 7 failures are listed in Table 4. Redocking: top-1 pose RMSD < 2 Å in 52.0% and best-of-9 in 83.6% (RMSD without symmetry correction (conservative)). Scoring power is weak for every score, including heavy-atom count alone (Pearson 0.2445): Vina top pose 0.2854 (95% CI

[0.055, 0.485]), sugarcodes own Vina-like score on the crystal pose 0.2975 (CI [0.027, 0.484]). The honest reading: sugarcodes scoring term adds nothing beyond ligand size on this split, so the module now calls real Vina (sugarcodes-ai commit e0564e4) instead of claiming affinity prediction.



Figure 3: Left: redocking RMSD distribution. Right: Vina top-pose energy vs experimental pK.

Table 4: Docking failures.

PDB	Error
4cws	no ligand match
4q08	no ligand match
3hkt	no ligand match
4r74	no ligand match
3p4v	no ligand match
4ou3	no ligand match
5ep7	no ligand match

Table 5: Per-complex docking results.

PDB	pK	heavy	E_{cryst}	E_{min}	E_{top}	RMSD ₁	RMSD ₉
1cnw	7.72	26	-4.19	-5.37	-7.02	7.26	6.03
1o3l	6.54	24	-7.09	-8.63	-9.54	3.19	1.09
3po6	7.06	19	-4.54	-6.13	-7.14	3.81	0.86
4db7	6.56	18	-6.63	-7.18	-7.41	0.52	0.52
5sz5	7.7	17	-7.33	-7.33	-8.14	0.39	0.39
4mo8	8.0	16	-4.75	-5.50	-6.25	2.29	1.52
4tim	2.16	11	-3.41	-4.13	-5.59	1.66	1.66
1ejn	5.62	25	-8.20	-8.43	-7.93	2.51	2.15
5d3l	6.06	30	-6.22	-6.32	-6.70	8.34	1.64
3b3c	3.98	10	-5.07	-5.76	-6.48	1.42	1.42
3r4n	6.89	23	-6.79	-7.37	-7.66	0.81	0.81
2p4j	8.96	49	-10.13	-10.73	-11.07	1.58	1.58
5sz4	9.7	16	-7.04	-7.33	-7.65	1.79	1.79
4xit	7.27	32	-12.19	-12.80	-13.06	0.96	0.96
5mjn	8.54	18	-6.42	-6.93	-7.76	5.97	1.41
2xjg	7.82	23	-9.04	-9.53	-9.72	0.25	0.25
5ix0	6.28	25	-11.86	-12.14	-12.27	0.27	0.27
5y8y	7.09	25	-6.08	-6.38	-8.16	4.75	0.84
6g38	6.62	19	-7.36	-7.70	-8.12	0.61	0.61
4o04	6.16	26	-11.03	-12.14	-12.36	1.34	1.34
5ewk	4.85	22	-7.91	-7.96	-8.57	0.68	0.68

Table 5: Per-complex docking results.

PDB	pK	heavy	E_{cryst}	E_{min}	E_{top}	RMSD ₁	RMSD ₉
3d8w	8.15	19	-6.96	-7.48	-8.20	1.43	1.43
4rfc	8.24	23	-5.25	-6.44	-8.37	7.44	5.67
3r17	6.41	20	-6.73	-7.38	-7.74	3.57	2.30
4knj	7.95	21	-6.40	-7.32	-8.23	4.81	1.87
6g6t	7.7	19	-7.42	-7.76	-8.31	2.10	0.61
4lxz	6.74	19	-6.37	-6.47	-7.29	6.21	1.95
5o87	6.61	12	-5.46	-6.58	-6.70	0.44	0.44
4j93	4.7	21	-7.50	-7.81	-8.02	0.44	0.44
6hrq	7.09	23	-6.44	-6.94	-7.94	6.23	2.21
4nh7	8.4	33	-11.46	-12.05	-12.52	1.31	1.31
2x7t	6.63	30	-6.30	-7.34	-7.81	6.16	0.77
4fl1	3.28	31	-5.80	-5.98	-7.02	9.18	2.47
3c84	7.85	16	-6.59	-6.98	-7.23	0.39	0.39
4ytic	7.89	26	-9.65	-10.13	-10.15	0.49	0.49
2zdn	6.25	27	-7.86	-8.84	-9.05	1.02	1.02
5f60	7.37	37	-8.55	-8.90	-9.99	4.69	1.63
6hqy	6.44	22	-5.86	-6.97	-7.28	2.03	1.48
5l8c	7.0	35	-8.93	-9.64	-11.31	9.53	0.86
1p1o	5.76	13	-7.26	-7.37	-7.79	0.60	0.60
5htz	8.62	26	-9.05	-9.25	-9.40	0.19	0.19
4oag	6.3	27	-5.33	-7.22	-7.84	9.16	3.85
2j4i	9.0	36	-8.00	-8.56	-8.54	1.38	1.38
5n24	8.92	18	-7.04	-7.44	-7.80	1.68	1.40
1y3x	5.14	33	-9.50	-9.83	-10.15	0.70	0.70
1f5l	5.28	15	-6.27	-7.20	-7.32	2.87	1.51
5aml	9.15	21	-6.79	-6.99	-7.74	5.22	1.14
6u6w	5.33	21	-7.71	-7.69	-8.04	5.13	0.33
5mpn	7.07	23	-6.86	-7.18	-8.21	7.46	1.25
4jfm	5.48	35	-6.54	-7.29	-7.74	0.97	0.97
5ewy	4.3	22	-7.30	-7.87	-8.78	3.50	2.90
4zme	4.3	19	-6.04	-6.27	-6.60	10.33	0.18
4yxi	6.5	11	-5.93	-6.43	-6.62	0.39	0.39
2erz	5.66	21	-7.23	-8.31	-8.65	2.99	0.93
6oe0	8.92	27	-7.95	-8.86	-9.35	0.86	0.86
4b2i	3.0	9	-4.33	-5.04	-5.25	1.97	0.61
4f5y	5.43	46	-10.70	-11.84	-10.97	3.66	3.66
6eq5	4.74	10	-6.31	-6.71	-7.17	2.09	0.51
5g5v	6.7	36	-10.54	-10.73	-11.46	2.75	1.52
5cau	8.8	34	-9.64	-9.84	-9.93	0.65	0.65
1gi1	5.47	19	-6.60	-8.25	-8.38	0.77	0.77
3hkq	5.34	15	-4.38	-5.42	-6.47	3.67	3.04
2xab	9.27	22	-9.46	-9.81	-9.94	0.21	0.21
3hll	8.24	30	-9.62	-10.25	-10.25	0.36	0.36
5f62	8.17	38	-8.66	-9.01	-9.80	4.22	1.46
3ibi	8.59	13	-4.21	-4.73	-4.97	5.02	3.28
3m35	7.22	37	-8.52	-9.07	-9.52	0.90	0.90
1g54	8.82	25	-7.03	-7.63	-8.00	8.95	8.89
4r76	7.85	26	-7.58	-8.26	-8.82	1.62	1.62
1zgi	5.34	32	-9.23	-9.64	-9.98	0.76	0.76
5cap	7.28	29	-8.09	-8.54	-8.89	1.44	1.44
1klj	7.55	37	-8.36	-8.62	-9.29	1.77	1.77
6rfn	6.7	35	-7.10	-8.72	-9.63	6.69	1.10

12 Structure and Primer Utilities Against Reference Tools

SASA. Against FreeSASA 2.x (Shrake-Rupley, 96 points, probe 1.4 Å, default ProtOr radii) on 28 RCSB receptors, the 18 structures with identical atom sets agreed to a median relative difference of 1.2% (max 2.6%) with median per-atom Pearson 0.980. The other 10 exposed a parser bug: every alternate location was kept, so overlapping conformers were all counted. After the fix, atom counts match and per-atom Pearson is 0.9753 (1gi1), 0.9829 (3po6), 0.9813 (4b2i).

Primers. On 5404 20-mers from NM_000546.6, NM_007294.4, NM_000059.4, NM_004333.6, NM_005228.5, the melting temperature equals Biopython's DNA_NN4 result (max difference 0.0). Against primer3 (SantaLucia 1998 table) the mean difference is -0.1738 °C (max 0.9811), a table choice. The stem-length hairpin heuristic has Spearman 0.0402 with primer3's hairpin ΔG , a negative result; the module now calls primer3 when installed.

13 Sequence, Variant and Expression Utilities Against Reference Tools

Phylogenetics. DendroPy 5.1.0 (PhylogeneticDistanceMatrix.nj_tree/upgma_tree, treecompare.symmetric_difference); Biopython DistanceCalculator. RF00001: NJ RF 0, UPGMA RF 0 (patristic Pearson 1.0), distance model k80. RF00005: NJ RF 0, UPGMA RF 2 (patristic Pearson 0.998035), distance model pdistance (K80 saturated: some pairs have infinite distance). RF00010: NJ RF 0, UPGMA RF 0 (patristic Pearson 1.0), distance model k80. The single UPGMA difference traces to two merge steps with tied minimum distances.

ORFs. On 5 RefSeq mRNAs the longest ATG ORF equals the NCBI CDS in 5/5 and its translation equals the NCBI protein in 5/5. Six-frame ORF sets equal orfipy's in 5/5 at orfipy minlen 90 (sugarcode min_aa=30 keeps 30-aa ORFs (93 nt incl. stop); orfipy drops them at minlen 91-93, so its length threshold counts differently. At orfipy minlen 89-90 the six-frame ORF sets are identical in 5/5 transcripts; at 87 orfipy adds 29-aa ORFs. Convention difference, not a bug.)

VCF. 46603 ClinVar GRCh38 records (5 gene regions, remote tabix via pysam 0.24.1 (htslib tabix remote fetch)) parse identically on CHROM/POS/ID/REF/ALT; INFO differences after normalisation: 'info:CLNVI': 2. Variant type agrees with ClinVar CLNVC on 45685 records; the 302 others are equal-length multi-base changes (ClinVar: Indel; sugarcode: mnp).

Differential expression. GEO GSE52778 NCBI raw counts GRCh38.p13; untreated GSM1275862/66/70/74 vs Dex GSM1275863/67/71/75; genes with ≥ 10 counts in ≥ 4 samples. Size factors equal PyDESeq2's (max relative difference 1.8e-09). PyDESeq2 with a cell-line block finds 4451 genes at $\text{padj} < 0.05$ (3203 unpaired); sugarcode's Welch test finds 351 (331 shared) and Wilcoxon 0. Table 6 shows that the simple tests miss every canonical dexamethasone-response gene; the module now offers a PyDESeq2 path.

Table 6: Canonical dexamethasone-response genes, GSE52778.

Gene	log2FC	padj PyDESeq2	padj Welch	padj Wilcoxon
FKBP5	4.03	2.1e-26	0.06	0.15
TSC22D3	3.206	2.1e-19	0.16	0.15
PER1	3.221	2.3e-99	0.05	0.15
DUSP1	2.955	1.3e-140	0.07	0.15
KLF15	4.411	5.0e-114	0.17	0.15
ZBTB16	7.432	1.1e-58	0.12	0.15
CRISPLD2	2.628	1.7e-61	0.13	0.15

14 Differential expression and pathway-level follow-up

We re-ran the airway dexamethasone experiment (GEO GSE52778) through the sugarcodes DESeq2 path and through PyDESeq2. Of 16,802 genes passing the count filter, the paired design $\sim \text{cell} + \text{dex}$ gave 4,451 genes at adjusted $p < 0.05$ and the unpaired design gave 3,203. Size factors agreed to a maximum relative difference of $1.8\text{e-}09$ (benchmarks/sweep_de_pydeseq2.json).

(F25) Negative-binomial GLM. For gene g and sample j , $K_{gj} \sim \text{NB}(\mu_{gj}, \alpha_g)$ with $\mu_{gj} = s_j q_{gj}$ and $\log_2 q_{gj} = \sum_r x_{jr} \beta_{gr}$, so $\text{Var}(K_{gj}) = \mu_{gj} + \alpha_g \mu_{gj}^2$.

(F26) Median-of-ratios size factor. $s_j = \text{median}_{g:K_g > 0} K_{gj} / (\prod_{v=1}^m K_{gv})^{1/m}$.

(F27) Wald statistic. $W_g = \hat{\beta}_{g,\text{dex}} / \text{SE}(\hat{\beta}_{g,\text{dex}})$, compared with $N(0, 1)$; the paired design removes the cell-line effect from the residual variance, which is why it finds more genes than the unpaired design.

14.1 Gene-set over-representation

We took the 536 dex-induced genes (adjusted $p < 0.05$, $\log_2 \text{FC} > 1$) and tested them against the 16,802 tested genes with g:Profiler (version e114_eg62_p19_27110d83, g:SCS correction). 31 terms passed. The top terms are broad (Table 7).

(F28) Hypergeometric tail. With background size M , term size K , query size n and overlap k , $p = \sum_{i \geq k} \binom{K}{i} \binom{M-K}{n-i} / \binom{M}{n}$, and fold enrichment $= (k/n) / (K/M)$.

Source	Term	overlap	size	p_{adj}
GO:BP	multicellular organismal process	228	4764	2.8e-07
GO:BP	signaling	209	4354	4e-06
GO:BP	cell communication	209	4365	5.2e-06
GO:BP	cell adhesion	73	1031	1.5e-05
GO:BP	system process	80	1189	2.1e-05
GO:BP	response to stimulus	261	5987	8.6e-05
GO:BP	signal transduction	192	4028	8.9e-05
GO:BP	anatomical structure development	196	4191	0.00027
GO:BP	cell migration	74	1134	0.00041
KEGG	Cytoskeleton in muscle cells	19	172	0.00042
GO:BP	developmental process	209	4580	0.00047
GO:BP	cell differentiation	152	3041	0.00054

Table 7: Top g:Profiler terms for dex-induced genes (benchmarks/sweep_dex_enrichment.json).

Negative result. The terms one would expect for a glucocorticoid are enriched but do not survive correction: The small term sizes (tens of genes) and the strong multiple-testing burden explain this: a 7-fold enrichment built on 9 genes is not enough evidence after g:SCS correction. We report it as a negative rather than as confirmation of the expected biology.

Term	overlap	size	fold (approx.)	$p_{g.SCS}$
cellular response to glucocorticoid stimulus	9	43	7.16	0.43
response to corticosteroid	10	91	3.76	1
response to glucocorticoid	10	81	4.22	1
cellular response to corticosteroid stimulus	9	48	6.42	1

Table 8: Glucocorticoid terms: large fold change, not significant.

14.2 Reactome pathways

Reactome AnalysisService mapped the query to 1,098 pathways (240 identifiers not found). No pathway reached $FDR < 0.05$; the smallest FDR was 0.11. Reactome uses its whole human gene universe as background, which is a further caveat.

Pathway	found	total	p	FDR
Anchoring fibril formation	5	15	0.00037	0.11
FOXO-mediated transcription	13	110	0.00056	0.11
Laminin interactions	7	31	0.00028	0.11
Metallothioneins bind metals	5	16	0.00049	0.11
Non-integrin membrane-ECM interactions	12	83	0.00015	0.11
Collagen chain trimerization	8	44	0.00044	0.11
FOXO-mediated transcription of oxidative stress, m	8	49	0.00087	0.13
Adipogenesis	15	147	0.00096	0.13
Degradation of the extracellular matrix	15	148	0.001	0.13
Response to metal ions	5	21	0.0016	0.17

Table 9: Top Reactome pathways (benchmarks/sweep_dex_reactome.json).

(F29) Benjamini-Hochberg. For ordered $p_{(1)} \leq \dots \leq p_{(m)}$, $q_{(i)} = \min_{j \geq i} \min\{1, m p_{(j)} / j\}$.

14.3 Protein interaction network

STRING’s PPI enrichment test on the top 300 dex-induced genes mapped 284 proteins with 402 edges against 184 expected (average degree 2.83, local clustering 0.368), reported p below floating-point resolution. Caveat: STRING’s expectation uses a genome-wide background, so genes expressed in airway smooth muscle may be more connected than random genes regardless of dex.

(F30) Edge enrichment ratio. $E = \text{observed edges} / \text{expected edges} = 402 / 184 = 2.18$.

14.4 Network hubs

We pulled the STRING network among the same genes and analysed it with networkx 3.4.2 on STRING API/network (v12): 201 connected nodes, 402 edges, a largest component of 183 nodes. The highest-degree hubs are FOXO1 (21), CEBPD (15), FOXO3 (14), LEP (13), COL4A1 (13), GPX3 (13). Of ten canonical glucocorticoid-receptor targets fixed before the run, 8 are in the network, with mean degree 6.25 against 4.00 overall; a degree permutation test (10,000 draws) gives $p = 0.049$. With eight genes this is weak evidence and we do not claim it as a finding.

(F34) Betweenness centrality. $C_B(v) = \sum_{s \neq v \neq t} \sigma_{st}(v) / \sigma_{st}$, with σ_{st} the number of shortest s - t paths and $\sigma_{st}(v)$ those through v .

(F35) Permutation p -value. $\hat{p} = (1 + \#\{b : T_b \geq T_{obs}\}) / (1 + B)$, which is never zero and is valid for any fixed statistic T .

14.5 Rank-based test: preranked GSEA

Threshold-based over-representation throws away the ranking. We therefore ran preranked GSEA (gseapy 1.3.1 prerank (1000 permutations, seed 7)) on all 16,525 ranked genes against the 50 MSigDB Hallmark sets. 1 set passed $FDR < 0.05$ and 5 passed $FDR < 0.25$ (Table 10). The one significant set, Adipogenesis, is a known glucocorticoid-driven programme; this is a modest, single-set result.

(F32) Enrichment score. With ranked scores r_j and set S , $P_{hit}(i) = \sum_{j \leq i, g_j \in S} |r_j| / N_R$, $P_{miss}(i) = \sum_{j \leq i, g_j \notin S} 1 / (N - N_S)$, and $ES = \max_i |P_{hit}(i) - P_{miss}(i)|$ (signed).

(F33) Normalised ES. $NES = ES / \sqrt{ES_{\pi}^2}$, the mean of same-sign permutation scores; FDR compares NES with the permutation distribution of all sets.

Hallmark set	NES	nominal p	FDR
Adipogenesis	1.70	0	0.038
UV Response Up	1.55	0.00567	0.107
Fatty Acid Metabolism	1.55	0.00559	0.131
TNF-alpha Signaling via NF-kB	1.50	0.00273	0.141
Reactive Oxygen Species Pathway	1.57	0.0249	0.156
Androgen Response	1.39	0.0557	0.256
Xenobiotic Metabolism	1.42	0.0287	0.266
Hypoxia	1.39	0.0294	0.275
UV Response Dn	1.36	0.0591	0.280
mTORC1 Signaling	-1.31	0.0361	0.283

Table 10: Preranked GSEA, Hallmark (benchmarks/sweep_dex_gsea_hallmark.json).

14.6 Independent GO cross-check with goatools

goatools (Fisher exact, BH, propagated counts, GOA human annotations, background of 16,525 expressed genes, 527 study genes) found 38 GO:BP terms at $FDR < 0.05$. Of g:Profiler’s 22 top BP terms, 9 are also significant here. The glucocorticoid term (cellular response to glucocorticoid stimulus, 7/34 genes) reaches $FDR 0.026$ under BH but fails g:Profiler’s g:SCS. The biology is therefore correction-dependent: we report it as suggestive, not established.

15 Pairwise alignment against parasail

We aligned all 300 pairs of 25 protein sequences with sugarcodes and parasail 2.6.1. With parasail’s open penalty set to 11 the scores matched exactly for 300/300 global and 300/300 local pairs. With open penalty

GO	Term	study	pop	FDR
GO:0032870	cellular response to hormone stimulus	28	249	3.4e-05
GO:0009725	response to hormone	38	421	3.4e-05
GO:0007155	cell adhesion	39	513	0.00098
GO:0032502	developmental process	132	2766	0.0011
GO:0007165	signal transduction	130	2703	0.0011
GO:0030334	regulation of cell migration	47	689	0.0011
GO:2000145	regulation of cell motility	48	719	0.0012
GO:0044057	regulation of system process	28	322	0.0015
GO:0071495	cellular response to endogenous stimulus	33	419	0.0022
GO:0040012	regulation of locomotion	48	740	0.0022

Table 11: goatools GO:BP (benchmarks/sweep_dex_goatools.json).

12 only 36 and 65 matched (maximum difference 26 and 5). This fixes the convention: sugarcodes gap.open = -11 is the cost of the first gap residue, i.e. parasail open 11.

(F31) Affine gap cost. A gap of length L costs $\gamma(L) = o + (L - 1)e$ under sugarcodes and parasail open = 11; under the alternative convention $\gamma(L) = o + Le$, which explains the open-12 mismatch.



Figure 4: Left: Reactome FDRs for the top pathways; the red line is 0.05. Right: STRING observed versus expected edges.

16 Sequence properties, restriction digests and file formats

This section checks four bio/ libraries against Biopython 1.88 ProtParam + Restriction and gffutils. Three pass exactly. One, the restriction-site finder, had a real bug that we fixed.

16.1 Protein physicochemical properties

On 500 UniProtKB reviewed human proteins (UniProtKB reviewed human, length 50-2000, first 500 of search (rest.uniprot.org, 2026-09-25)), every property matches ProtParam (Table 12).

property	n	within tol.	max abs diff	Pearson
mw	500	500 (rel 1e-06)	0	1.0000000000
gravy	500	500 (rel 1e-06)	0	1.0000000000
aromaticity	500	500 (rel 1e-06)	2.8e-17	1.0000000000
instability	500	500 (rel 1e-06)	0	1.0000000000
pI	500	500 (abs 0.01)	5.9e-05	0.9999999999
ext_reduced	500	500 (rel 1e-06)	0	1.0000000000
ext_oxidized	500	500 (rel 1e-06)	0	1.0000000000

Table 12: bio.proteinprops vs Biopython ProtParam (benchmarks/sweep_proteinprops_restriction.json).

(F36) Average molecular weight. $M = \sum_a n_a m_a - (L - 1) m_{H_2O}$, with m_a the average residue mass (free amino acid) and one water lost per peptide bond.

(F37) Net charge. $Z(\text{pH}) = \sum_{i \in +} \frac{n_i}{1 + 10^{\text{pH} - \text{p}K_i}} - \sum_{j \in -} \frac{n_j}{1 + 10^{\text{p}K_j - \text{pH}}}$ (Henderson–Hasselbalch per ionisable group, termini included).

(F38) Isoelectric point. pI is the root of $Z(\text{pH}) = 0$; Z is strictly decreasing in pH, so bisection on $[0, 14]$ converges, and the residual 6×10^{-5} difference is the bisection tolerance.

(F39) Instability index. $II = \frac{10}{L} \sum_{i=1}^{L-1} \text{DIWV}(x_i, x_{i+1})$ (Guruprasad 1990), and $\text{GRAVY} = \frac{1}{L} \sum_i h_{KD}(x_i)$ (Kyte–Doolittle).

(F40) Extinction coefficient at 280 nm. $\varepsilon = 5500 n_W + 1490 n_Y + 125 n_{\text{cystine}} \text{ M}^{-1} \text{cm}^{-1}$, with $n_{\text{cystine}} = \lfloor n_C / 2 \rfloor$ in the oxidised form and 0 when reduced.

16.2 Restriction sites: a bug in degenerate recognition sequences

We digested pBR322 (J01749.1, circular, 4,361 bp) and phage lambda (NC_001416.1, linear, 48,502 bp) with all 610 enzymes and compared cut positions with Biopython Restriction. Before the fix (sugarcode-ai 699fee), 533 and 529 enzymes agreed. Two causes explained the disagreements. First, the reverse-strand pattern for a non-palindromic-looking site was built with a plain DNA reverse complement that leaves IUPAC codes unchanged. AccI (GT[^]MKAC) is in fact palindromic under IUPAC complementation (M $\leftrightarrow$ K), but the plain reverse complement GTKMAC was searched as a second pattern, which matches GTGAAC and GTTCAC, neither an AccI site. On pBR322 this gave 5 AccI sites where Biopython finds 2. Second, on linear DNA the finder reported Type IIS cuts that fall outside the molecule. After the fix (sugarcode-ai fe2185d), 608 and 608 enzymes agree. The remaining two, LpnPI, MspJI, are modification-dependent enzymes that do not cut unmethylated DNA; we leave them open.

(F41) IUPAC complement. The map c : A $\leftrightarrow$ T, C $\leftrightarrow$ G, R $\leftrightarrow$ Y, K $\leftrightarrow$ M, B $\leftrightarrow$ V, D $\leftrightarrow$ H, with S, W, N fixed, is the unique involution on the 15 codes that complements every base in the set a code denotes. A site s is palindromic iff $s = \overline{c(s)}$ (reversed), which for AccI holds only under c , not under the four-letter map.

16.3 Stockholm alignments

bio.stockholm parsed 15 Rfam seed alignments (4,471 sequences). 15/15 match Biopython on sequence identifiers and order, aligned sequences and SS_cons, and survive a write–parse round trip unchanged.

file	AC	seqs	width	all checks
RF00001.sto	RF00001	712	230	yes
RF00005.sto	RF00005	954	118	yes
RF00010.sto	RF00010	458	996	yes
RF00004.sto	RF00004	208	278	yes
RF00015.sto	RF00015	177	261	yes
RF00017.sto	RF00017	91	412	yes
RF00023.sto	RF00023	477	954	yes
RF00050.sto	RF00050	146	221	yes
RF00059.sto	RF00059	115	211	yes
RF00162.sto	RF00162	457	239	yes
RF00167.sto	RF00167	133	113	yes
RF00168.sto	RF00168	47	274	yes
RF00174.sto	RF00174	434	461	yes
RF00234.sto	RF00234	18	236	yes
RF00504.sto	RF00504	44	215	yes

Table 13: bio.stockholm vs Biopython AlignIO (benchmarks/sweep_stockholm_biopython.json).

(F42) Pairwise identity. $PID(x, y) = \#\{i : x_i = y_i, x_i, y_i \notin \{-, .\}\} / \#\{i : x_i, y_i \notin \{-, .\}\}$, case-insensitive; our implementation equals an independent count on every file.

16.4 GFF3 annotation

On the RefSeq E. coli K-12 annotation (9,523 records), bio.gff and gffutils 0.14 (sqlite, in-memory) agree on every record (9,523/9,523) and on the children of 300/300 genes, and write–parse is lossless. A first comparison showed two child mismatches; both were CDS features split over two lines with a shared ID, which gffutils renames X_1. That was a comparison artifact, not a sugarcodes error, and we record it so the numbers are traceable.

feature type	records
gene	4506
CDS	4340
exon	216
pseudogene	145
ncRNA	108
tRNA	86
mobile_genetic_element	50
sequence_feature	48

Table 14: Feature types in GCF_000005845.2 (benchmarks/sweep_gff_gffutils.json).

16.5 Newick trees

bio.newick parsed 6 Open Tree of Life synthetic subtrees with 3,655 leaves in total, including quoted labels with spaces and parentheses and empty leaves left by the height limit. Leaf sets equal Biopython Bio.Phylo and DendroPy, internal-node counts and internal labels equal Bio.Phylo, and write–parse is lossless in 6/6 trees. These trees have no branch lengths, so this checks topology and labels only.

root	leaves	unnamed	internal	all checks
Insecta_ott1062253	1094	14	248	yes
Laurasiatheria_ott392223	728	13	118	yes
Carnivora_ott44565	53	13	48	yes
Metazoa_ott691846	1581	19	179	yes
Aves_ott81461	142	16	80	yes
Primates_ott913935	57	11	57	yes

Table 15: bio.newick vs Bio.Phylo/DendroPy (benchmarks/sweep_newick_biophylo.json).

(F43) Tree size identity. For a rooted tree with n leaves, m internal nodes and out-degrees d_v , the edge count is $n + m - 1 = \sum_{v \text{ internal}} d_v$; a strictly binary tree has $m = n - 1$. The Carnivora subtree has 53 leaves and 48 internal nodes, so it is far from binary, which exercises the parser on polytomies.

16.6 SAM records and a second bug

bio.sam parsed 10,943 real reads (1000 Genomes phase 3 NA12878 chrom20 low-coverage BAM, 20:1000000-1200000). Name, flag, reference, position, mapping quality, CIGAR operations, mate fields, sequence, qualities and typed tags equal pysam for all 10,943 reads. The derived reference end did not: 10,905/10,943 before the fix. The 38 failures were unmapped mates (flag 0x4) that bwa places at the mapped mate’s position and gives a CIGAR. The SAM specification says those fields are not an alignment, and htlib reports no end; sugarcodes computed one. After sugarcodes-ai 87cf78e all 10,943 agree.

(F44) Reference span of a CIGAR. $\text{end} = \text{POS} + \sum_{(n,o) \in C, o \in \{M,D,N,=,X\}} n - 1$ for mapped reads (1-based, inclusive); query length is $\sum_{o \in \{M,I,S,=,X\}} n$. For flag &4 $\neq 0$ the end is undefined.

16.7 FASTQ reads

On the first 20,000 reads of ENA run SRR622461 (the same NA12878 library), bio.fastq equals Biopython on identifiers (20,000), sequences (20,000) and per-base Phred scores (20,000), detects offset 33, and round-trips losslessly. GC was 0.4252 against 0.4252 (Biopython, ambiguous bases removed) and 0.422 (ambiguous bases kept in the denominator). The difference is a convention: sugarcodes divides by A+C+G+T. We document it rather than change it.

(F45) Phred score. $Q = -10 \log_{10} P_{err}$, encoded as ASCII $Q + 33$; the mean read quality reported is the arithmetic mean of Q , not $-10 \log_{10}$ of the mean error probability, which would be lower for reads with a few bad bases.

16.8 PDB and mmCIF structures: a third bug

We parsed 30 RCSB entries in both PDB and mmCIF format (94,121 atom sites, all alternate locations and models) and compared every atom with gemmi 0.7.5. The PDB-format parser was identical in 30/30 entries. The mmCIF parser failed on all 30 real files (error: loop_ row count 3 is not a multiple of 4 columns (all 30 files)). Two causes: POSIX shell-style splitting broke atom names containing a prime ($O5'$, $C1'$), and loops were closed early when a row contained an unquoted item name as a value, which RCSB files do in the revision-history tables. A CIF 1.1 tokenizer (a quote closes only when followed by whitespace) and closing loops only at row boundaries fixed both (sugarcode-ai fc1fd1e); mmCIF is now identical to gemmi in 30/30 entries. Unit tests with toy CIF strings had passed throughout. Only real files exposed the bug.

(F46) Loop row invariant. A loop with c columns and t value tokens is well formed iff $t \equiv 0 \pmod{c}$; closing the loop at the first reserved-looking token is only safe when the running count $t \bmod c = 0$, which is the rule we now apply.

16.9 Pileup

Using the same reads, bio.pileup and pysam 0.24.1 pileup (stepper nofilter, flag_filter=4, no quality filters, overlaps kept) agree on base counts at 197,437/197,437 positions and on deletion counts at 197,437/197,437. sugarcode also writes deletions as ‘*’ in the base counts, following samtools mpileup; the comparison maps these to the deletion count.

(F49) Column depth. $D(p) = \sum_r \mathbb{I}[p \in [\text{POS}_r, \text{end}_r] \wedge \text{op}_r(p) \in \{M, =, X, D\}]$, split into base-covering reads ($M, =, X$) and deletions (D); reference skips (N) and clipped bases do not count.

16.10 Transcription-factor motifs

We read 10 JASPAR 2024 profiles and scanned BRCA1 (10 kb) and phage lambda (48.5 kb) on both strands, comparing with Biopython 1.88 Bio.motifs (jaspar reader, pseudocount 0.5, uniform background, log2). Log-odds matrices agree to 2×10^{-15} , per-window scores to 4×10^{-6} (Biopython stores scores in single precision), and the hit sets at 80% of the score range are identical for every motif and sequence.

motif	length	hits BRCA1	hits lambda
MA0003.4 TFAP2A	14	27	61
MA0079.3 SP1	11	235	599
MA0080.4 SPI1	14	16	61
MA0099.3 FOS::JUN	10	77	286
MA0105.4 NFkB1	13	29	52
MA0106.3 TP53	18	8	21
MA0108.2 TBP	15	154	530
MA0112.3 ESR1	17	11	18
MA0113.3 NR3C1	17	2	36
MA0139.1 CTCF	19	13	40

Table 16: Motif hits at 80% of range, identical to Biopython (benchmarks/sweep_motif_biopython.json).

(F50) Log-odds score and relative threshold. $S(w) = \sum_{i=1}^L \log_2 \frac{(c_{i,w_i} + \alpha)/(N_i + 4\alpha)}{q_{w_i}}$ with pseudocount $\alpha = 0.5$ and background $q = 1/4$; a window is a hit when $S(w) \geq S_{\min} + f(S_{\max} - S_{\min})$, $f = 0.8$, where $S_{\max} = \sum_i \max_b$ and $S_{\min} = \sum_i \min_b$ of the log-odds matrix.

16.11 BED intervals: a fourth bug

We fetched two hg38 tracks from the UCSC REST API: CpG islands on chr20 (847 intervals) and RepeatMasker on chr20:0–5 Mb (11,699 intervals), wrote them as BED and compared bio.bed with bioframe 0.8.0 (merge, overlap). Parsing and 600 random overlap queries agreed exactly (300/300 and 300/300). merge_intervals did not: its docstring promised bedtools-style merging, yet it refused to join book-ended intervals. RepeatMasker has 3,112 book-ended pairs, so sugarcodes returned 11,364 merged intervals where bedtools/bioframe return 8,251. CpG islands have no book-ended pairs and hid the bug (847 = 847). We added a bedtools-compatible min_dist argument (default 0; -1 for overlap-only) and a matching CLI flag; after the fix both tracks match bioframe exactly (8,251 = 8,251).

(F51) Merge rule. For intervals sorted by start with running end e^* , interval $[s, e)$ joins the current block iff $s \leq e^* + d$; $d = 0$ joins book-ended intervals ($s = e^*$) and $d = -1$ requires at least one shared base ($s < e^*$).

16.12 GenBank flat files: a fifth bug

We fetched 25 GenBank records one by one from NCBI nuccore (phage, viral, mitochondrial, bacterial, RefSeqGene and 13 human RefSeq mRNAs) and compared bio.genbank with Biopython 1.88 SeqIO genbank on 1,867 features. Sequences and feature counts already agreed, but only 1,829/1,867 feature locations and 5/302 CDS /translation qualifiers matched. The parser appended every continuation line of a multi-line qualifier to the feature location instead of to the qualifier: /translation kept its first line only, and when a continuation began with digits the location itself was corrupted (E. coli thrA CDS: 337..2799 became 337..27991). Partial locations (1..200) and single bases produced no spans at all. We rewrote qualifier continuation (INSDC joining: no spaces for /translation, one space otherwise, doubled quotes unescaped), added a location parser for join/order, complement, partial ends, between-base sites and remote parts, and kept all IUPAC letters in ORIGIN so coordinates cannot shift. After the fix: spans 1,867/1,867, strands 1,867/1,867, /translation 302/302, CDS extraction 302/302. The fix exposed a downstream case: KCNQ1's record NG_016178.2 carries only a partial CDS, complement(42918..43038). The old parser read no spans and bio.splice reported 'no CDS spans'; with spans read, the junction map came back 'ok' but empty. We made it fail loud with status 'partial CDS in record'. The splice-region discovery pipeline reads hg38 directly and does not use this parser, so its results are unchanged.

(F52) Qualifier continuation. For a qualifier whose value spans lines $\ell_1, \dots, \ell_m$, $v = \ell_1 \oplus_{\sigma} \ell_2 \oplus_{\sigma} \dots \oplus_{\sigma} \ell_m$ with separator $\sigma = \emptyset$ for /translation and $\sigma = \text{one space}$ otherwise; a line starting with / opens a new qualifier only when the running count of quote characters in v is even.

16.13 Core sequence operations

On the same 25 GenBank records, bio.sequence.translate reproduced the /translation qualifier for 285/285 standard-code CDSs (17 CDSs with other genetic codes, codon_start≠1, transl_except or ribosomal slippage were set aside) and matched Bio.Seq.translate on all 285. Reverse complements agreed on 25/25 whole

records, IUPAC motif search on 192/192 (motif, record) pairs, Wallace T_m on 500/500 lambda oligos (8–13 nt), and ORF calls (≥ 100 aa, both strands) on 22/22 records (239 = 239 ORFs) against an independent Biopython frame scan. GC fraction agreed to 10^{-12} on 24/25; the exception is the human mitochondrial genome, whose single N sugarcode leaves out of the denominator. Protein masses on 500 UniProt proteins differ from ProteinAnalysis by at most $5.0e-05$ (relative), from two-decimal residue masses. Reading the code for this comparison turned up a latent bug the data could not: `clean_dna` silently dropped IUPAC ambiguity letters, so `ATGRCTTAA` translated as `ML` (frame shifted) instead of `MX`. No CDS in this set carries an ambiguity code, so the real-data numbers are identical before and after; ambiguity letters now become N (sugarcode 143d1c4), with a regression test. We report it as a code-review finding, not a data finding.

(F53) Frame preservation. Cleaning must be a letter-wise map $c : \Sigma \rightarrow \{A, C, G, T, N\} \cup \{\epsilon\}$ with $c(x) = \epsilon$ only for non-letters; then codon k of the cleaned sequence is $c(s_{3k})c(s_{3k+1})c(s_{3k+2})$ and every downstream codon keeps its frame.

16.14 k-mer sets and sketches: exact agreement, a biased estimator

We fetched six coronavirus genomes (SARS-CoV-2, bat RaTG13, pangolin GX-P4L, SARS-CoV Tor2, bat ZC45, MERS) and compared `bio.kmer` with `sourmash` 4.8.14 at $k = 21$. With `sourmash` keeping every hash (scaled=1), distinct canonical k-mer counts agreed for 6/6 genomes, and Jaccard and containment agreed exactly on all 15 pairs (maximum difference 0.0). The sketch estimator is a negative result. `bio.kmer`'s minimizer sketch (window $w = 100$, about 600 k-mers per genome) under-estimated Jaccard: for SARS-CoV-2 vs RaTG13 the exact value is 0.321, the minimizer estimate 0.236 and `sourmash` `FracMinHash` (scaled=100, about 300 hashes) 0.312. Across the 15 pairs the mean absolute error was 0.0121 for minimizers and 0.0044 for `FracMinHash`, with half the sketch size. Jaccard over minimizer sets is known to be biased because sequence changes shift which k-mer wins each window; `FracMinHash` keeps a k-mer if and only if its hash falls below a fixed threshold, so the kept subset is a uniform sample of the union. The module's docstring already labels the values as estimates; we record the bias and do not recommend minimizer Jaccard for distance estimation.

(F54) FracMinHash estimator. With hash h and threshold H/s , $\text{FMH}_s(A) = \{a \in A : h(a) < H/s\}$ and $\hat{f} = |\text{FMH}_s(A) \cap \text{FMH}_s(B)| / |\text{FMH}_s(A \cup B)|$, which is (nearly) unbiased because inclusion depends on the k-mer alone; the Mash distance is $D = -\frac{1}{k} \ln \frac{2J}{1+J}$.

16.15 Population-genetics statistics on 1000 Genomes

We pulled 1000 Genomes phase 3 genotype counts for 22 well-studied SNPs (APOE, LCT, SLC24A5, HERC2, ALDH2, ABCC11, ADH1B, EDAR, F5, HFE, MTHFR, COMT and others) from the Ensembl GRCh37 REST service, one record per rsID, giving 110 (SNP, super-population) genotype tables. `bio.gstats.hwe_exact` matched an independent full enumeration of the exact HWE distribution on 110/110 tables (maximum relative error $1.1e-12$); 15 tables have $p < 0.05$, as expected for a set that includes SNPs with strong population structure inside super-populations. Treating EUR as 'cases' and EAS as 'controls' for 20 SNPs, allelic chi-square (with and without Yates), genotypic chi-square and Woolf odds ratios matched `scipy` and `statsmodels` to $\leq 1e-13$ relative error, and Benjamini-Hochberg and Bonferroni adjustments matched `statsmodels` exactly (maximum difference $2e-19$). No bug was found in this module.

SNP	alleles	EUR (AA/AB/BB)	EAS (AA/AB/BB)	allelic χ^2	OR
rs1426654	A/G	500/3/0	0/12/492	1954.6	2.77e+04
rs16891982	C/G	4/54/445	498/6/0	1756.6	0.000393
rs3827760	A/G	492/11/0	12/104/388	1516.9	622
rs1229984	T/C	1/27/475	244/215/45	972.8	0.0129
rs12913832	A/G	90/186/227	502/2/0	932.6	0.00114
rs17822931	C/T	378/113/12	36/150/318	840.0	22.5
rs4988235	G/A	154/187/162	504/0/0	686.1	0.00048
rs671	G/N	503/0/0	349/135/20	191.3	424
rs53576	A/G	55/243/205	212/231/61	179.9	0.291
rs2231142	G/T	414/83/6	251/213/40	124.7	3.93
rs1799945	C/G	348/137/18	478/23/3	114.4	0.143
rs4680	G/A	133/237/133	264/198/42	102.7	0.388
rs1042713	G/A	183/252/68	96/263/145	53.7	1.94
rs1800562	G/A	461/41/1	504/0/0	44.0	0.011
rs429358	T/C	358/134/11	421/79/4	22.4	0.515
rs1801133	G/A	204/231/68	259/192/53	10.9	0.731
rs7412	C/T	441/61/1	407/93/4	9.5	1.67
rs10427255	C/T	131/236/136	94/257/153	5.8	1.24
rs1815739	T/C	90/257/156	98/251/155	0.2	0.964
rs762551	C/A	58/206/239	55/220/229	0.1	0.967

Table 17: EUR vs EAS allele-frequency contrasts from 1000 Genomes phase 3, computed by bio.gstats and matching scipy/statsmodels (benchmarks/sweep_gstats_scipy.json).

(F55) Exact HWE distribution. For n individuals with n_A copies of allele A and $n_B = 2n - n_A$, the probability of h heterozygotes is $P(h) = \frac{n!}{n_{AA}!h!n_{BB}!} \frac{2^h n_A!n_B!}{(2n)!}$ with $n_{AA} = (n_A - h)/2$, $n_{BB} = (n_B - h)/2$; the two-sided p-value is $\sum_{h:P(h) \leq P(h_{obs})} P(h)$.

16.16 UniProt lookup: a wrong protein and a cache that never hit

bio.uniprot.search maps a gene symbol to its UniProt entry, and several modules (binding sites, mutation docking) build on it. We checked it against HGNC’s curated symbol-to-UniProt mapping for 60 human genes, fetching each HGNC record separately. Before the fix 59/60 matched. The miss was CDKN2A: the query gene:CDKN2A also matches CDKN2A-AS1, and taking the top hit returned Q9UH64 (the antisense-locus protein) instead of P42771 (p16-INK4a). We switched to gene_exact and prefer the entry whose primary gene name equals the symbol; after the fix 60/60 match and all 60 returned sequences have the length of the entry fetched by accession. Reading the connector turned up a second bug. The on-disk cache named each file by abs(hash(url)); Python salts string hashes per process, so a new process never found its own cache and offline=True always failed. The same pattern was in the ChEMBL, gnomAD and structure connectors. All four now use SHA-256, and a new process running offline reads TP53 and CDKN2A from cache (P04637, P42771).

(F56) Stable cache key. A cache key must be a pure function of the request, $k(u) = \text{SHA256}(u)$; a per-process salted hash $k_s(u)$ gives $\Pr[k_{s_1}(u) = k_{s_2}(u)] \approx 2^{-64}$ for independent salts, so the hit rate across processes is effectively zero.

16.17 NCBI Gene lookup: Huntington resolved to the serotonin transporter

`bio.entrez.gene_id` resolves a symbol to an NCBI Gene ID and feeds the gene-analysis module. Against HGNC’s curated Gene IDs for the same 60 symbols it scored 58/60. HTT returned 6532, which is SLC6A4 (the serotonin transporter, alias HTT), not huntingtin (3064); TTN returned 7276, which is TTR (transthyretin, alias TTN), not titin (7273). The query field [sym] also matches aliases and the search took the top hit. The function now fetches up to ten hits and returns the one whose official symbol equals the query; after the fix 60/60 match (sugarcode 93d5b12). For a user asking about Huntington disease or titin cardiomyopathy the old answer would have been about the wrong gene.

(F57) Symbol resolution rule. Given candidate IDs $u_1, \dots, u_r$ from an alias-aware search, return $u^* = u_{\min\{i: \text{sym}(u_i)=q\}}$ if that set is non-empty, else u_1 ; an alias match is used only when no official-symbol match exists.

16.18 Reactome pathway mapping

`bio.reactome.pathways_for_uniprot` asks Reactome’s live ContentService which lowest-level pathways contain a protein. As an independent check we used Reactome’s own bulk file UniProt2Reactome.txt for the 61 HGNC-curated UniProt accessions of our 60 genes. A first comparison against canonical-accession rows matched only 59/61: KRAS (P01116) has 9 canonical rows but the service returned 71 pathways, and LMNA (P02545) 2 vs 8. The service pools the isoform rows (P01116-1, P01116-2); pooling them in the file gives 61/61 identical sets (775 = 775 protein-pathway pairs). No bug, but a caveat for users: KRAS4A and KRAS4B pathways come back together.

16.19 ChEMBL activities: “best first” from an arbitrary page

`bio.chembl.activities_for_target` promises measured IC50/Ki/Kd values ‘best first’. It fetched the first 50 rows in ChEMBL’s default order and sorted those, so ‘best’ meant best of an arbitrary page. On 12 cancer and CNS targets (between 1,731 and 21,238 activities each) the old top-10 shared at most one molecule with the true top-10, and the reported best was far from the real one; for ABL1 it was 1,200 nM when ChEMBL holds exact measurements at 0.015 nM. The function now asks ChEMBL to sort all activities that carry a pChEMBL value by pChEMBL, descending (sugarcode c6db085). The best value now equals the best pChEMBL-valid measurement on 12/12 targets; this check is partly circular because the reference uses the same ordering, so the evidence for the bug is the before-fix column. Restricting to pChEMBL also drops data-entry artefacts that sort first by raw value: 0.0 nM (JAK2, IDH1), 5×10^{-9} nM (EGFR) and -0.63 nM (DRD2).

(F58) pChEMBL. $\text{pChEMBL} = -\log_{10}(A \cdot 10^{-9})$ for an activity A in nM with an exact relation; ranking by pChEMBL descending equals ranking by A ascending on valid records, and $A \leq 0$ has no pChEMBL.

16.20 gnomAD constraint: a version-2 cutoff on version-4 data

`bio.gnomad.gene_constraint` reads pLI and LOEUF from gnomAD r4 and flags a gene as strongly loss-of-function constrained when $\text{LOEUF} < 0.35$. That cutoff comes from gnomAD v2. gnomAD’s own help page says v4 o/e and LOEUF values run higher than v2, that v2 thresholds do not transfer, and suggests $\text{LOEUF} < 0.45$ (the most constrained 15% of 17,063 MANE Select transcripts) for Mendelian interpretation. Under

target	activities	best before (nM)	best after (nM)	pChEMBL
EGFR	21,238	0.45	0.01	11.00
ABL1	6,290	1200	0.015	10.82
BRAF	9,702	0.3	0.02	10.70
JAK2	18,707	1	0.0106	10.97
FLT3	6,684	0.8	0.01	11.00
ERBB2	3,534	5	0.04	10.40
PIK3CA	9,583	1	0.016	10.80
IDH1	9,132	70	0.04	10.40
ALK	4,300	0.4	0.07	10.15
KIT	6,127	4	0.02	10.70
CDK4	1,731	10	0.026	10.59
DRD2	13,215	0.031	0.0149	10.83

Table 18: Best reported activity per target before and after the fix (benchmarks/sweep_chembl_order.json).

the old cutoff TP53 (LOEUF 0.418, pLI > 0.999) was called ‘not strongly constrained’. On our 60 genes the old rule flagged 13 and the v4 rule flags 19; the newly flagged genes are APC, FMR1, LMNA, NF1, SCN5A, TP53. Agreement with the independent pLI > 0.9 call rose from 51/60 to 57/60 (sugarcode 0a265e7). The sweep also hit gnomAD’s API rate limit (HTTP 400, ‘try again in a few minutes’); the connector used to retry three times within about 1.5 s, which only burns quota, and now fails fast with a distinct rate-limit error so callers can wait or fall back to the cache.

(F59) LOEUF. With observed pLoF count O and neutral expectation E , $\text{LOEUF} = q_{0.95}(\lambda \mid O)/E$ where the posterior of the true count is $\lambda \mid O \sim \text{Gamma}(O + 1, 1)$ under a flat prior and Poisson likelihood; a gene is flagged when $\text{LOEUF} < \tau_v$ with version-specific $\tau_{v4} = 0.45$.

16.21 Splice-junction maps against ClinVar canonical-site variants

bio.splice.junction_map builds donor and acceptor windows, keyed by cDNA position, from each gene’s RefSeqGene record, taking the CDS with the most exons as canonical. ClinVar names canonical splice-site variants on the MANE Select transcript (c.N+1, c.N+2, c.N-1, c.N-2), which gives an independent test of both the numbering and the sequence: the named position must be a junction in the map, and the base at the named intronic offset must equal ClinVar’s reference allele. We queried 12 genes; CDH1 is not in the module’s vendored RefSeqGene list and gets an explicit ‘no RefSeqGene junction map’ status rather than a guess (a coverage limit, noted for follow-up). Across the other 11 (TP53, BRCA1, BRCA2, MLH1, MSH2, APC, PTEN, ATM, PALB2, CFTR, NF1) ClinVar returned 161 such SNVs. The map had the junction for 159 and the right reference base for 159; the two misses are named on other transcripts (KCNK12, a neighbouring gene returned by the MSH2 search, and a non-MANE APC isoform). The ‘most exons’ rule picked the MANE transcript for every gene here, NF1 included. A first run of our own checker scored 96 because it assumed 15 intronic bases in the acceptor window; the windows hold 14 intronic bases plus the first exonic base, consistently across all three map builders. No module bug.

(F60) Window offsets. For a donor at exon end b the window is $x_{b-2} \dots x_{b+6}$, so intronic position $+k$ is index $2 + k$; for an acceptor at exon start c the window is $x_{c-14} \dots x_c$, so intronic position $-k$ is index $14 - k$ (0-based); on the minus strand both windows are reverse-complemented before indexing.

16.22 CFD off-target scores

modules.cfd_offtarget scores guide/off-target pairs with the Cutting Frequency Determination model. We re-implemented the scoring loop of the authors' published calculator on their released matrices (mismatch_score.pkl, pam_scores.pkl) and scored the 600 BRCA1 guide/off-target pairs against all 16 PAM dinucleotides: 9,600/9,600 scores agree, maximum difference 2.2e-16. No bug.

(F61) CFD score. $\text{CFD}(g, o, p) = \pi(p_2 p_3) \prod_{i: g_i \neq o_i} m(\text{rg}_i: d\overline{o}_i, i)$, where $\overline{o}_i$ is the complement of the off-target base, m the position-specific mismatch activity and π the PAM activity.

16.23 Short tandem repeats: a flank homopolymer hid compound repeats

modules.str_scope.find_strs finds tandem repeats with 1–6 bp units. We ran it and pytrf 1.5.0 (the finder behind Krait and TRF-style pipelines) with pytrf's default minimum repeat counts (12/7/5/4/4/4 for mono- to hexanucleotides) on all 25 GenBank records in our dataset set, from phiX174 to a 193,689-bp RefSeqGene region. Before the fix sugarcode found 208 of pytrf's 211 repeats and nothing pytrf missed. The three losses had one cause: the scan moved left to right and, after taking a short homopolymer, jumped to the end of that run, past the start of an overlapping longer repeat. In HSV-1 (NC_001806.2) a seven-base G run swallowed the start of (GGGGTT)₄; in NG_005905.2 T and A runs did the same to (TTTTG)₄ and (AAAT)₄. The finder now takes every maximal periodic run at every unit length and resolves overlaps longest first, trying the other phases of a run before dropping it (sugarcode 727344a). After the fix it finds 211/211, with 0 extra calls; 209 have identical coordinates and the other 2 are the same span reported in the disease-motif phase (CGG rather than GCG in PTEN's 5' UTR), which the module does on purpose. The same module labels repeat genotypes. The old rule used only the distance from a normal upper bound (normal if ≤ 2 above it, intermediate up to 10 above, else 'pathogenic-range candidate'). Compared with the GeneReviews allele tables it mislabels HTT CAG counts 27, 28, 36 (27 and 28 are intermediate, 36 is reduced-penetrance pathogenic) and 154 FMR1 CGG counts between 5 and 300, from 47 to 200: every premutation allele (55–200) was called a pathogenic expansion, which in counselling terms confuses a carrier with an affected full-mutation allele. expansion_call now takes an optional locus and uses the published bands for HTT and FMR1 (0 disagreements on HTT 5–120); without a locus the old behaviour is unchanged.

(F62) Maximal periodic runs. For unit length u , position j is periodic if $x_j = x_{j+u}$; a maximal run of periodic positions $[a, b]$ defines the span $[a, b + u]$ holding $\lfloor (b + u - a - k)/u \rfloor$ whole units from phase $k \in \{0, \dots, u - 1\}$. Candidates over all u are kept greedily by length (ties: smaller u , then leftmost) subject to disjoint whole-unit intervals, which is the rule pytrf-style finders apply implicitly by reporting the longest repeat at each locus.

16.24 Bayesian ACMG scoring against ClinGen expert panels

modules.acmg_bayesian scores ACMG/AMP criteria with the Tavtigian et al. (2018) Bayesian model in its point form (supporting 1, moderate 2, strong 4, very strong 8; pathogenic ≥ 10 , likely pathogenic 6–9, likely benign -1 to -6 , benign ≤ -7). The ClinGen Evidence Repository publishes, for every variant a Variant Curation Expert Panel (VCEP) has classified, the criteria it applied and its final call. We used 7,929 current classifications from 33 panels across 117 genes (the server closed the bulk download after about 18 MB, so only complete rows were used). ClinGen writes modified strengths as suffixes: PM2.Supporting, PVS1.Strong, PM3.Very Strong, BS1.Stand Alone. The module accepted a strength only as a separate field,

so in 5,699 rows at least one code was rejected, and it reproduced 4,583/7,929 panel calls. The rejections were listed in the output rather than hidden, but anyone pasting ClinGen notation got a wrong class. The parser now reads the suffix, and a stand-alone benign strength acts like BA1 (sugarcode cdf160b). The module now reproduces 7,515/7,929 calls (94.8%), with no rejected codes. As a baseline we coded the original ACMG/AMP 2015 combining rules (Richards et al., Table 5), counting each criterion at its applied strength and calling conflicting evidence uncertain. They reproduce 7,267/7,929 (91.7%); the point model is closer to what the panels actually do. The largest remaining disagreement is 229 variants the panels call uncertain and the model calls likely benign; 177 of them net to exactly -1 point, typically a lone BP4. The 2015 rules need two benign supporting criteria for likely benign, and the panels follow that; the 2018 posterior bands do not. We report this as a known difference and did not change the bands. BA1 with pathogenic criteria also met is rare and panels treat it differently (21 benign, 3 pathogenic, 3 uncertain significance, 4 likely benign), so the module keeps BA1 as an override but flags the conflict.

(F63) Point form of the Bayesian model. With prior $\pi = 0.1$ and $O_{PVSt} = 350$, a criterion of strength k points has odds $350^{k/8}$; summed points S give posterior $P(S) = \frac{\frac{\pi}{1-\pi} 350^{S/8}}{1 + \frac{\pi}{1-\pi} 350^{S/8}}$, so $P(10) = 0.994$, $P(6) = 0.900$, $P(-1) = 0.051$ and $P(-7) = 0.0007$.

16.25 ClinVar interpretation: exon-spanning ranges and review stars

modules.openclinvar infers a variant’s molecular consequence from its HGVS name when the name alone decides it, and abstains otherwise. We pulled 720 ClinVar records (60 per gene for BRCA1, BRCA2, TP53, MLH1, MSH2, CFTR, APC, PTEN, LDLR, MYH7, SCN5A and RYR1) and scored the module’s call as correct when ClinVar lists that consequence for the record. Of 682 records with a ClinVar consequence the module got 646 right, abstained on 32 and was wrong on 4. All four were ranges judged by their first coordinate only: c.453+625_545+920delinsT in MLH1 starts in an intron but removes the exon between two introns, c.5468-64_5480dup in BRCA1 runs from the intron into the exon, and c.-19_*21del in TP53 deletes the whole coding sequence; the module called them intronic or UTR, the lowest-weight classes in its evidence model. Ranges are now read at both ends: ends in different exon/intron segments mean a boundary is crossed (splice disruption) and a range from the 5’ UTR into the CDS removes the start codon (sugarcode 7067655). After the fix: 651 right, 30 abstentions, 1 wrong; the remaining one (MSH2 c.2459-2_2472del) is annotated by ClinVar on an older transcript, and c.2459-2 is an acceptor-site base, so we kept the call. clinvar_stars maps a review status to ClinVar’s 0–4 gold stars, which weight ClinVar evidence in the module’s trace. Against the 9 statuses in ClinVar’s review-status table it was wrong once: ‘criteria provided, multiple submitters’ (the status used for somatic clinical-impact and oncogenicity classifications) got one star instead of two, because the rule required the germline suffix ‘no conflicts’. Now 9/9.

(F64) Segment rule for c. ranges. Write each end as (a, s, k) : anchor a (with region 5’ UTR, CDS or 3’ UTR), intron side $s \in \{+, -, \emptyset\}$ and offset k . Two ends lie in the same segment iff both are exonic, or both are intronic with $(a_1, s_1) = (a_2, s_2)$ or $s_1 = +, s_2 = -, a_2 = a_1 + 1$. A range whose ends lie in different segments crosses at least one splice site.

16.26 CRISPR on-target scores against measured guide activity

modules.crispr_opt has four on-target scorers: Rule Set 2 (the Azimuth gradient-boosted model, with or without the peptide-position features), Rule Set 1 (Doench 2014 logistic weights), and a hand-built heuristic

used only when a guide sits too close to the sequence edge for a 30-mer context. We scored all 5,310 guides in the Doench 2016 FC+RES set (17 genes; file from the Azimuth repository) and compared with the measured, within-gene rank-normalised activity. Spearman correlations: Rule Set 2 with position 0.7106, sequence-only 0.6693, Rule Set 1 0.3948, heuristic 0.1125; the repository’s own prediction column gives 0.7148. Median per-gene values are 0.6965, 0.6599, 0.468 and 0.1357. Both published models were trained on these guides, so their numbers are in-sample and flattering; the heuristic was not fitted and is close to useless ($\rho \approx 0.11$). Two consequences for users. `design_guides` ranks with Rule Set 1 even though Rule Set 2 ships in the same module; we have not switched it, because the case for switching needs a held-out screen and we have not run one. And the edge fallback should be read as ‘no reliable score’, which the module already labels in its `on_target_model` field (`heuristic_edge_fallback`). No code bug was found.

(F65) Spearman correlation. For scores s and measurements y with ranks $R(s_i), R(y_i)$, $\rho = \text{corr}(R(s), R(y))$, which for untied data equals $1 - \frac{6 \sum_i d_i^2}{n(n^2 - 1)}$ with $d_i = R(s_i) - R(y_i)$; the per-gene median guards against between-gene differences in dynamic range.

16.27 Replicate planning: a two-entry z table

`modules.gene_analysis.power_estimate` tells a user how many replicates per group an experiment needs. It added 0.84 to 1.96 when the target power was at most 0.8 and 1.28 otherwise, which is right only at 0.8 and 0.9. Against `statsmodels`’ `NormalIndPower` solver on 41 combinations of power (0.5–0.99), effect and variance, the old rule matched 7 times and over-asked by up to $2.03\times$ (power 0.5 needs 8 replicates for a one-SD effect, it said 16); at 0.95 and 0.99 it under-asked. The function now uses exact normal quantiles and takes `alpha` as a parameter (`sugarcode 4defcdb`); it matches `statsmodels` in 41/41 cases. This is a formula check, not a real-data sweep, and the rest of `gene_analysis` is not yet validated.

(F66) Two-sample size. For a two-sided test at level α with power $1 - \beta$, difference δ and per-observation variance σ^2 , $n = \lceil 2\sigma^2(z_{1-\alpha/2} + z_{1-\beta})^2 / \delta^2 \rceil$ per group.

16.28 QSAR descriptors against RDKit on approved drugs

`modules.qsar_bench` computes its own 2D descriptors from a small SMILES parser, so a QSAR model can be built without a chemistry toolkit. We compared them with RDKit 2026.03.6 on the 993 ChEMBL phase-4 molecules RDKit reads. Before the fix the parser rejected 110 of them: every SMILES with a / or \ stereo bond (88 molecules, including tamoxifen and diethylstilbestrol) and every sodium, potassium or silver salt (22). On the 883 it did read, molecular weight matched in 843/883 (charged bracket atoms such as the [O-] of a nitrate got an implicit hydrogen), rotatable bonds in 96/883 (the rule subtracted the ring rank from all non-terminal single bonds, so ring bonds counted as rotors: nicotine got 4 instead of 1) and fraction sp^3 in 296/883 (carbonyl and alkene carbons counted as sp^3). All four are fixed (`sugarcode bf8665b`): the parser reads directional bonds and common counter-ions, bracket atoms carry only their written hydrogens, a bond is rotatable only if it is single, acyclic, between non-terminal atoms and not next to a triple bond, and sp^3 carbons are non-aromatic carbons with only single bonds. Now all 993 molecules parse and molecular weight, heavy atoms, heteroatoms, formal charge, ring rank, rotatable bonds and fraction sp^3 agree on 993/993 each; donors agree on 992/993 (the miss is a dihydrate whose water RDKit does not count). Rotatable bonds match RDKit’s NonStrict definition; RDKit’s default Strict count, which

also drops amide C–N bonds, agrees on 655/993. The acceptor count stays a labelled heuristic (264/993 against RDKit’s Lipinski acceptors) and should not be read as Lipinski HBA.

(F67) Ring bond test. A bond (i, j) lies on a ring iff j is reachable from i in $G \setminus \{(i, j)\}$; the number of independent rings is the cyclomatic number $r = |E| - |V| + c$ for c connected components, which counts rings but not ring bonds. The module’s Morgan fingerprint is meant to follow ECFP. Its starting atom label used only hydrogens written inside brackets, so CH₃, CH₂ and CH carbons, and ring and chain atoms, began with the same identifier. On 5,000 random pairs of 300 drugs its Tanimoto similarities ranked with RDKit’s radius-2, 2048-bit Morgan at Spearman 0.7815, and the nearest neighbour agreed for 202/300 molecules. The label now uses RDKit’s default invariants (element, heavy degree, total hydrogens, charge, ring membership; sugarcodes 9a33eb4): Spearman 0.9005, nearest neighbour 236/300, mean absolute Tanimoto difference 0.0146. The bits are not identical to RDKit’s (different hashing, no removal of duplicate environments), so fingerprints from the two should not be mixed in one model.

(F68) Tanimoto on bit vectors. For fingerprints $a, b \in \{0, 1\}^L$, $T(a, b) = \frac{|a \wedge b|}{|a| + |b| - |a \wedge b|}$; two fingerprint schemes are compared by the rank correlation of T over sampled pairs and by agreement of each molecule’s arg-max neighbour. Split by expert panel (Table 19), the point model reproduces more calls than the 2015 rules for 21 of 33 panels, fewer for 11 and the same for 1. The rules do better for these panels: Myeloid Malignancy; Phenylketonuria; Platelet Disorders; Rett and Angelman-like Disorders; Familial Hypercholesterolemia; Mitochondrial Diseases; FBN1; Hereditary Breast, Ovarian and Pancreatic Cancer; Coagulation Factor Deficiency; Thrombosis; VHL. We did not check why against each panel’s written specification; for these panels a user should read the module’s class next to the rule-based one.

16.29 Elastic network modes: a doubled Hessian diagonal

modules.evofold_4d.anm_modes computes anisotropic network model (ANM) modes from C α coordinates; its fluctuation profile is offered as a B-factor proxy and feeds hinge detection. We built the ANM Hessian for the first chain of all 30 PDB structures in our set with ProDy 2.4.1 (cutoff 10 Å, $\gamma = 1$; ProDy’s own eigen-solver fails with the installed SciPy, so both Hessians were diagonalised with numpy). Before the fix no structure matched: the module looped over ordered residue pairs and subtracted each contact block from both diagonal blocks on both visits, so every diagonal block was doubled, rows of the Hessian no longer summed to zero and the six rigid-body zero modes disappeared. The ‘lowest modes’ it reported were therefore not the collective motions of the protein; the fluctuation profile correlated with ProDy’s at r as low as 0.19, and with crystallographic B-factors at median 0.28. Each contact is now added once (sugarcodes feced35). All 30/30 structures now give ProDy’s six lowest non-zero frequencies to the module’s four reported decimals, with fluctuation profiles correlated at $r \geq 1.0000$; the median B-factor correlation is 0.38, the same as ProDy’s (0.38). 2 chains have more than six zero modes because of chain breaks; the module skips all zero modes, which is the right behaviour.

(F69) ANM Hessian. For contacts $r_{ij} < r_c$ with unit vector $\hat{e}_{ij}$, $H_{ij} = -\gamma \hat{e}_{ij} \hat{e}_{ij}^\top$ for $i \neq j$ and $H_{ii} = -\sum_{j \neq i} H_{ij}$, so each block row sums to zero and H has six zero eigenvalues (three translations, three rotations) for a connected network; mean-square fluctuations are $\langle \Delta r_i^2 \rangle \propto \sum_k \|u_k^{(i)}\|^2 / \lambda_k$ over non-zero modes.

expert panel	variants	point model	2015 rules
Myeloid Malignancy VCEP	810	740	776
Monogenic Diabetes VCEP	801	780	723
Phenylketonuria VCEP	770	748	765
Platelet Disorders VCEP	582	560	562
Severe Combined Immunodeficiency Disease VCEP	375	368	328
Rett and Angelman-like Disorders VCEP	367	346	353
Familial Hypercholesterolemia VCEP	343	328	335
Malignant Hyperthermia Susceptibility VCEP	335	324	311
CDH1 VCEP	331	325	278
RASopathy VCEP	329	319	288
Lysosomal Diseases VCEP	308	296	273
Cerebral Creatine Deficiency Syndromes VCEP	266	256	249
Glaucoma VCEP	265	240	228
Mitochondrial Diseases VCEP	225	181	204
ACADVL VCEP	225	220	149
Hearing Loss VCEP	201	184	172
PTEN VCEP	198	186	183
Cardiomyopathy VCEP	171	168	165
DICER1 and miRNA-Processing Gene VCEP	125	117	116
Leber Congenital Amaurosis/early onset Retinal Dystrophy VCEP	125	124	116
TP53 VCEP	124	114	103
FBN1 VCEP	104	93	101
Hereditary Breast, Ovarian and Pancreatic Cancer VCEP	93	86	87
Coagulation Factor Deficiency VCEP	69	52	56
Thrombosis VCEP	62	53	59
InSiGHT Hereditary Colorectal Cancer/Polyposis VCEP	59	58	53
von Willebrand Disease VCEP	57	57	55
Brain Malformations VCEP	43	39	34
Pulmonary Hypertension VCEP	42	38	31
ENIGMA BRCA1 and BRCA2 VCEP	40	34	33
Hereditary Hemorrhagic Telangiectasia VCEP	32	32	30
VHL VCEP	31	28	30
Epilepsy Sodium Channel VCEP	21	21	21

Table 19: Panel calls reproduced by the Bayesian point model and by the ACMG/AMP 2015 combining rules, per ClinGen expert panel (benchmarks/sweep-acmg_erepo_per_panel.json).

16.30 Solvent-accessible surface area against FreeSASA

modules.structural_biophysics.shrake_rupley has its own Shrake–Rupley routine for per-atom and per-residue solvent-accessible surface area (SASA, 960 points per atom), separate from the bio.structures SASA checked earlier. We parsed 14 PDB structures from our set (1622–2058 protein heavy atoms) with the module’s own parser and gave the same atoms and radii to FreeSASA’s Lee–Richards algorithm (100 slices, probe 1.4 Å), a different numerical method. Total SASA agreed within 0.17% for every structure, and per-residue areas correlated at $r \geq 0.99994$ with a mean absolute difference of about 0.3 Å² per residue. No bug. The module’s Vina-style score is checked in the next subsection; its pose refinement is not yet validated.

(F70) Shrake–Rupley estimate. Atom i with expanded radius $R_i = r_i + r_p$ carries N near-uniform sphere points x_{ik} ; its area is $A_i = 4\pi R_i^2 \frac{1}{N} \sum_k \mathbf{1}[\|x_{ik} - c_j\| \geq R_j \forall j \neq i]$, with sampling error $O(N^{-1/2})$ per atom that largely cancels in residue and molecule totals.

16.31 Vina-form docking score against AutoDock Vina

The same module’s vina_score claims the AutoDock Vina 1.1 functional form on a fixed pose. Reading Vina’s source showed three departures. It measured surface distances with Bondi radii (C 1.70, N 1.55, O 1.52 Å) where Vina uses its X-Score radii (C 1.9, N 1.8, O 1.7, S 2.0 Å); it counted sulfur as hydrophobic, which Vina does not; and it added $w_{rot}N_{rot}$ to the energy, whereas Vina divides the intermolecular energy by $1 + w_{rot}N_{rot}$. To test the pair terms numerically with no atom-typing ambiguity we scored stacked all-carbon rings, where every atom is a Vina hydrophobic carbon, with AutoDock Vina 1.2.7 in score-only mode. Before the fix the module gave -1.5646, -0.8557 and -0.3662 kcal/mol at 3.6, 4.0 and 4.6 Å separation where Vina gives -2.31, -1.65 and -0.634. After the fix (sugarcode 28b8590) it matches Vina within 0.0046 kcal/mol at all 8 separations from 3.4 to 6.0 Å; the residual is Vina’s grid interpolation and three-decimal output. The rotor division is checked against the source only. Element-level typing still cannot separate polar from apolar carbon or place donors without hydrogens, so on real complexes the score remains an approximation, as the module says.

(F71) Vina score. With surface distance $s_{ij} = r_{ij} - R_i^{XS} - R_j^{XS}$, $e = \sum_{ij} [w_1 e^{-(s/0.5)^2} + w_2 e^{-((s-3)/2)^2} + w_3 s^2 \mathbf{1}[s < 0] + w_4 h(s) + w_5 b(s)]$ over pairs within 8 Å, and $\Delta G = e / (1 + w_{rot}N_{rot})$ with $w = (-0.0356, -0.0052, 0.840, -0.03)$, $w_{rot} = 0.0585$; h is 1 for $s < 0.5$ falling to 0 at 1.5, b is 1 for $s < -0.7$ falling to 0 at 0.

16.32 Library-evaluation graph keys against RDKit canonical SMILES

molecule_eval deduplicates generated libraries with an order-invariant graph digest (four rounds of Weisfeiler–Lehman relabelling over element, aromaticity and charge, then a sorted edge list) rather than canonical SMILES. A digest like this must give the same key for every SMILES spelling of a molecule and different keys for different molecules. We tested both on 993 ChEMBL phase-4 drugs, comparing against RDKit 2026.03.6 canonical SMILES with stereochemistry removed, and wrote three randomised SMILES per drug. Before the fix 3 drugs changed key under re-spelling (acetazolamide ChEMBL19 among them). All three were written by ChEMBL with a directional bond marker between two aromatic ring atoms, as in /N=c1/sc. The shared SMILES parser read every / or \ as an explicit single bond, which broke the aromatic ring; RDKit keeps such a bond aromatic. After the fix (sugarcode 15a8ba7), where the marker only sets direction and bond order follows the implicit rule, there are 0 invariance failures, 0 keys shared by molecules RDKit calls different, and 0 RDKit molecules split across keys; both give 980 unique structures.

The directional-bond change left the QSAR descriptor and Morgan benchmarks unchanged when re-run. Internal diversity (mean pairwise Tanimoto distance, radius-2 2048-bit fingerprints, 300 drugs, all pairs) is 0.9055 with the module’s fingerprint and 0.8994 with RDKit’s. Four WL rounds can in principle merge graphs that differ only beyond radius four; no such case occurred in this set.

(F72) Internal diversity. For a library of n unique molecules with fingerprints f_i , $D = \binom{n}{2}^{-1} \sum_{i < j} (1 - T(f_i, f_j))$ with $T(a, b) = |a \wedge b| / (|a| + |b| - |a \wedge b|)$; the module samples at most 10^4 pairs and bootstraps a 95% interval.

16.33 Nussinov folding against an independent implementation, exhaustive oracle and ViennaRNA

riboswitch uses an exact Nussinov dynamic program (maximum number of nested base pairs, minimum loop 3) to score designed switches. We checked the recursion three ways. On 100 short sequences (length up to 16) we enumerated every non-crossing matching by brute force; the module matched the exhaustive optimum on all 100 (the oracle itself needed two fixes during the check - reused positions and a stale pruning bound - which is why we keep it in the repository). On 145 real Rfam seed sequences from four riboswitch families (TPP RF00059, FMN RF00050, glycine RF00504, cobalamin RF00174), an independently written Nussinov recurrence (pair- i -with- k decomposition instead of the module’s bifurcation form) agreed on the optimal pair count for all 145 (0 mismatches). Against ViennaRNA 2.7.2 minimum-free-energy folds the median base-pair F1 is 0.194 (family means 0.159 FMN to 0.278 glycine), with the module placing $1.289 \times$ as many pairs: maximising pair count is not the same as minimising free energy, and we do not claim MFE agreement. The module’s estimated_energy_kcal_mol field (-3 per GC pair, -2 per AU/GU) is a heuristic ranking score, not a thermodynamic energy; designs needing energies should go through ViennaRNA, as in the rna_nussinov module check.

(F73) Nussinov recursion. $N_{ij} = \max\{N_{i+1,j}, N_{i,j-1}, \max_{k > i+m, (s_i, s_k) \in \mathcal{P}} [1 + N_{i+1,k-1} + N_{k+1,j}]\}$ with pair set $\mathcal{P} = \{AU, UA, GC, CG, GU, UG\}$ and minimum loop $m = 3$; the module evaluates the equivalent bifurcation form $N_{ij} = \max\{N_{i+1,j}, N_{i,j-1}, [j - i > m](s_i, s_j) + N_{i+1,j-1}, \max_{i < k < j} (N_{ik} + N_{k+1,j})\}$.

16.34 Prime-editing guide design against published, validated pegRNAs

prime_design builds pegRNAs: a 20-nt spacer, a 3’ extension of reverse-transcription template (RTT) plus primer-binding site (PBS). Three departures from the published anatomy were found by replaying two experimentally validated pegRNAs through the module. (i) The PBS was reverse-complemented out of the RT-template region, so PBS and RTT overlapped; the PBS must anneal to the 3’ flap of the nicked strand, i.e. it is the reverse complement of the protospacer’s PAM-proximal end. (ii) For plus-strand PAMs the RTT started at the edit, not at the nick. (iii) When the RT template was truncated to the chosen length, the module kept the nick-distal end; reverse transcriptase reads from the PBS, so the RTT’s 3’ end must pair with the nick base. After the fixes (sugarcode 947b55e) the module reproduces both published extensions exactly: HEK3 +1 CTT insertion (Anzalone et al. 2019; Addgene 132778; also validated by pegFinder, Chow et al. 2021) RTT TCTGCCATCAAAGC + PBS GTGCTCAGTCTG, and RNF2 +5 G>T RTT AACGAACACATCAGG + PBS TAATGACTAAGATG. Before the fix the ‘PBS’ came out as TCTGCCATCAAAA - RTT sequence, not flap. End to end, from the HEK3 genomic context (Anzalone substrate AVA024) with the +1 CTT edit, the top design uses the published spacer and a PBS from the correct flap (10 nt auto-chosen by the Wallace- T_m rule vs the published 12 nt; same annealing tract), and its RTT contains the published template at the nick. A crash on

insertion edits with empty ref was fixed along the way. The PBS/RTT boundary convention (spacer positions 17/18, one nt PAM-proximal of the canonical Cas9 nick) was adopted to match every published validated pegRNA we could locate, including Anzalone’s own in-vitro pegRNAs.

(F74) pegRNA extension. Spacer $g = g_1 \dots g_{20}$ with PAM at positions 21-23; PBS = $\overleftarrow{g_{17-L} \dots g_{16}}$ (length $L \in [10, 16]$, chosen to bring Wallace $T_m = 2N_{AT} + 4N_{GC}$ near 32 °C); RTT = $\overleftarrow{e_{n+1} \dots e_{n+\ell}}$ for edited strand e , nick at n , $\ell \in [10, 20]$; extension = RTT • PBS appended 3’ of the scaffold.

16.35 Hill occupancy, burst noise and Wright-Fisher drift against theory

Two modules were checked at the formula level. promoter_lib’s thermodynamic occupancy agrees with the Hill form $1/(1 + (K_d/c)^h)$ to 2.2e-16 over a grid of Hill coefficients, K_d and concentrations (half-occupancy at $c = K_d$ holds exactly; the $h = 1$ case also matches a logistic/expit evaluation to 1.1e-16), and its expression-noise summary matches Poisson-burst theory ($\text{Var} = \mu(1 + B)$, $\text{Fano} = 1 + B$) on all 4 tested settings. stability_ai’s seeded Wright-Fisher simulation of construct loss (per-generation plasmid-loss and mutation terms, then viability selection $f' = a(1 - s)/(1 - as)$ with $a = f(1 - L - m)$, binomial resampling) tracks an independently written deterministic recursion within 0.00204 at population 2×10^6 across plasmid and genomic constructs, is reproducible under a fixed seed, and gives the expected extremes (0% extinction for a low-burden plasmid, 100% for a high-burden toxic one). No bug in either; both are theory checks, so the parameters themselves (loss rates, selection coefficients) remain literature-free defaults and are not validated against experimental stability data - a limitation we state rather than a defect.

(F75) Wright-Fisher with loss and selection. $a_g = f_g(1 - L - m)$, $f_{g+1} \sim \text{Bin}(N, a_g(1 - s)/(1 - a_g s))/N$, with plasmid-loss rate $L = (2 \times 10^{-3})(1 + 4b)$ for burden b , mutation $m = 10^{-4}k$ for size k kb ($\times 2$ if toxic), selection $s = 0.08e$ for expression load e .

16.36 FASTA parsing against Biopython on Rfam seed files

bio.fasta was compared with Biopython 1.88 SeqIO on three real Rfam seed-alignment FASTA files (TPP RF00059: 16245 records, FMN RF00050: 6377, glycine RF00504: 9474) - 32096 records in total. Identifiers, full descriptions and sequences agree exactly on every record (0 mismatches), the streaming reader agrees with the in-memory parser on all three files, and a write-read round trip over 200 records is lossless. One deliberate difference: Biopython rejects a file that starts with blank lines (‘comments at the beginning of the file’), while bio.fasta skips them; we treat this as leniency, not a defect, and record it here.

(F76) FASTA round trip. For records $r_i = (h_i, s_i)$, write-read gives

$\text{mathrm{parse}}(\text{mathrm{write}}(r)) = r$ with sequence lines wrapped at width w :
 $\text{mathrm{write}}(r) =$
 $> !h_i$
 $,$
 $|$
 $,$
 $s_i[kw : (k+1)w]_{k=0}^{\lceil \text{ceil}[s_i]/w \rceil - 1}$; identity holds for $w = 50$ on 200 RF00059 records.

16.37 CRISPRater guide-efficacy scores against the author-computed reference scores

crisprater vendors the published ten-feature linear model of Labuhn et al. 2018 (NAR 46:1375) with its intercept, coefficients and class thresholds. We scored all 3141 sgRNAs from the paper's Supplementary Table 3 (author-computed CRISPRater scores, PMC5814880) and compared. As vendored, 3139 of 3141 guides mismatched. The cause was the GC feature: the module summed GC over spacer positions 4-14, but the published model uses GC4-13 (Figure 4C of the paper; the narrower window also avoids double-counting position 14, which is its own negative feature). After the fix (sugarcode 55748fa) the module matches the authors' scores on 3138 of 3141 guides (99.90%) within 5×10^{-4} . The three residual outliers each differ from the published model by about one feature weight (exactly one weight in two cases), consistent with isolated anomalies in the supplementary table itself, which also contains two duplicate sequences; we document them rather than fit to them. Class thresholds (low <0.56 , high >0.74) were confirmed against the paper text.

(F77) CRISPRater score. $s = 0.6505037 + 0.14177$

,
 $\mathrm{GC}_{413} + 0.06966$
, $[b_{20}=G] + 0.04216$
, $[b_3 \text{ in } A, T] + 0.03303$
, $[b_{12} \text{ in } A, G] + 0.02355$
, $[b_6=G] - 0.04746$
, $[b_4 \text{ in } A, T] - 0.04878$
, $[b_{18} \text{ in } A, G] - 0.06982$
, $[b_5 \text{ in } A, C] - 0.07088$
, $[b_{14}=G] - 0.08161$
, $[b_{15}=A]$, with
 GC_{413} the GC fraction of spacer positions 4-13.

16.38 Platform performance analytics against independent recomputation

omega_stats aggregates recorded metric observations into dashboards, release comparisons and 51 diagnostics. Every computed quantity was recomputed independently with numpy/statistics on synthetic series (4 modules, accuracy and latency, 30 observations each): per-series mean/standard-deviation/min/max match to the module's rounding (0 mismatches over 8 series), subnetwork mean-of-means matches, least-squares drift slopes match numpypolyfit exactly (0 mismatches), pass rates use the correct direction per kind (latency $\leq$ target, others $\geq$; 0 mismatches), bootstrap intervals are seed-deterministic and contain the mean, release-comparison effect sizes match the pooled-variance Cohen form (0 mismatches), and the diagnostics block has exactly 51 entries. The release gate behaved correctly on a planted mixed case: a 5% candidate improvement in accuracy with a 5% latency regression gates fail. No bug.

(F78) Standardized release effect. For baseline a and candidate b , $d = (\bar{b} - \bar{a}) / \sqrt{(s_a^2 + s_b^2)/2}$ with population variances; relative improvement is $(\bar{b} - \bar{a}) / |\bar{a}|$, sign-flipped for lower-is-better kinds, and the gate fails when any series drops more than the tolerance (5%).

16.39 16S diversity statistics against scipy and explicit Gower centering

microbiome_exp computes per-sample alpha diversity (Shannon, Gini–Simpson, richness, Pielou evenness), a Bray–Curtis beta-diversity matrix on relative abundances, classical PCoA ordination, and permutation-based differential abundance. On a 12-sample synthetic cohort every alpha quantity matches scipystatsentropy and direct forms (0 mismatches across both entry points), the Bray–Curtis matrix matches scipyspatialdistancebraycurtis to 0.0, and a planted control-enriched taxon is recovered with the correct sign and a significant permutation p (1.968 log2 fold, $p = 0.034$; unplanted cohort median $p = 0.51$). The PCoA had a real bug: the Gower centering subtracted means of the unsquared distances, shifting the leading eigenvalue by 0.80. After the fix (sugarcode 2de74ce) the first two eigenvalues match an explicit $-\frac{1}{2}JD^2J$ computation to 8.3e-16. Disease associations in the module are heuristic flags over a small built-in table, and the module itself reports them as hypotheses, not diagnoses; we did not validate the association table against cohort literature.

(F79) Gower-centered PCoA. With Bray–Curtis matrix D , $G = -\frac{1}{2}JD^{\circ 2}J$ for $J = I - n^{-1}\mathbf{1}\mathbf{1}^\top$; coordinates $U_k \text{diag}(\sqrt{\lambda_k})$ from the top eigenpairs of G . Centering must use means of $D^{\circ 2}$, the elementwise square - using means of D does not double-center the squared matrix.

16.40 m6A site prediction against the SRAMP benchmark and single-nucleotide miCLIP sites

rna_decoder scores candidate m6A sites by a DRACH motif search weighted by region priors (highest near the stop codon and in the 3'UTR) and a local AU-richness exposure term. We validated it on two independent real datasets. First, motif biology: of 9536 single-nucleotide miCLIP CIMS sites in HEK293 cells (GSE63753, Linder et al. 2015), 77.3% match the DRACH consensus the module searches for, and the five most common site 5-mers (GGACT, GGACA, AGACT, GAACT, TGACT) are all DRACH. Second, discrimination: the SRAMP mature-messenger benchmark (Zhou et al. 2016) labels 1210 transcript positions on the 40 transcripts we fetched live from Ensembl (positions are 1-based adenosine coordinates: 135 of 258 positives land on A, vs 36 under a 0-based reading). Because the benchmark predates the current Ensembl build, 708 labels no longer sit at a DRACH-centred A and were excluded, leaving 99 positives and 403 negatives. The check found a real defect: with the module's original defaults the whole sequence was treated as coding, so the final 100 nt of any transcript inherited the near-stop prior; scored that way the module achieved AUC 0.5496, below its own exposure feature alone (0.562). The default now auto-detects the coding region from the first ATG (AUC 0.5886), and with true Ensembl CDS coordinates the module reaches AUC 0.6702 on 88 positives and 361 negatives over 38 CDS-aligned transcripts. The regional prior direction is validated by the data: 49 of 88 positives (55.7%) fall in near-stop CDS or 3'UTR against 130 of 361 negatives (36.0%), the established m6A topology. This remains a heuristic scorer, far below trained predictors, and the m5C/psi/m1A motif maps are placeholders the module itself flags; we report the AUCs, not a discovery.

(F80) m6A site score. $s = 0.55r + 0.45(1 - \text{GC}_{31})$ with region prior $r \in \{0.35 \text{ (5'UTR)}, 0.55 \text{ (CDS)}, 0.85 \text{ (< 100nt from stop)}, 0.75 \text{ (3'UTR)}\}$ over a 31-nt flank; sites are DRACH occurrences, modification at the cen-

tral A. Discrimination is measured by Mann–Whitney $AUC = (\sum_{i \in \text{pos}} R_i - P(P+1)/2) / (PN)$ with midranks R_i .

16.41 Live evidence mining against PubMed E-utilities and ClinicalTrials.gov

evidence_mining queries two public APIs - NCBI E-utilities (esearch then efetch) and the ClinicalTrials.gov v2 studies endpoint - and normalises both into one evidence table. We replayed its traffic against independent live calls made seconds apart with the same query ('ivacaftor cystic fibrosis', limit 10 per source). The module's PubMed identifiers match the independent esearch idlist exactly and in order (10/10), its trial identifiers match the independent v2 call exactly and in order (10/10), the record_count field equals the emitted table length, all PubMed URLs are correctly formed, all PMIDs are numeric, all trial IDs are well-formed NCT identifiers, and 20 of 20 records carry a non-empty title. Failure semantics are as documented: strict mode is fail-closed (an offline cache miss raises), non-strict mode returns both source errors in-band with zero records rather than silently discarding a failed source, and offline replay from the content-addressed cache reproduces the live record order exactly. No bug. This validates transport, parsing and bookkeeping; it does not validate the mined effect-direction/sample-size text heuristics, which remain unverified extraction guesses.

(F81) Evidence-table merge. Records $r = (s, i, t, u, d)$ with source $s \in \{\text{PubMed}, \text{CTG}\}$, identifier i , title t , URL u , date d ; $\text{mine}(q) = \{r\}_{s,i}$ with $\text{record_count} = |\text{mine}(q)|$, strict mode $\Rightarrow$ any source error aborts, non-strict $\Rightarrow$ errors $\subseteq$ result. Verified: $i_{\text{PubMed}}^{\text{module}} = i_{\text{PubMed}}^{\text{research}}$ and $i_{\text{CTG}}^{\text{module}} = i_{\text{CTG}}^{\text{v2}}$ elementwise on 10+10 live records.

16.42 Rare-disease signatures against the HPO annotation corpus

rarenet.ai ranks six rare diseases from symptom sets, variants and omics over a curated table of symptom signatures, causal genes and prevalences. We audited the table against the Human Phenotype Ontology gene-to-phenotype corpus (333,984 gene–phenotype–disease rows, current release). The audit found one stale curation: Gaucher disease was keyed to the retired gene symbol GBA, which appears nowhere in the current corpus - any gene-keyed lookup would silently miss it; the current symbol GBA1 restores the mapping. After the fix, all 6/6 gene–disease pairs are confirmed (CFTR–OMIM:219700, HTT–143100, FBN1–154700, PAH–261600, GBA1–230800, DMD–310200), and of the 24 curated symptoms 12 match the disease's annotated HPO term names directly while 21 match once clinical synonyms are applied (lens dislocation = Ectopia lentis, poor growth = Failure to thrive, calf hypertrophy = Calf muscle hypertrophy). The three residue are annotation absences in this HPO release, not curation errors: no adult-onset row for HTT/OMIM:143100, no musty-odor row for PAH/OMIM:261600, no fatigue row for GBA1/OMIM:230800. Behaviourally, the module ranks the correct disease first for all 6/6 complete signatures. Limitations recorded: the symptom vocabulary is lay-oriented, the prevalence fields are unverified literature defaults, and six diseases is a demo-scale table.

(F82) Signature score. $c = \min\{0.99, \min(1, 0.25|S \cap S_d| + 0.1 \sum_{s \in S \cap S_d} f(s)^{-1}) + 0.4[g_d \in V] + 0.15[g_d \in O]\}$ for input symptoms S , disease signature S_d , causal gene g_d , variant genes V , omics-disrupted genes O , and symptom-frequency weight f ; candidates rank by c . Table audit: g_d must appear in the HPO corpus under the OMIM identifier of the disease.

16.43 Tumour-promoter selectivity table against Human Protein Atlas pathology data

oncocircuit designs tumour-selective kill circuits over a five-entry promoter table with cancer associations and numeric tumour-selectivity values. We audited the table against the Human Protein Atlas v23 pathology resource (403,241 gene–cancer patient-sample rows). PSA/KLK3 is confirmed outright: detected in all ten prostate-cancer samples and in none of the 89 samples across the other 19 cancers (detection and High-expression shares in the claimed cancer both 1.0). AFP holds at High-expression level (liver is the maximum-High cancer; High share in claimed tissue 0.5), although low-level AFP detection is pan-cancer. HER2/ERBB2 is directionally right with breast the maximum-High cancer (0.364 High rate) and gastric much weaker (0.083) than the table implies. Two entries do not survive as written: Survivin/BIRC5 is near-ubiquitous across tumours (breast 1.0, lung 0.917, colorectal 0.917 detection; its High expression concentrates in testis), so it works as a proliferation-associated second input but its 0.80 selectivity figure is unsupported; and hTERT/TERT protein is essentially undetectable by immunohistochemistry in every cancer, so the pan-cancer hTERT claim - which rests on promoter-reactivation and promoter-mutation biology, not protein abundance - can be neither confirmed nor refuted with this evidence class. The circuit behaviour itself is internally consistent: the dual-input AND design expresses the payload at 9.5168 in tumour against 0.0502 in normal and 0.1463 in single-positive tissue, while the single-input comparator leaks (single-positive equals tumour output). We kept the table’s selectivity numbers as uncalibrated literature-free defaults and recorded the evidence class each claim actually needs.

(F83) Detection specificity. For gene g and claimed cancer set C , $\text{detshare} = \sum_{c \in C} \text{det}_{g,c} / \sum_c \text{det}_{g,c}$ with $\text{det} = n_{\text{High}} + n_{\text{Med}} + n_{\text{Low}}$ per HPA patient counts; PSA/KLK3 attains 1.0. Dual-input gate: payload $\propto \prod_i a_i$ vs single-input $\propto a_1$; selectivity requires output in single-positive tissue $\ll$ tumour.

16.44 Educational DNA-to-code utilities against Ensembl proteins and consequence rules

`dna_to_code` maps molecular-biology concepts to code analogies and exposes small executable helpers. We ran the helpers on 38 real mouse Ensembl coding sequences. Translation reproduced the Ensembl protein exactly for 38 of 38 transcripts, and the codon table matched NCBI translation table 1 on all 64 codons. The mutation helper had a real gap: it only reported silent and nonsense flags, so start-codon and stop-codon substitutions were silently lumped with missense changes. On 220 real substitutions the old flags agreed with Sequence Ontology consequence terms on 158; every start-lost and stop-lost change was misreported. The fixed helper returns a consequence term and agrees on 220 of 220. A second error sat in the teaching snippet for transcription, which swapped T for U on the template strand; on real cDNAs that gives 0 of 38 correct mRNAs, against 38 of 38 for the corrected reverse-complement form. Negative note: the Ensembl VEP service returned server errors throughout the run, so the reference classifier is an independent Biopython codon-level implementation of the same rules, not VEP itself.

(F84) Codon-level consequence. For substitution at CDS position p (0-based), codon index $k = \lfloor p/3 \rfloor$, reference codon r and mutant codon m with $a = T(r)$, $b = T(m)$ under table T : start_lost if $k = 0 \wedge r = \text{ATG} \wedge m \neq \text{ATG}$; stop_lost if $a = * \wedge b \neq *$; stop_retained if $a = b = *$; stop_gained if $b = *$; synonymous if $a = b$; else missense. Template identity: $\text{mRNA} = \text{rev}(t)$ with $A \rightarrow U$, i.e. the coding strand with $T \rightarrow U$.

16.45 Proteochemometric drug–target baseline against live ChEMBL and UniProt

dti_bench trains a target-conditioned ridge model over ligand fingerprints, protein composition and explicit interaction terms, and ingests exact-relation pChEMBL pairs from the live ChEMBL API. The audit found that the ingestion client could never succeed on the live API: it read component sequences inline from the target resource, which embeds no sequences, so every call raised before any data was retrieved. The fixed client fetches the component resource and fails closed on multi-component targets. On six kinases it retrieved 1,466 exact-relation records; all six target sequences match UniProt canonical exactly, and the median aggregation reproduces 481 of 481 independently recomputed molecule medians. On the pooled real set a warm-random split gives RMSE 0.953 against 1.403 for a naive-mean baseline (R^2 0.539, seed-deterministic split, all 293 test records inside the applicability domain). A chronological split degrades to R^2 0.134, and a cold-target split holding out ABL1 and EGFR collapses to R^2 -0.883 - but the module’s applicability gate flags all 405 of those predictions as outside-domain, which is the intended honest behaviour for a composition baseline. Ligand similarity rank-agrees with the RDKit Morgan generator (Spearman 0.864 over 1,770 pairs). Caveat: the year-granularity time split lets the boundary year straddle train and test.

(F85) Cold-start applicability gate. Ridge on standardised features z : $\hat{y} = b + \sum_j \beta_j z_j$, $\beta = (Z^\top Z + \alpha I)^{-1} Z^\top y$. Applicability: inside iff $\max_{s \in T} \text{Tan}(f(x), f(s)) \geq 0.35$ and $\max_{q \in Q} \text{pid}(p, q) \geq 0.30$ with positional identity $\text{pid} = n^{-1} \sum_i [p_i = q_i]$. Cold-target holdout: $R^2 = -0.883$ with 405/405 flagged outside.

16.46 Mutation docking against Platinum experimental affinity changes

mutdock predicts how a point mutation in a binding pocket changes small-molecule binding. We validated it against the Platinum database of experimentally measured affinity changes (1,008 single-point records). For 36 entries we rebuilt the true binding pocket from the fetched wild-type PDB structure (residues within 6 Å of the ligand) and compared against $\Delta\Delta G = RT \ln(K_{mt}/K_{wt})$. Direction is right in 72.2% of cases, but the rank correlation with experiment is only 0.036. The cause is visible in the predictions: the docking rescore is composition-based, so a same-class substitution leaves it unchanged and the output collapses onto the class-penalty table. Auditing that table on 776 Platinum entries shows every covered class pair has the correct sign, since binding-site mutations hurt binding on average, but the magnitudes are miscalibrated and five of sixteen pairs use an undocumented default. We kept the table and documented the gap rather than refit it to the validation data. Formula checks pass: the reported fold-change equals $e^{\Delta\Delta G/0.593}$ exactly.

(F86) Mutation $\Delta\Delta G$ heuristic. $\Delta\Delta G = 0.6 (G_{mt}^{dock} - G_{wt}^{dock}) + 0.4 P(c_{wt} \rightarrow c_{mt})$ with class-penalty table P ; fold-change = $e^{\Delta\Delta G/RT}$, $RT = 0.593$ kcal/mol at 298 K. Platinum audit: sign correct for 11/11 covered pairs; Spearman $\rho = 0.036$.

16.47 Biomaterial degradation model against its own reported half-lives

bio_material designs production and predicts degradation for four biomaterials. The audit found an internal contradiction: the degradation curve decayed with half-life $1/k$ while the reported half-life was $\ln 2/k$ (for PHA in soft tissue, 42.0 d versus the reported 28.9 d). The fixed module uses the first-order form $M_0 e^{-kt}$; all 16 material-by-site cases now agree with their reported half-lives and track e^{-kt} to within 0.15 percentage points. The remaining physics helpers match independent recomputation: rubber elasticity $G = \nu RT$, $E = 3G(1 - \nu)^2$; Herschel–Bulkley stress; the analytic cosh profile of steady reaction–diffusion (agreement 3.5×10^{-18}); and a deterministic polymerization integration. Table audit: PHA modulus and tensile match the

Goodfellow PHB datasheet exactly; PLA tensile sits inside the datasheet range while its modulus is 0.3 GPa below it; spider silk matches classic dragline values; bacterial cellulose is unverified. Degradation rates and site factors remain heuristic.

(F87) First-order degradation. $M(t) = M_0 e^{-kt}$, $t_{1/2} = \ln 2/k$; effective modulus of a porous network $E = 3\nu RT(1-p)^2$. Pre-fix curve used 2^{-kt} ($t_{1/2} = 1/k$) against a reported $\ln 2/k$.

16.48 Cell-free expression optimizer grid alignment

cell_free_opt grid-searches a Gaussian response surface for cell-free protein synthesis yields. The audit found that the default grid never evaluated the documented optimal template concentration, capping the reported optimum at 1.879 g/L against the model maximum 2.0 g/L. After the fix the search returns the exact optimum for grid steps 1-4 and agrees with an independent brute-force argmax. Cost, batch normalisation, sensitivity elasticities and Pareto filtering all match independent recomputation. Caveat: the Euler-integrated kinetic trace overshoots the analytic integral (final 2.034 vs 1.9 g/L at 6 h), a declared discretisation.

(F88) CFPS response surface. $y = 2 \prod_i e^{-((x_i - o_i)/w_i)^2}$ g/L over reagents x_i with optima o_i and widths w_i ; kinetic trace $r(t) = v_{max} e^{-t/tau}$, cumulative $v_{max} tau(1 - e^{-t/tau})$; grid must contain o_i for all i .

16.49 CAR-T construct curation against approved products

car_t_designer recommends CAR architectures from curated antigen, scFv and costimulatory tables. Its CD19 recommendation (FMC63, CD8 α hinge, 4-1BB, CD3 ζ) is exactly the Kymriah architecture, and its BCMA recommendation (C11D5.3, CD8 α , 4-1BB) matches the Abecma FDA label; the alternative CD28 CD19 product is acknowledged in the rationale. The costimulatory choice rule, safety-switch inclusion, toxicity formula and all helper functions (selectivity ratios, logic gates, exhaustion and killing curves, trial persistence) reproduce independent recomputation. Predicted response for CD19 is 0.6, in a plausible range for the declared class-level heuristic; the module states it is not fitted to per-antigen clinical data.

(F89) CAR heuristic scores. CRS risk = $\min(0.3 + 0.2 e_c, 0.9)$, off-tumour = $0.15 |N|$ for costim expansion e_c and normal tissues N ; exhaustion $E(d) = 1 - e^{-rd/14}$; kill fraction = $f(1 - e^{-hoR})$ at E:T ratio R with antigen-positive fraction f ; persistence $p(0.85)^m$.

16.50 Biofoundry scheduler against workflow dependencies

biofactory_1_a generates robotic protocols and schedules them. The audit found that the scheduler ignored workflow order: steps started whenever their resource was free, so the golden-gate thermocycle ran at $t = 0$ in parallel with its own input dispenses and the reported makespan was 120 min against the true 274 min. After the fix every step starts at the later of resource-free and predecessor-end; all three workflows now give ordered schedules whose makespan equals the sequential sum (golden-gate 274, gibson 126, pcr-screen 173 min). The Bayesian condition update, PID trace, reagent half-life compensation, Zhang Z' factor and DAG rerouting all match independent recomputation exactly. Protocol steps sit within standard NEB Golden Gate and Gibson method bounds.

(F90) Assay and update identities. Zhang $Z' = 1 - 3(\sigma_s + \sigma_b)/|\mu_s - \mu_b|$; conjugate update $\pi^{-1} = \pi_0^{-1} + n/u$, $\mu = (\mu_0/\pi_0 + \sum o_i/u)\pi$ for noise variance u ; reagent activity $a = \ell 2^{-t/t_{1/2}}$ with compensation $1/a$; list schedule $s_i = \max(r_{free}, e_{i-1})$.

16.51 Metabolic pathway designer against KEGG and fermentation stoichiometry

metabodesigner searches a curated reaction graph for glucose-to-product routes and scores them with an LP flux model. Every reaction was checked against its KEGG ENZYME record (name and substrate/product compound IDs) and every metabolite formula against KEGG COMPOUND. The audit found that the glycolysis graph had no triose-phosphate isomerase, so the aldolase product DHAP was a dead end and the LP returned 1.0 mol lactate and 1.0 mol ethanol per mol glucose against the textbook 2. The thiolase step also consumed one acetyl-CoA instead of two, and by cancellation the 3-hydroxybutyrate yield stayed at the correct 1.0. After adding EC 5.3.1.1, a byproduct-recycling closure, the 2:1 thiolase stoichiometry and the PDH CO_2 , the yields are 2.0, 2.0 and 1.0 mol/mol; 11 of 17 reactions match KEGG exactly, 5 are correctly named lumped steps and one has a class-level EC; 15 of 16 checkable reactions are carbon balanced. The remaining imbalance is citrate synthase, which treats oxaloacetate as a cofactor. Free-energy and kcat defaults are placeholder priors and were not validated.

(F91) Pathway yield by LP. $\max v_T$ s.t. $S_{int}v = 0$, $0 \leq v \leq 1$; molar yield $Y = (Sv)_T / (-(Sv)_{glc})$; carbon balance $\sum_s n_C(s) = \sum_p n_C(p)$ with acyl-CoA counted as $n_C - 21$; glycolysis requires $\text{DHAP} \rightleftharpoons \text{GAP}$ for $Y_{lac} = 2$.

16.52 Biosensor design against PubChem descriptors and binding literature

bio_switch couples curated ligand-binding domains to reporters and models dose-response, noise and processing. Its descriptor table was checked against PubChem for all five small-molecule analytes: molecular weights, TPSA and donor counts match, acceptor counts differ only by definition (Lipinski N+O counting versus PubChem’s refined rules), and logP values differ by method (clogP versus XLogP3). The audit found that the Lipinski violation count was cast to a boolean, so lactose (two violations: 8 donors, 11 acceptors) was classed moderate instead of low under the module’s own rule, and the returned field read True instead of 2. A second curation error put the TetR-tetracycline K_d at 10 nM against the measured $K_a = 3 \pm 2 \times 10^9 \text{ M}^{-1}$ (0.2-1 nM, Takahashi 1991, PMID 1812784); both are fixed. The fluoride riboswitch K_d of 60 μM matches the crcB literature exactly; the theophylline aptamer value (0.3 μM) sits within the 0.1-0.32 μM literature range. Hill response, tunable-sensor c10/c90 (grid-quantized, 7.94 versus continuous 9.0 for $n = 2$), gate logic, temporal filtering, feedback curves and degradation kinetics all reproduce independent recomputation, and the Gillespie simulation converges to the deterministic occupancy (0.1664 versus 0.1667 bound fraction) with seed reproducibility and a correct copies-per-fL conversion. Documented caveats: the reaction-diffusion profile uses a semi-infinite slab ($\exp(-x/\lambda)$) where a finite cell gives flatter cosh profiles, off-target cross-reactivity is an order-of-magnitude proxy, and the remaining curated K_d values are plausible curations, not individually verified.

(F92) Sensor dose-response and noise. Hill $R(c) = V_{max}c^n / (K_d^n + c^n)$, threshold ratio $c_{90}/c_{10} = 81^{1/n}$; occupancy $\bar{b} = L / (L + K_d)$ from birth-death propensities $k_{on}L(N - b)$, $k_{off}b$; copies per fL = $6.022 \times 10^2 C_{\mu\text{M}}$; intracellular gradient $c(x) = c_0 e^{-x/\lambda}$, $\lambda = \sqrt{D/k}$.

16.53 Dark-genome regulatory decoder against JASPAR and UCSC CpG islands

dark_genome decodes the regulatory landscape of non-coding sequence: TF motif hits, enhancer clusters, CpG islands and lncRNA candidates, plus biophysical surrogates. Its eight TF consensus motifs were checked against JASPAR position weight matrices: five match the matrix argmax sequence exactly (AP-1, NF- κ B, TATA-box, E-box, CCAAT), while p53, GC-box and HNF4 follow the canonical literature consensus (El-Deiry RRRCWWGYYY, matching the p21/WAF1 site GAACATGTCC; GC-box GGGCGG; HNF4 DR1) that differ from the assay-specific JASPAR versions fetched. The audit found that motif_binding_energy credited any base at a degenerate motif position, so R matched C/T: on poly-C with motif RRR the module reported match 1.0 against the true 0.0, and the p53, TATA-box and HNF4 minimum energies on a real promoter were all inflated. After the fix to set-based IUPAC matching all eight energies equal independent recomputation. Against the UCSC cpGIslandExt track on hg38 the module detects both real promoter islands tested (MLH1, CDKN2A) but calls 3 false positives in a 13 kb desert window because it omits UCSC's post-merge length filter; the tradeoff is documented rather than refit. All 64 motif hits reported for the 10 kb MLH1 locus pass an independent IUPAC regex check. The weighted coordinate field, damped graph propagation, nucleosome occupancy, enhancer-gene causality, logic gates, temporal trajectories and the precision-weighted perturbation update all match independent recomputation exactly.

(F93) Regulatory field and propagation. Potential $\phi = 0.3g + 0.3a + 0.25h + 0.1(1-m) + 0.05 \min(1, 20\rho)$ for GC g , ATAC a , H3K27ac h , methylation m and motif density ρ ; contacts $w_{ij} = (1 + |i - j|)^{-0.8}$; damped propagation $s_{t+1} = (1 - \delta)s_0 + \delta A_{row}^{-1} A s_t$; CpG $O/E = n_{CGN}/(n_{CG}) > 0.6$ with GC > 0.5 ; energy $E = -8 \text{ match} + 0.8 \text{ AT shape}$ with set-based IUPAC matching.

16.54 Cell-fate GRN engine against HGNC and reprogramming cocktails

cellfatenet simulates a curated 18-node lineage GRN with Hill dynamics, epigenetic gating and optimal-control reprogramming. Curation checks: 19 of 20 gene symbols are approved HGNC symbols (OCT4 is an alias of POU5F1); the cardiac driver edges cover the full Ieda GMT cocktail (GATA4, MEF2C, TBX5; PMID 20691899), while OSKM lacks KLF4/MYC and the neuronal BAM set lacks POU3F2 (coverage caveats). The audit found two defects. First, production was $\sigma(2s)$, a 0.5 basal floor that left every gene half-expressed with no regulators (quiet TNNT2 steady state 1.43); a thresholded $\sigma(6(s - 0.5))$ brings the quiet state to 0.20 with single-driver activation preserved. Second, the optimal-control stack was self-defeating: controls could force structural markers, unspecified targets counted as zeros so driver activation was penalized - the pre-fix fibroblast-to-cardiomyocyte recipe repressed MEF2C and TBX5, the opposite of the GMT protocol - the finite-difference step sat below solver noise, and negative concentrations turned the x^2 Hill term into a sign exploit. After restricting controls to transcription-factor nodes, scoring only specified targets, setting $\varepsilon = 10^{-3}$ and clamping the Hill input, the cardiac search overexpresses MEF2C and reaches TNNT2 1.98 / MYH6 2.01 with stochastic success 1.0. Deterministic trajectories match an independent integrator to 0.0; the Langmuir chromatin occupancy, noisy-ensemble statistics and attractor stability ($\max |\dot{x}| = 3.1 \times 10^{-8}$) all check out.

(F94) Fate dynamics and control. $\dot{x} = g \odot \sigma(6(W\phi(x) - 0.5)) + u - \gamma x$, $\phi(x) = x_+^2 / (0.25 + x_+^2)$, $\gamma = 0.35$; occupancy $= K_a c / (1 + K_a c)$ with $K_a = e^{-\Delta G/RT}$; objective $J = \sum_{i \in S} (x_i(T) - t_i)^2 + 0.08 \|u\|_2^2 + 0.04 \|u\|_1$ over specified targets S with controls restricted to TF nodes; success $= e^{-\|x(T) - t\|_S}$.

16.55 Literature knowledge graph against live PubMed, iCite and statsmodels

biogpt_lit builds an evidence-weighted claim graph from the literature: a precision-biased relation extractor, citation/sample-size/replication edge weights, contradiction detection, temporal consensus, multi-hop latent paths, gap prioritization and a DerSimonian-Laird meta-analysis engine. The sweep ingested 8 live PubMed abstracts (query: SIRT1 resveratrol activation, all 2026) through the module's own entrez client and merged NIH iCite citation counts (only one PMID has an iCite record so far; the query returns very recent papers - documented). The audit found one defect: the extractor accepted any token before a relation word as the subject, producing the garbage triple (and, inhibits, Neuronal) from PMID:42746974 - one of four live-extracted claims. A function-word subject filter fixes it; post-fix all three extracted triples are valid ((formulations, activates, SIRT1), (SIRT1, activates, AMPK), (AMPK, inhibits, mTORC1)) and the negation/loss-of-function unit suite passes 7/7. The meta-analysis engine matches statsmodels combine_effects (DL) to below 10^{-9} on fixed effect 0.29756, random effect 0.32299 and $\tau^2 = 0.08470$. Evidence weights, consensus/confidence, contradiction detection, temporal agreement (0.692), multi-hop path sets, bottleneck path strength (0.515), gap leverage and the update_validation flip all equal independent recomputation exactly.

(F95) Evidence weighting and DerSimonian-Laird pooling. $w = 0.3 + 0.3 \min(\log_{10}(1 + c)/3, 1) + 0.25 \mathbb{I}_{\text{rep}} + 0.15 \min(n/100, 1)$; fixed effect $\bar{\theta} = \sum w_i \theta_i / \sum w_i$ with $w_i = 1/v_i$; $Q = \sum w_i (\theta_i - \bar{\theta})^2$; $\tau^2 = \max(0, (Q - (k - 1)) / (S_1 - S_2 / S_1))$; random weights $1/(v_i + \tau^2)$; temporal agreement $a_t = w_t^+ / (w_t^+ + w_t^-)$; path strength = $\min_{e \in p} w_e$.

16.56 Gene digital twin against NCBI Gene, UniProt, RefSeq and PubMed

gene_analysis profiles a gene locus (composition, longest ORF, regulatory landscape, CRISPR targets, variant interpretation) and merges live NCBI Gene, UniProt, ClinVar and Reactome records. For TP53, BRCA1, CFTR, HBB and EGFR the live profile returned the correct NCBI Gene identifier and chromosome, the Swiss-Prot canonical accession and canonical protein length in 5 of 5 genes. On RefSeq mRNAs NM_000518.5, NM_000546.6 and NM_000492.4 the longest ORF equals the annotated CDS (147, 393 and 1480 residues) and molecular weights agree with Biopython to within 4.2×10^{-5} relative. Publication counts match an independent PubMed query exactly and replicate numbers equal statsmodels NormalIndPower in 4 of 4 designs. Bug 44 was found on real transcripts: HGVS c. coordinates were indexed into the transcript without the 5' UTR offset, and full ClinVar names were parsed from the accession digits, so the sickle variant c.20A>T read codon TTC or ACT instead of GAG. After the fix, 6 of 6 clinical hotspots (HBB E7, Q40; TP53 R175, R248, R273; CFTR G551) map to the correct reference base and residue.

(F96) HGVS coordinate mapping and sample size. Locus position $p = s_{CDS} + c$ for $c > 0$ and $p = s_{CDS} + c + 1$ for $c < 0$, with s_{CDS} the 0-based start of the longest forward ORF; codon start $s_{CDS} + 3 \lfloor (p - 1 - s_{CDS}) / 3 \rfloor$; replicates $n = \lceil 2\sigma^2(z_{1-\alpha/2} + z_{1-\beta})^2 / \delta^2 \rceil$; Boltzmann populations $p_k = e^{-E_k/RT} / \sum_j e^{-E_j/RT}$ with $RT = 0.593$ kcal/mol.

16.57 Liquid-biopsy stack against TCGA PanCancer and cfDNA fragment literature

liquid_biopsy combines ctDNA calling, a cfDNA fragment-length model, Dirichlet-mixture haplotype EM, tumor architecture assembly and longitudinal tracking. Its tumor-type marker panels were scored against TCGA PanCancer Atlas mutation calls from cBioPortal for six cancers. TP53 was the most frequently mutated checked gene in lung adenocarcinoma (52.1% of 566 samples) and was mutated in 59.8% of 179

pancreatic samples, yet was missing from both panels; adding it raised panel coverage from 46.3% to 76.1% (lung) and 66.5% to 74.9% (pancreatic). Bug 45: haplotype candidates were built by filling unobserved loci with 0, so a haplotype observed only as partial fragments was never a candidate; with truth 111/000 the EM returned 000, 011 and 110. After the fix it returns 111 and 000 at 0.479 each. The fragment model's docstring claimed entropy rises with tumor fraction; the model itself shows entropy peaking near balanced mixtures (4.60 bits at 0.3, 4.55 at 0.6), whereas KL divergence and the 100-150 bp fraction rise monotonically. Anchors match Underhill et al. (134-144 bp versus 167 bp).

(F97) Fragment mixture and haplotype EM. Gamma component with mode m and spread s : $k = 1 + (m/s)^2$, $\theta = m/(k-1)$; entropy $H = -\sum_b p_b \log_2 p_b$; $D_{KL}(p||q_h) = \sum_b p_b \log(p_b/q_{h,b})$; fragment likelihood $P(f|h) = (1-\epsilon)^{n_o-m} \epsilon^m$ over n_o observed loci with m mismatches; update $\pi_j = (\sum_i r_{ij} + \alpha)/(N + \alpha K)$; panel coverage = $|\bigcup_{g \in P} S_g|/N$.

16.58 AAV capsid engineer against real VP1 sequences and scoring recomputation

vector_opt designs capsid variants by trading receptor-affinity gain against NAb escape and conservation penalties over five variable regions. Its scoring was recomputed exactly from the stated formulas with the module's own random sequence for three seeds, and directed evolution round r equals a fresh engineering run seeded $1+r$. The region annotations were checked against the real capsid sequences of AAV2 (UniProt P03135, 735 residues) and AAV9 (Q6JC40, 736 residues): a global alignment gives 82.3% identity, and window identity confirms VR-VIII (0.33) and VR-IV (0.43) as the most variable regions. Two annotation limits were found and documented: receptor contact is serotype-dependent (AAV2's heparan-sulfate contact R585/R588 sits inside VR-VIII, PMID 9445046, while the table follows the AAV9 default), and the GH-loop conservation constant 0.35 reflects the loop tip, since the measured identity over the 590-620 span is 0.81.

(F98) Capsid variant scoring. Receptor gain $g = 0.15 m c_r u_1$ with $c_r \in \{1, 0.2\}$, $u_1 \sim U(0.7, 1.3)$; NAb loss $l = 0.2 m u_2 \mathbb{1}_{\text{epitope}} + 0.02 m u_2 (1 - \mathbb{1}_{\text{epitope}})$; integrity $I = 1 - 0.08 m (1 - C_r)$; transduction $T = \text{clip}(0.5 + g - (1 - I), 0, 1)$; ranking score $0.5 T + 0.5 \min(l, 1)$; global identity $604/734 = 0.823$.

16.59 Neurosurgical planner against an admissible-path oracle

neuroplan_ai plans tumor approaches: corridor scoring over an eloquent-region risk field, weighted A $\times$ pathfinding, MRI segmentation, tractography clearance and a NeuroTwin resection simulator. The corridor algebra, NeuroTwin probabilities, segmentation threshold and centroid all recompute exactly. Two bugs were fixed. Bug 47: tract clearance was measured to streamline vertices only, so a fiber passing near the tumor between samples was reported 7.30 mm clear when the segment comes within 6.0 mm; clearance is now point-to-segment. Bug 46: the A docstring claimed an admissible heuristic, but the frontier uses a 50-fold inflated weight; against an admissible oracle with the identical cost, the returned path cost 640.6 versus the 340.1 optimum (88% gap) on an oblique entry. The docstring now states the greedy bound, and four dead duplicate definitions were removed. The normalized atlas places brainstem lowest, motor cortex highest and visual cortex most posterior, consistent with anatomy.

(F99) Risk field, path cost and tract clearance. $R(p) = \sum_i w_i e^{-d_i^2/2\sigma^2}$, $\sigma = 12$; step cost $c = \ell(1 + \lambda \min(R, 1.25))$, $\lambda = 8$; frontier $g + 50h$; segment clearance $\min_t \|p_0 + t(p_1 - p_0) - c\| - r$, $t = \text{clip}(\frac{(c-p_0) \cdot (p_1-p_0)}{\|p_1-p_0\|^2}, 0, 1)$; deficit $p = \min(0.95, w k f / (1 + \text{clr}/5))$; aggregate $1 - \prod(1 - p_i)$.

16.60 Structure predictor against 1UBQ, AlphaFold DB and the 1978 Chou-Fasman rules

alpha_fold.ui implements Chou-Fasman prediction, idealized backbone building, pocket detection and refinement, and can analyze real structures. On real data: ubiquitin 1UBQ parses to 76 residues with X-ray resolution 1.8 Å and exactly recomputed B-factor statistics; the AlphaFold model of HBB (P68871) has mean pLDDT 97.18 with no low-confidence residues. Bug 48: the helix-extension rule admitted any residue with $P_\alpha > 100$, so on ubiquitin the prediction was 92% helix while 1UBQ has one helix and five strands. The 1978 breaker rule (stop when the mean propensity of the next tetrapeptide drops below 100) with per-residue P_α/P_β conflict resolution predicts the 23-34 helix plus strand regions, $Q_3 = 0.513$ against the literature assignment - the expected range for this method. The parameter table matches the canonical 1978 values for all 20 amino acids. Generated backbones round-trip through a standard parser with consecutive C α -C α distances of 3.799-3.801 Å, and restrained refinement lowers the energy from 100.96 to -8.99 with exact 3.8 Å bonds.

(F100) Chou-Fasman nucleation, breaker and conflict rules. Helix nucleus: ≥ 4 of 6 residues with $P_\alpha \geq 100$; strand nucleus: ≥ 3 of 5 with $P_\beta \geq 100$; extension stops when $\bar{P}$ over the next tetrapeptide < 100 ; overlaps resolved by $\bar{P}_\alpha$ vs $\bar{P}_\beta$ over the ± 3 window; $Q_3 = \frac{1}{n} \sum_i \mathbb{I}[\hat{s}_i = s_i] = 0.513$ on ubiquitin; refinement energy $E = E_{LJ} + E_{elec} + k_b \sum (r - 3.80)^2$.

16.61 Fragment molecule generator against PubChem and live ChEMBL

chemgpt_engine assembles molecules from a 12-fragment library and scores fragment-additive ADMET. Seven assembled molecules - phenol, aniline, benzoic acid, fluorobenzene, 2-aminopyridine, 4-methylimidazole and phenethylamine - each resolved to the correct real compound in PubChem (queried by SMILES), with molecular weights within 0.03 Da. The scoring formulas recompute exactly, and the Pareto front (20 of 24 candidates) matches a brute-force non-dominated filter over an independently regenerated candidate list. Novelty claims were checked against the live ChEMBL similarity service: phenol’s nearest neighbor is phenolate sodium at 100% similarity, while the top generated design has no neighbor at or above 80%, so its “novel scaffold” label is measured, not asserted. No defect was found in this module.

(F101) Fragment-additive assembly and Pareto filtering. $MW = \sum_f MW_f - 2.016(k - 1)$ for k fragments; $\log S = 0.5 - \log P - 0.01(MW - 200)/50$; dominance $a \succeq b \iff \forall j o_j^a \geq o_j^b \wedge \exists j o_j^a > o_j^b$; Boltzmann weights $w_i = e^{-E_i/0.593} / \sum_j e^{-E_j/0.593}$ sum to 1.

16.62 Circuit stability scorer against recomputation and real restriction sites

syn_stab.ai estimates the evolutionary stability of synthetic circuits: expression burden, plasmid-loss and mutation survival, a Wright-Fisher-like population simulation and sequence-level risk. All formulas recompute exactly, and the population simulation matches the closed selection recursion in the deterministic limit. Bug 49: the sequence-risk palindrome counter compared each 6-mer with its plain reverse, so real inverted repeats - EcoRI GAATTC, HindIII AAGCTT, SnaBI TACGTA - all counted zero; the reverse-complement check counts them, while textual palindromes such as GATTAG correctly remain at zero. Strategy comparison ranks genomic integration above promoter weakening, consistent with the loss-rate model.

(F102) Burden, retention and selection recursion. Burden $b = \min(1, 0.25ge)$; plasmid retention $(1 - \lambda)^t$ with $\lambda = \lambda_0(1 + 4b)$; mutation survival $e^{-\mu Lt}$; update $f \leftarrow f(1 - \mu)(1 - c)/(1 - fc)$ with diffusion

noise $\sqrt{f(1-f)/N}$; inverted repeat: $x = \overline{\text{comp}(x)}$.

16.63 Phage designer against recomputation and adsorption kinetics

phage_designer plans tail-fiber retargeting, chimeric endolysins, adsorption, escape risk and cocktails. The fiber and lysin formulas, escape probability $1 - e^{-\mu s N}$ and host-range scoring all recompute exactly. Bug 50: adsorption_kinetics accepted time_min but never used it - bound phage was identical at $t = 1$ and $t = 60$ minutes. Adsorption is a first-order process (Schlesinger 1932), so the function now returns $P_0(1 - e^{-k B t})$; post-fix, bound phage grows with time and matches the closed form to machine precision.

(F103) Fiber adsorption score and first-order kinetics. Adsorption score $a = \min(0.99, 0.55 + 12/K_d)$; lysin potency $\bar{\alpha}$ with MIC = $\max(0.1, 8(1.1 - \bar{\alpha}))$ $\mu\text{g/mL}$; escape $P = 1 - e^{-\mu s N}$; kinetics $B(t) = P_0(1 - e^{-k C t})$.

16.64 CRISPR phage forge against the real NDM-1 plasmid and a host-chromosome screen

phageforge designs CRISPR-armed phages against resistance genes: backbone selection, guide design, payload budget, kill curves, escape risk and cocktail coverage. On the real blaNDM-1 CDS (813 bp, from plasmid pKpANDM-1, FN396876.1, 4318 bp) the module selects the P1 backbone (15 kb capacity, lytic, the eligible maximum for the 4.5 kb payload) and all three guides verify as 20-mers inside the CDS. Bug 51 couples two defects: the off-target screen ran against the target gene itself while the report claimed a screen against a supplied phage genome - an input that does not exist - and the CFD scanner skipped every perfect-match site, making the worst off-target class invisible (a planted exact match returned zero hits pre-fix). Perfect matches are now reported at CFD 1.0, design_phage accepts background_genome, and the report states exactly what was screened; against a real 40 kb window of the K. pneumoniae MGH 78578 chromosome (NC_009648.1) all three blaNDM-1 guides show zero CFD hits. Bug 52: the cocktail coverage fraction was always 1.0 whenever any host was covered (union denominator); an optional target panel gives a real fraction (0.75 with one uncovered target in the sweep case). Kill curve, escape risk, payload budget and microbiome-sparing recompute exactly; backbone capacities were checked against the NCBI-verified genome sizes (T7 39,937 bp; lambda 48,502; T4 168,903; P1 94,800; M13 6,407) and are payload headroom, plausible but not individually literature-sourced (documented).

(F104) Phage infection, escape and screening formulas. Infected fraction $f = 1 - e^{-a \text{MOI}}$; surviving $S = (1 - f)(1 - r) + r$; effective burst $f B$; expected escapees $E = N \mu^k$ for k independent targets with $P_{\text{esc}} = 1 - e^{-E}$; guide off-target risk $R = \sum_s \text{CFD}(g, s)$ over background sites within three mismatches, perfect matches included; cocktail coverage $|\mathcal{T} \cap \mathcal{C}|/|\mathcal{T}|$; microbiome sparing $0.6 + 0.3(1 - 0.5 e)$.

16.65 Variant copilot against a live ClinVar panel and UniProt

bio_copilot is a variant-interpretation workbench: a mutation-to-phenotype computation graph (variant annotation, structural perturbation, pathway effect, phenotype inference with inconsistency flags), CRISPR guide enumeration, FASTA/PDB structured output, a UniProt-grounded question router, Monte-Carlo uncertainty propagation and beta-binomial experiment feedback. The curated gene slice was checked against live UniProt: all three lengths match exactly (TP53 P04637 393 aa, BRCA1 P38398 1863 aa, KRAS P01116 189 aa) and the TP53 DNA-binding domain 102-292 equals the UniProt feature, while the BRCA1 RING and BRCT windows are wider local approximations (documented against the UniProt zinc finger 24-65 and

the BRCT domains 1642-1736 and 1756-1855). An eight-variant panel was scored against live ClinVar records. Bug 53: a premature stop set pathway flux to 1.0, but the risk formula $0.15 + 0.5f$ capped at 0.65, so TP53 R213* and BRCA1 Q563* - both Pathogenic in ClinVar, one expert-panel reviewed - were returned as “uncertain significance”. The fix adds a per-gene loss-of-function flag with a PVS1-style floor $r \geq 0.75$ (ACMG/AMP guideline, PMID 25741868) and deliberately withholds it from gain-of-function oncogenes: KRAS L168* is Likely benign in ClinVar and other KRAS truncating records are VUS or conflicting, so the module stays in the uncertainty band and raises an explicit caveat flag. Exact-band agreement rose from 4/8 to 6/8; the two remaining calls are one band away with documented rationale (oncogene-nonsense conservatism, and a KRAS hotspot-coverage caveat - the curated hotspot list was not refit to the validation panel). The guide enumerator returns a set identical to crispr_opt on the real 813 bp blaNDM-1 CDS (137 guides, both strands, same forward-coordinate convention), and to_pdb output on the real 189-residue KRAS sequence parses cleanly (189 C α atoms, residue names matching, consecutive C α -C α 3.8297 Å against the documented 3.8 Å ideal-helix geometry). Uncertainty propagation matches the analytic product-of-Gaussians mean and the beta-binomial update matches scipy exactly.

(F105) Variant risk with a mechanism-aware nonsense floor. $r = 0.15 + 0.5f + 0.25\mathbb{I}_h$; flux $f = \min(1, \frac{\Delta\Delta G}{3}c)$ with $c = 1.5$ at hotspots and 0.8 otherwise, and $f = 1$ for nonsense; $\Delta\Delta G = 0.8|\Delta H| + 1.2\mathbb{I}_h + 0.3(1 - \mathbb{I}_h) + 0.5\mathbb{I}_d$ over hydrophobicity change ΔH , hotspot h and domain d indicators; the floor $r \leftarrow \max(r, 0.75)$ applies exactly when the variant is nonsense and the gene carries the loss-of-function flag (PVS1-style); bands $r > 0.7$ likely pathogenic, $r > 0.4$ uncertain, else likely tolerated.

(F106) Product of independent Gaussians. $E[\prod_i X_i] = \prod_i \mu_i$ by independence; for $(\mu, \sigma) = (0.8, 0.1)$ and $(0.7, 0.05)$ the analytic mean is 0.56, matched by a 2×10^4 -sample Monte-Carlo mean whose 95% interval contains it.

(F107) Beta-binomial experimental feedback. Prior mean μ_0 with strength s : $a = \mu_0 s + n_s$, $b = (1 - \mu_0)s + (n - n_s)$; posterior mean $a/(a + b)$, equal to the scipy beta mean ($\mu_0 = 0.5$, $s = 10$, 9/10 successes $\rightarrow 0.7$).

16.66 Cell-image analyser against analytic shapes and BBBC001 ground truth

bioimage.ai segments microscopy frames (flat-field correction, Otsu or adaptive threshold, distance-transform splitting), measures per-cell morphology and computes 50 screening diagnostics. Two defects were found on geometric ground truth. Bug 54: the perimeter was the boundary-pixel count, which measures a 45-degree edge at $1/\sqrt{2}$ of its length - circularity scored 1.27 for a perfect disk and 1.69 for a rotated square. The Freeman chain convention (orthogonal steps $\times 1$, diagonal $\times \sqrt{2}$) gives 0.92 for the disk - the standard region-props value, the residual being pixelation - and exact perimeters for cardinal shapes (76 for a 20-pixel square, $56\sqrt{2}$ for the diamond). Bug 55: distance-maximum markers fired on mid-ramp pixelation plateaus, so one convex elongated cell was split into three fragments; regional maxima by morphological reconstruction leave a single marker (the same cell now measures aspect 3.74 against nominal 4), while a genuine two-peak touching pair still splits. The Otsu threshold matches an independent between-class implementation exactly, and on all six real BBBC001 images (Broad benchmark, human HT29 cells) the module counts sit 4-9% above the inter-observer mean and track the manual counts with Pearson $r = 0.9983$ - inside the noise band of the two human counters, who disagree by up to 27% on one image.

(F108) Otsu thresholding. Class probabilities $\omega(k) = \sum_{i \leq k} p_i$, means $\mu(k) = \sum_{i \leq k} i p_i$; pick $k^* = \arg \max_k (\mu_T \omega(k) - \mu(k))^2 / (\omega(k)(1 - \omega(k)))$.

(F109) Freeman chain perimeter and circularity. $P = N_{\text{ortho}} + \sqrt{2} N_{\text{diag}}$ over the traced outer boundary chain; circularity $C = 4\pi A / P^2$; a filled disk scores ≈ 0.91 , an $n \times n$ square $4\pi n^2 / (4(n-1))^2$.

(F110) Regional maxima by reconstruction. Reconstruct $d - 1$ under the mask d by iterated geodesic dilation $r \leftarrow \min(\delta(r), d)$ to stability; maxima $\{d > r\}$; markers label the maxima and each foreground pixel takes its nearest marker (Voronoi split).

16.67 Evolution sandbox against population-genetics theory

bioplayground simulates a Wright-Fisher construct locus (selection, symmetric mutation, multinomial drift, bottlenecks), directed evolution, Lotka-Volterra competition and diversity metrics. One defect was found. Bug 56: the reported generation count was ten times the number of trajectory snapshots - 310 for a 300-generation run, 100 for a construct lost at generation 92 - and the playground report computed diversity on the last ten-generation snapshot rather than the final state ($h = 0.0198$ reported when the construct was already lost, true $h = 0$). The fix counts generations directly and exposes the final frequency vector, which the report now uses. Validation is against analytic theory: the two-allele mutation equilibrium $p_t = \frac{1}{2} + \frac{1}{2}(1 - 2\mu)^t$ holds (0.5001 vs 0.5012 at $N = 4 \times 10^4$), a single-species Lotka-Volterra run tracks the closed-form logistic within 8×10^{-4} mid-course and exactly at carrying capacity, coexistence calls match competition theory in both regimes, and the Shannon/heterozygosity metrics equal scipy exactly. A design limitation is documented: the construct starts at frequency 1, so with the default mutation rate every run fixes at generation 1 - loss requires mutation strong enough to seed competing alleles.

(F111) Selection-mutation recursion and equilibrium. $p' = p w_0 / \bar{w}$ with $\bar{w} = \sum_i p_i w_i$; symmetric mutation $m_i = s_i(1 - \mu) + \mu(1 - s_i)/(n - 1)$ conserves $\sum_i m_i = 1$; two-allele equilibrium $p_t = \frac{1}{2} + (p_0 - \frac{1}{2})(1 - 2\mu)^t$.

(F112) Logistic competition. $\dot{x}_i = g_i x_i (1 - \sum_j c_{ij} x_j)$; single-species solution $x(t) = K / (1 + A e^{-gt})$, $A = K/x_0 - 1$, $K = 1/c$; coexistence requires interspecific competition below intraspecific for both species.

(F113) Diversity metrics. Heterozygosity $h = 1 - \sum_i x_i^2$; Shannon $H = -\sum_i x_i \ln x_i$; effective alleles e^H .

16.68 Bioprint multiphysics against constitutive-equation ground truth

bioprint_pro calibrates bioinks (Cross-model rheology, Hagen-Poiseuille pressure, first-order crosslinking) and runs a multiphysics chain: extrusion, reaction-diffusion gelation, poro-viscoelastic mechanics and a maturation ODE, with 51 computed diagnostics and a seeded global optimizer. Two defects were found. Bug 57: the extrusion flow used the viscosity at a guessed shear (6.7 s^{-1}) while the resulting wall shear was 33.9 s^{-1} ; the shear-thinning ink is far thinner there, so the flow rate came out $4.36 \times$ low (0.213 vs $0.928 \text{ mm}^3/\text{s}$). The fixed point of $q = \pi r^4 \Delta p / (8 \eta (4q / \pi r^3) L)$ is now solved by bisection and matches an independent root finder. Bug 58: the stress model was labeled Oldroyd-B while returning a shear-thinning stress; an Oldroyd-B fluid has constant shear viscosity, so the label now reads Giesekus-family. The calibrator recomputes

exactly for all 15 windows (alginate at 8 mm/s: 160 s^{-1} , $6.512 \text{ Pa}\cdot\text{s}$, 104.2 kPa), the gelation PDE matches $1 - e^{-kt}$ at the surface, and the maturation ODE matches an independent RK4 integration to 10^{-7} .

(F114) Cross/Carreau rheology and pipe flow. Cross: $\eta = \eta_\infty + (\eta_0 - \eta_\infty)/(1 + (\lambda \dot{\gamma})^n)$; Carreau-Yasuda: $\eta_\infty + (\eta_0 - \eta_\infty)(1 + (\lambda \dot{\gamma})^a)^{(n-1)/a}$; wall shear $\dot{\gamma}_w = 4Q/\pi r^3$; self-consistent rate $Q = \pi r^4 \Delta p / (8 \eta(\dot{\gamma}_w) L)$.

(F115) Gelation, relaxation and maturation. Conversion $x = 1 - e^{-kt}$, $t_{95} = -\ln 0.05/k$; modulus $E_0(1 + 9x)$; stiffness $2 + 48 g^2 \text{ kPa}$; standard-solid strain $\varepsilon(\tau) = \varepsilon_0(0.35 + 0.65e^{-\tau/\tau})$; Euler buckling $E\pi^2/(12n^2)$; maturation $\dot{c} = 0.18c(1 - c/5)o - 0.03c$ with oxygen $\dot{o} = 0.25(1 - o) - 0.08co$.

16.69 Virtual-lab sessions and mechanistic experiments

biosim_vr pairs a scripted virtual bench (move, pipette, incubate, measure) with mechanistic experiments: rigid-pose Lennard-Jones docking by differential evolution, CRISPR guide enumeration, and a stochastic protocol simulator with 51 asserted diagnostics. Two defects were found and fixed. Bug 59: the scripted transfection session set the plate signal to 0.72 but measured the plate-reader object, which has no signal state, so the log read 0.0 while the conclusion claimed 0.72 RFU; the measurement now targets the plate. Bug 60: the docking session crashed outright, calling dock with swapped arguments and a nonexistent seed keyword; it now reports -2.94 kcal/mol . The guide enumerator reproduces crispr_opt's forward-strand set exactly on the real $bla_{\text{NDM-1}}$ CDS (65 guides), the rotation is a proper rotation, the returned docked pose reproduces the returned score exactly with rigid bonds, and on a spherical ground-truth pocket the ligand settles at the LJ well ($3.00 \text{ \AA}$).

(F116) Virtual-lab kinetics and scoring. Incubation gain $g = (1 - e^{-t/240})e^{-((T-37)/12)^2}$; shifted LJ well $u(r) = (3/r)^{12} - 2(3/r)^6$ with minimum -1 at $r = 3$; guide score $\max(0, 1 - 2|GC - 0.5| - 0.15 \max(0, h - 3))/(1 + n_{\text{dup}})$.

16.70 Digital cell twins and virtual drug trials

cell_twin builds a multi-omic patient-cell state and screens drug responses with a viable/apoptotic/proliferating/damage ODE. Two defects were found and fixed. Bug 61: the reported maximum apoptosis was taken at the last-listed dose, so a descending dose list silently returned the minimum-dose value. Bug 62: the combination screen claimed a mechanistic combined-exposure rerun but merely added a constant $+0.08$ synergy bonus to the Bliss expectation for every pair; combinations are now genuinely rerun with the Bliss-combined effect scalar $1 - (1 - e_a)(1 - e_b)$ in the response ODEs, so the Bliss excess is solved, not assumed. The pathway activities $s/(s + 100)$, the effect scalar $k \cdot d/(d + 1) \cdot (1 - r)$, and the Bliss expectation all recompute exactly, and the single-drug ODE matches an independent RK4 integration to 10^{-3} in terminal viability.

(F117) Twin response model. Pathway activity $A = s/(s + 100)$; effect $e = kd/(d + 1)(1 - r)$; ODE $\dot{v} = g - d_{\text{kill}}, \dot{a} = d_{\text{kill}} - 0.02a, \dot{p} = -0.05ep + 0.02A_p(1 - p), \dot{u} = i - 0.03A_{\text{rep}}u - 0.02u$ with $g = 0.04A_p v(1 - v/2), i = 0.08ev, d_{\text{kill}} = i(0.5 + A_a)(1 - 0.5A_s)$; Bliss combination $e_{ab} = 1 - (1 - e_a)(1 - e_b)$.

16.71 Module 91: cellpainter (Cell Painting morphological profiling)

Finding (BUG 63). Mechanism similarity was cosine over raw signature vectors whose entries sit near 1.0 (fold change vs vehicle), so the shared baseline dominated: distinct mechanisms scored about 0.999 (sugarcodes commit e6fb780) and the novel-mechanism hint could never trigger. After the fix, similarity is computed on vehicle-subtracted deltas: the three distinct mechanism pairs score 0.0, the same mechanism at two doses scores 1.0, and the novel-mechanism hint is reachable (benchmarks/sweep_cellpainter.json, 6/6 checks passed). The image path on real BBBC001 stacks is deferred to cellpainter_4d.

(F118) Delta-cosine and signature model. Signature feature $s_j = 1 + (\delta_j - 1) \log_{10}(1 + d) \min(1, h/24)$ for dose d and time h ; similarity $\cos(\mathbf{s}_a - \mathbf{1}, \mathbf{s}_b - \mathbf{1})$; plate normalisation $z = (x - \bar{x}) / (1.4826 \text{ MAD})$; vehicle ($d = 0$) gives $\mathbf{s} = \mathbf{1}$ exactly. Recomputed signature value 1.12 matched the module; QC CV 0.0816 matched.

16.72 Module 92: cellpainter_4d (temporal morphology and real nuclear images)

Finding (BUG 64). The morphology rate constant was set from the observation window ($\tau = T_{\text{obs}}/5$) while cell death used fixed hourly rates, so the same apoptosis run gave 12 h nuclear areas of 71.49, 120.84 and 152.29 μm^2 for 24, 72 and 144 h windows. Kinetics now use a fixed program clock and the 12 h area is 120.84 for every window (benchmarks/sweep_cellpainter_4d.json).

Real images. New segment_nuclei on BBBC001 (6 HT29 Hoechst fields, two human counters): mean count deviation 8.18% (inter-human 11%, CellProfiler 6.2%). Negative result: two earlier drafts scored 7.0% and 5.8% but leaked across the watershed background (synthetic 121-px disks measured 145/109/109, then 81 px); the shipped version is the geometry-correct one and nothing was tuned to the ground truth.

(F119) Program-clock relaxation and segmentation geometry. $a(t) = a_0 t_a + a_0(1 - t_a)e^{-t/\tau}$, $\tau = T_{\text{prog}}/(5s)$; $n(t) = n_0 e^{-0.04t}$ (apoptosis); Otsu $t^* = \arg \max_t [\mu_T \omega(t) - \mu(t)]^2 / [\omega(t)(1 - \omega(t))]$; seed radius $r = \frac{1}{2} \sqrt{A/\pi}$; circularity $4\pi A/P^2$ with $P = \frac{\pi}{4} E$ for E exposed pixel edges; aspect ratio $\sqrt{\lambda_1/\lambda_2}$ of the pixel covariance.

16.73 Module 93: crispr_cargo (delivery vehicle selection and PK)

Finding (BUG 65). pk_model added a zero-width-time trapezoid $(c_0 + c_0)/2 \cdot dt$ at $t = 0$, overstating AUC by exactly $c_0 dt$ at every step size: +5.8% for LNP over 96 h at $dt = 1$ (608.317 vs analytic 574.824). The fixed grid matches the closed form $C_0/k_e(1 - e^{-k_e t})$ to 2.8×10^{-4} relative at $dt = 1$ and exactly at $dt = 0.01$ (benchmarks/sweep_crispr_cargo.json).

(F120) One-compartment PK and payload accounting. $C(t) = (D/V_d)e^{-k_e t}$, $k_e = \ln 2/t_{1/2}$; AUC = $C_0(1 - e^{-k_e T})/k_e$; CDS length = $3N_{aa} + 3$ bp (UniProt Q99ZW2 SpCas9 1368 aa $\rightarrow$ 4107 bp, J7RUA5 SaCas9 1053 aa $\rightarrow$ 3162 bp, fetched live); AAV's 4.7 kb cargo limit leaves under 0.6 kb once the SpCas9 CDS is in. Compartment mass balance error $< 10^{-6}$.

16.74 Module 94: crispr_muse (adaptive guide design)

Finding (BUG 66). enumerate_configurations only searched 3' of the spacer for the PAM, but Cas12a (TTTV) and CasX (TTCN) PAMs are 5' of the spacer, so every valid Cas12a/CasX site was missed (0

configurations on a 5'-TTTA + 23-nt guide construct). A `pam_side` field fixes this; Cas9 enumeration is unchanged (benchmarks/sweep_crispr_muse.json).

(F121) Posterior and policy quantities. Beta posterior $\alpha' = a + s$, $\beta' = b + f$, $\mu = \alpha' / (\alpha' + \beta')$, $\sigma^2 = \alpha' \beta' / [(\alpha' + \beta')^2 (\alpha' + \beta' + 1)]$ (matches Beta(9,3): 0.75, 0.1201); clonal selection $f_i^{t+1} = f_i^t w_i / \sum_j f_j^t w_j$ (b after 10 doublings: 102.4/103.3); policy entropy $-\sum p \ln p$, uniform = $\ln 4$; repair prior $[1.1 - 0.35m, 0.2 + 1.5m, 0.05 + 1.4d]$ with microhomology m , donor d , S-phase s .

16.75 Module 95: ecosystem (plugin, SDK and standards tooling)

Finding (BUG 67). `resolve_plugin_order` keyed dependencies by name, so two different manifests with the same name collapsed into one plugin (`plugin_count = 1`) and the second plugin's capabilities vanished. Duplicate names now raise; self-cycles, two-plugin cycles and missing dependencies are all rejected (benchmarks/sweep_ecosystem.json).

(F122) Plugin graph resolution. Kahn topological order over the dependency digraph $G = (V, E)$ with in-degree $d^-(v) = |\text{deps}(v)|$; acceptance iff $|\text{order}| = |V|$; dependency density $|E| / (n(n-1))$; integration readiness $(c/a + r/n)/2$ with c/a compliant assets and r/n parallel roots. Fifty deterministic diagnostics over the live registry (95 modules, 9 subnetworks).

16.76 Module 96: enterprise_bio (tiered entitlements and governed operations)

Finding. No defect found under adversarial probes: a single-field tamper breaks the audit chain; cross-tenant vault decryption fails (per-tenant Fernet keys); academic blocks module 41/vault/robot while allowing module 40; illegal robot transitions and budget overspend are rejected (benchmarks/sweep_enterprise_bio.json, 6/6 checks, 8 new tests).

(F123) Audit chain and usage forecast. Event digest $h_i = \text{SHA256}(\text{canon}(e_i))$ with e_i carrying `prev = h_{i-1}` ($h_0 = \text{genesis}$); verification replays links and digests. Per-tenant vault key $k_t = \text{HMAC-SHA256}(k_m, \text{sugarcodes-vault})$. Forecast $\hat{h} = h_{\text{used}} \cdot 30/d$; over-quota iff $\hat{h} > q$.

16.77 Module 97: epi_edit (CRISPRa/i epigenome perturbation design)

Findings (BUGs 68 and 69). `expression_trajectory` reported `steady_state = target/decay`, ignoring its own feedback term: it reported 1.577 while the ODE actually converged to 1.385. The fixed point now solves $\delta f y^2 + \delta y - T = 0$ and matches the 72 h trajectory endpoint. Separately, CRISPRa with the TSS too close to the sequence start silently fell back to designing guides $-300..+100$ (inside the gene, the CRISPRi window) while reporting `guide_window [0,0]`; CRISPRa is now strictly upstream-only and reports zero guides honestly (benchmarks/sweep_epi_edit.json).

(F124) Effector response and guide windows. Recruitment $r = o^n / (0.5^n + o^n)$ ($r = 0.5$ at $o = 0.5$); state $s_k = \text{clip}(b_k + r\Delta_k)$; expression $= 0.45a + 0.4ac + 0.15(1-m)$. CRISPRi window $[-50, +300]$ around TSS, CRISPRa $[-400, -50]$ upstream - verified on FN396876 (4318 bp): 5 CRISPRi guides ($0.395 \times$ repression), 5 CRISPRa guides ($4.46 \times$ activation).

(F125) Expression trajectory fixed point. $\dot{y} = T/(1+fy) - \delta y$, $\delta = \ln 2/t_{1/2}$; steady state $y^* = (\sqrt{\delta^2 + 4\delta fT} - \delta)/(2\delta f)$ for $f > 0$, else T/δ .

16.78 Module 98: fate_predictor (cell-fate reprogramming design)

Finding. No code defect; validated against live PubMed: fibroblast→neuron ASCL1/BRN2/MYT1L (Vierbuchen 2010, PMID 20107439), fibroblast→cardiomyocyte GMT (Ieda 2010, PMID 20691899), B-cell→macrophage CEBPA (Xie 2004, PMID 15163413), fibroblast→endothelial ETV2 (Morita 2015, PMID 25540418). The exact-subset optimizer rediscovers the published factor sets. Negative: the hermetic ODE peaks at 5.65% target fraction versus the curated 35% literature rate for the same protocol - documented as a calibration gap, not refitted (benchmarks/sweep_fate_predictor.json).

(F126) TF subset selection and state transitions. Score = $\sum_{g \in T} e_g^+ w_g + \sum_{g \in S} e_g^- w_g - 0.08k^{1.5}$ over all $\binom{n}{k}$ subsets; ODE $\dot{s} = -c$, $\dot{\tau} = c - m - 0.01x\tau$, $\dot{f} = m - 0.005f$, $\dot{x} = b(s + \tau) - 0.02x$ with conversion $c = 0.02 \sigma A s$ and maturation $m = 0.035 \tau e^{-x}$.

16.79 Module 99: gene_explorer (central-dogma trace)

Finding (BUG 70). explore() translated from frame 0 of the raw input, so on a real RefSeq mRNA with a 5' UTR (NM_000518.5 HBB, 628 nt) it reported a 20-aa UTR peptide (2260 Da) as the protein while its own longest_orf knew the true 147-aa hemoglobin. The protein block now annotates the longest ORF (translation_mode longest_orf), falling back to frame translation only when no ORF exists (benchmarks/sweep_gene_explorer.json).

(F127) ORF-anchored translation and variant cascade. ORF translation of NM_000518.5 matches UniProt P68871 exactly (147 aa); sickle-cell E6V (GAG→GTG) gives exactly one protein change (MVHLTPEE→MVHLTP); silent CCT→CTT gives zero. Cascade $\Delta\Delta G = 0.5 + 2c + 0.8i + 0.4n_{aa}$; expression ratio $\max(0.05, 1 - r - 0.1n_{aa})$; export probability $p_{cap}(1 - e^{-L_{polyA}/50})$.

16.80 Module 100: gene_tx_opt (gene-therapy program design)

Finding. No defect found: canonical programs verified (liver AAV8+TBG, CNS AAV9+Syn1, retina AAV2, muscle AAV9); an oversized transgene selects nothing; score and immune formulas recompute exactly; delivery mass declines monotonically over 720 h (benchmarks/sweep_gene_tx_opt.json, 5/5 checks, 8 new tests). Documented negative: the 168 h default window ends before the expression peak (true peak 198 h).

(F128) Vector-promoter scoring and delivery cascade. Expression = $\tau a s_p f_c (1 - p_i)$; score = $0.55E + 0.2\text{spec} + 0.15f_c + 0.1(1 - p_i)$; immune risk $p_i = 1 - (1 - p_{\text{nab}})(1 - p_{\text{route}})$; delivery $\dot{P} = -(k_e + u)P$, $\dot{T} = uP - 0.015T$, $\dot{I} = 0.015T - 0.008I$, $\dot{E} = 0.004I - 0.02E$ with uptake $u = 0.025 a \tau P$.

(F129) Splice-site scoring on the HBB locus. Donor log-odds $s_d = \sum_i \log_2 p_i(b_i)/0.25$ normalized by the consensus maximum (Shapiro-Senapathy MAG|GTRAGT, invariant GT at $p = 1$). On the real NG_000007.3 HBB locus (GenBank CDS join 70595..70686, 70817..71039, 71890..72018; translates to the exact 147-aa

P68871 protein): IVS2 donor scores 0.823 at the exact annotated junction, IVS2 acceptor 0.805, IVS1 acceptor 0.683 within 2 nt. Negative: this RefSeqGene haplotype carries a non-consensus IVS1 donor GTTGGT scoring 0.476, correctly below the 0.8 default threshold. Loop confidence $c = \min(1, 20000/\text{span})$; contact probability $p(d, \text{conv})$ reproduces 0.9103/0.354/0.1619/0.0629 at 1 kb convergent, 1 kb non-convergent, 10 kb convergent, 10 kb non-convergent.

16.81 Module 102: infinite_diagnosis (cross-domain case reasoning)

Findings (BUG 71, BUG 72). BUG 71: the raw case text went to the catalog ranker, so function words drove every angle; the query and alone ranks SynDroid (13.033), Liquid Biopsy (12.427), NeoHunter (11.215), Micro-Tx (10.457), while the content words tall stature lens dislocation match no module. The ranker now receives content words only. BUG 72: no module spec names a disease, symptom or gene, so cystic fibrosis and CFTR matched 0 of 95 catalog documents although RareNet AI holds curated records; case terms are now bridged to those records by whole-phrase match (Marfan via lens dislocation and tall stature, PKU via musty odor and seizures, CF via name or gene). `joint_case_view` validated on live ClinVar: VCV000007105 CFTR p.Phe508del (Pathogenic, practice guideline) is the lead hypothesis at support 2.5; VCV000015333 HBB p.Glu7Val (Pathogenic, 2-star) scores 1.5 but is correctly not a lead because HBB lies outside the CF symptom differential. Negatives: gnomAD skipped for HGVS-only input; the bridge covers only six curated diseases (benchmarks/sweep_infinite_diagnosis.json).

(F130) Catalog ranking and curated bridge. Score $S(d, q) = \sum_{t \in q \setminus W} \text{idf}(t) \text{tf}(t, d) \cdot (1.5 \text{ if } t \in \text{name}(d))$ with $\text{idf}(t) = \ln(1 + (N - n_t + 0.5)/(n_t + 0.5))$, $N = 95$, and W the function-word set; bridge $B(q) = \{D : \text{name}(D) \cup \text{sym}(D) \cup \text{genes}(D) \cap \text{phrases}(q) \neq \emptyset\}$ adds RareNet AI when $B(q) \neq \emptyset$. Variant support = $2.0w_*/0.8$ for a ClinVar pathogenic match (w_* the review-star weight), giving +2.5 for a practice guideline and +1.5 for 2-star.

16.82 Module 103: living_computer (boolean gene-circuit compiler)

Finding (BUG 73). Compiled circuits were not digital at the module's documented input normalization (fully induced = 1.0 REU). A repress edge with Hill coefficient $n = 4$ leaks $1/(1 + 2^4) = 5.9\%$ of $v_{\max}$ at a 1.0 input; with internal protein at $5 \times$ transcription (decay 0.2), one leaked edge leaves ≈ 0.33 REU - comparable to $K = 0.5$ - which re-activates downstream gates and compounds through cascades. Measured at 1.0/0.0 inputs, identical on both simulation engines: XOR(1, 1) = 7.452 (reads ON, should be OFF), XNOR(1, 1) = 0.148, AND-NOT(1, 1) = 1.208 (OFF above the 0.5 threshold), 3-input XOR has 3 wrong rows. The verify path masked it by testing only at HIGH = 3.0 (leak 0.08%). Fix: $n = 8$ (leak $1/(1 + 2^8) = 0.39\%$), one constant, timing untouched. After: 14 expressions $\times$ all input rows $\times$ both engines $\times$ both input conventions = 0 wrong rows; margins at documented levels min-ON 7.55 vs max-OFF 0.079 (benchmarks/sweep_living_computer.json).

(F131) Hill repression leak and cascade compounding. A repress edge passes $\ell(x) = 1/(1 + (x/K)^n)$ of $v_{\max}$; at documented input $x = 1$, $K = 0.5$: $\ell = 1/17$ for $n = 4$ vs $1/257$ for $n = 8$. Internal protein steady state $p = \text{tx}/\delta = 5 \text{tx}$ ($\delta = 0.2$), so a leaked edge deposits $5\ell v_{\max} \approx 0.33$ REU; a downstream gate reads it as $(p/K)^n/(1 + (p/K)^n) \approx 0.16$ each, and an OR of two such leaks re-opens the reporter at 7.45 REU. With $n = 8$: $5\ell v_{\max} \approx 0.02$ REU, downstream activation $(0.02/0.5)^8 \approx 6 \times 10^{-12}$ - the cascade stays shut.

16.83 Module 104: living_tx (engineered-probiotic design)

Finding. No defect; the gut-community ODE is a single-inoculum challenge whose only stable equilibrium has the engineered strain extinct: growth vanishes at the shared carrying capacity while the washout loss $0.08(1 - \text{engraftment})$ persists (per-capita rates -0.008 to -0.056 h^{-1} at the resident equilibrium, substrate 0.0385). All four chassis wash out by day 56 at doses 10^5 - 10^{12} CFU, matching the clinical need for repeated dosing of Nissle-class chassis (PMID 30102294). Two additive honesty fixes shipped: the verdict was window-blind (1-day simulations read “persists”/ready only because decay had not acted; 14 days reads washed_out for the same design), so it now carries the assessment window, the washout timescale, and a short-window caution; model_status now declares the single-inoculum semantics. Literature: PMID 10958782 (IL-10 *L. lactis*) and 30102294 (SYNB1618) match the top payload fits; PMID 25626737 (GLP-1 *L. gasseri*) contradicts the GLP-1 fit (0.65) - calibration gap documented, not refitted (benchmarks/sweep_living_tx.json).

(F132) Single-inoculum extinction and washout timescale. Engineered per-capita rate $g(E) = \frac{1}{2}eS(1 - T/10) - 0.08(1 - e)$ with T the total biomass; at the resident-only equilibrium ($R=10$, $S^* = \frac{0.02}{0.5+0.02} = 0.0385$), $g = -0.08(1 - e) < 0$ for every chassis ($e < 1$), so $E \rightarrow 0$ exponentially with timescale $\tau = 1/[0.08(1 - e)] \text{ h}$ (41.7 h for *Lactobacillus*). The verdict caution fires when the assessment window $w < 3\tau$: a “persists” verdict inside 3τ only means decay has not finished. Combined containment risk is a probability union $1 - \prod_i (1 - r_i)$ over payload, persistence, transfer and shedding components.

16.84 Module 105: micro_tx (microbe-prebiotic therapeutic design)

Findings (BUG 74, BUG 75). BUG 74: SCFA_YIELD credited butyrate to Bifidobacterium (0.10), Akkermansia (0.15) and *L. reuteri* (0.10) - none produce butyrate. Bifidobacterium makes acetate and lactate and reaches butyrate only by cross-feeding partners (PMID 38126785); colonic butyrate producers are specific clostridial clusters (PMIDs 26925050, 19807780), here *F. prausnitzii* (0.75) and Roseburia (0.70), left intact. Deltas: resistant-starch ecology butyrate $0.5425 \rightarrow 0.5141$; Bifidobacterium flux butyrate $0.29 \rightarrow 0$. BUG 75: compartment migration used a periodic (np.roll) Laplacian, so the proximal compartment exchanged directly with the distal one as if the gut were a ring; replaced with a zero-padded chain Laplacian (identical mass conservation, column sums 0); final abundances shift $\leq 2 \times 10^{-5}$ because compartments start identical - a structural fix with near-zero numeric effect. Verified exact: the LP flux equals its analytic optimum $b = Y_i \sum_j d_j A_{ij}$ for all five strains; Michaelis-Menten fermentation conserves mass to 3.6×10^{-15} ; host-outcome formulas reproduce exactly (benchmarks/sweep_micro_tx.json).

(F133) Constraint flux, fermentation and chain migration. Biomass flux $b^* = Y_i \sum_j d_j A_{ij}$ (uptake saturates its enzyme-access bounds; LP and analytic optima agree to 10^{-6}). Fermentation $\dot{S} = -v_{\max} S / (K_m + S)$ with $v_{\max} = 0.12 + 0.18 \bar{A}_j$, products $(0.55, 0.25, 0.20) \dot{S}$ so $S + \sum \text{products}$ is conserved to solver tolerance. Migration is the chain Laplacian $\mu(\uparrow x + \downarrow x - \deg x)$, $\mu = 0.04$, $\deg = (1, 2, 1)$; the periodic form it replaces coupled compartment 1 to compartment 3 directly. Ecology: $\dot{x}_c = x_c \odot (\rho_c r + A x_c)$ with retention $\rho = (0.92, 1, 1.08)$, competition $A_{ii} = -0.85$, $A_{ij} = -0.18$, cross-feeding $A_{21} = 0.22$ (acetate $\rightarrow$ butyrate producers).

16.85 Module 106: microaiverse (cultivation solver)

Findings (BUG 76, BUG 77). BUG 76: predicted_success was unbounded - $0.3 + 0.2$ per supplement + 0.2 for partners reached 1.5 with five gaps, and duplicate gaps stacked identical supplements to inflate it further (five copies of the same B12 gap also scored 1.5 with five duplicate supplements). Gaps are now deduplicated and the score capped at 0.95; one- and two-gap scores are unchanged (0.7, 0.9). BUG 77: optimize_medium crashed inside linprog with a raw solver error for a target with no requirements; it now returns the zero-cost empty medium. Verified exact: the LP medium equals the analytic optimum $x_i = \text{req}_i$; partner coverage = $\sum \min(\text{sec}, \text{req}) / \sum \text{req}$, competition and net score recomputed exactly. Anchors: PMID 12004133 (diffusion-chamber isolation of “uncultivable” microbes), PMID 35536023 (Bacteroides metabolic synergy) (benchmarks/sweep_microaiverse.json).

(F134) Medium LP and cross-feeding score. The medium problem $\min_x c^\top x$ s.t. $x_i \geq \text{req}_i$, $x \geq 0$ has the closed-form optimum $x^* = \text{req}$, cost $c^\top \text{req}$ - the LP reproduces it exactly, and the empty-requirements case is the zero vector at cost 0. Partner score = $\text{cov} - 0.4 \text{comp}$ with $\text{cov} = \sum_m \min(s_m, r_m) / \sum_m r_m$ over secreted versus required metabolites and $\text{comp} = \sum_m \min(u_m^i, u_m^p) / \sum_m u_m^i$ over shared uptake. Success heuristic $\min(0.95, 0.3 + 0.2 n_{\text{supp}} + 0.2 \lceil \text{partners} \rceil)$ - bounded after BUG 76.

16.86 Module 107: microbiome_rx (community-metabolite intervention design)

Finding (BUG 78). optimize_intervention’s input penalty (0.02 per gram) was $\sim 10^3$ times larger than the squared metabolite errors this model produces ($\sim 10^{-5}$ at its natural $\sim 10^{-2}$ output scale), so the penalty always won: every target received fiber=sugar=0 - even reachable ones (butyrate 0.005 $\rightarrow$ diet {0,0}, achieved 0.00201; butyrate 0.5 $\rightarrow$ {0,0} with converged=True and no reported gap). The penalty is now a 10^{-6} /g tie-breaker and the result reports target versus achieved per metabolite. After: butyrate 0.005 $\rightarrow$ fiber 1.011 g, achieved 0.00474 (error 2.6×10^{-4}); butyrate 0.5 $\rightarrow$ fiber 3 g best-effort, achieved 0.00731, error 0.493 visible. Metabolic sim responds to diet (butyrate 0.00212 $\rightarrow$ 0.01133 for fiber 0.2 $\rightarrow$ 2.5); diagnostics exactly 50 for 4/5/6-species communities (benchmarks/sweep_microbiome_rx.json).

(F135) Penalty scaling in intervention optimization. Objective $J(z) = \sum_m (M_m(z) - v_m)^2 + \lambda \sum z$; the coupled ODE yields $M_m = O(10^{-2})$, so errors are $O(10^{-5})$ - 10^{-1} while $\lambda \sum z$ with $\lambda = 0.02$ is $O(10^{-2})$ per gram - the penalty swamps the signal and $\arg \min J = \{0,0\}$ for all v . A tie-breaker needs $\lambda \ll \min_z \|\Delta M(z)\|^2 / \|z\|$, here $\lambda = 10^{-6}$: reachable targets are hit (error 2.6×10^{-4}), unreachable ones return the best-effort boundary $z = 3$ with the gap reported. Coupled dynamics: $\dot{x}_i = x_i(g_i \text{mean}_{m \in C_i} \frac{M_m}{1+M_m} + \sum_j a_{ij} x_j - 0.08 \sum x - \text{abx}_i)$, $\dot{M}_m = 0.02(d_m - M_m) + \sum_i (\frac{0.12}{|P_i|} [m \in P_i] - \frac{0.15}{|C_i|} [m \in C_i]) [\dot{x}_i]^+$.

(F136) Path-count ranking versus embedding noise. The repurposing resilience index $\text{tri} = .45 \min(1, .4 n_{\text{paths}}) + .25 \min(n_{\text{targets}}/3, 1) + .3 d_{\text{target}}$ reproduces the scan exactly: aspirin $\rightarrow$ arthritis gives $.45(.8) + .25(2/3) + .3 = 0.8267$ ($n_{\text{paths}} = 2$, targets COX1/COX2). The graph-convolution score $\text{tri}_{\text{graph}} = .4(\cos + 1)/2 + .35 \text{path_support} + .25 d_{\text{target}}$ with untrained random embeddings $\cos \sim U(-1, 1)$ has $\mathbb{E}[\text{tri}_{\text{graph}} | n_{\text{paths}} = 0] = .4 \cdot .5 + .25d$ - a drug with zero mechanistic paths can outrank one with a path purely by embedding luck, which is BUG 79: filtering $n_{\text{paths}} = 0$ removes 7/16 diseases whose entire candidate lists were such draws (T2D top: thalidomide 0.338, 0 paths).

(F137) Okapi BM25 length normalisation for catalog search. With term frequency $f_{t,d}$, document length $|d|$ and mean length $\overline{|d|}$, the fixed ranker scores $s(q, d) = \sum_{t \in q} \text{idf}_t \frac{f_{t,d}(k_1+1)}{f_{t,d} + k_1(1-b+b|d|/\overline{|d|})}$ with $k_1 = 1.2$, $b = 0.75$ and $\text{idf}_t = \ln(1 + (N - n_t + .5)/(n_t + .5))$. The previous score $\sum_t \text{idf}_t f_{t,d}$ grows linearly in $|d|$, and catalog documents span 11 to 2132 tokens, so a long spec outranks a module queried with its own summary: self-retrieval $R@1$ rose from 67/95 to 93/95 by summary (MRR 0.8224 $\rightarrow$ 0.9895) and from 74/95 to 93/95 by name (BUG 80).

(F138) Order-invariant flux trend. With per-sample flux $\phi_i = n_i^{\text{online}} / \max(1, n_i^{\text{total}})$, the trend is defined on the time-ordered series, $\Delta\phi = \phi_{\arg \max_i t_i} - \phi_{\arg \min_i t_i}$, which is invariant to the order samples arrive in. The previous $\phi_{\text{last}} - \phi_{\text{first}}$ in input order gave +0.0526 for a platform falling from 95/95 at $t = 1$ to 90/95 at $t = 2$ when the samples arrived reversed; the ordered definition gives -0.0526 (BUG 84). Search relevance is now $\sum_{t \in q} \#\{t \in \text{tokens}(d)\}$ over whole tokens; substring counting matched the query a in 95 of 95 registry records (BUG 85).

(F139) Stratified holdout and generalisation gap. For labels y with classes c , the holdout draws round($h n_c$) indices per class ($h = 0.2$), so the E. coli promoter set (UCI 67, 53+53) splits 84/22 with 11 promoters held out. The reported gap is $g = \text{acc}_{\text{train}} - \text{acc}_{\text{holdout}}$. One-hot input $x \in \{0, 1\}^{4L}$ ($L = 57$, 228 features, 84 training sequences) is position-specific and memorises: $g = 1.0 - 0.727 = 0.273$, 10-fold CV 0.765; normalised 3-mer frequencies $x_w = c_w / \sum_{w'} c_{w'}$ (64 features, position-free) give $g = 0.94 - 0.773 = 0.168$, CV 0.842. The previous interface reported only $\text{acc}_{\text{train}}$ (BUG 86); on the default synthetic task that was 0.965 against 0.64 on fresh sequences.

(F140) Exact support-queue selection. Given tickets i with hours h_i , priority p_i and SLA risk r_i , the queue solves $\max_{x \in \{0,1\}^n} \sum_i p_i r_i x_i$ subject to $\sum_i h_i x_i \leq H$ by enumerating all 2^n subsets ($n \leq 22$), breaking ties by $\sum_i x_i$ then ticket ID. The returned set must be $\{i : x_i^* = 1\}$ by index; keying rows by an ID that defaulted to the position within the subset returned the priority-1 ticket for $p = (1, 2, 9)$, $H = 1$ while reporting objective 9, and returned both rows sharing one ID at $\sum h = 2 > H$ (BUG 89).

(F141) Half-viability dose of a scaled Hill curve. For $V(d) = 100/(1 + k(d/b)^n)$ with kill scale k , curve scale b and $n = 1.5$, setting $V = 50$ gives $k(d/b)^n = 1$, so $\text{IC}_{50} = b k^{-1/n}$. Reporting b as the IC_{50} is correct only at $k = 1$; at the curated $k \in [0.6, 0.75]$ it left $V(b) = 100/(1 + k) \in [57.1, 62.5]\%$ and tied all compounds. With the identity, olaparib ($k = .75$) gives $1.211 \mu\text{M}$ and cisplatin ($k = .7$) $1.268 \mu\text{M}$, rising to $2.791 \mu\text{M}$ ($b = 2.2$) in a TP53-mutant organoid (BUG 92).

(F142) Bliss expectation and log-linear IC50 interpolation. Under Bliss independence the expected combination viability is $V_{ab} = V_a V_b$; synergy is the excess $V_{ab} - V_{ab}^{\text{obs}} > 0$ and requires a measured V_{ab}^{obs} . The replaced index $\frac{a}{a+b} + \frac{b}{a+b} - 1 \equiv 0$ scored every pair as maximally synergistic (BUG 95). For doses $c_i < c_{i+1}$ bracketing $V = 0.5$, $\log_{10} \text{IC}_{50} = \log_{10} c_i + \frac{V_i - 0.5}{V_i - V_{i+1}} (\log_{10} c_{i+1} - \log_{10} c_i)$; on the cisplatin curve ($V(1) = 0.588$, $V(3) = 0.216$) this gives $1.297 \mu\text{M}$ against the exact 1.268 , where the nearest-dose rule gave 1.0 (BUG 97).

(F143) Renormalised multi-omic fidelity over measured layers. With layer weights $w = (.35, .3, .2, .15)$ for mutation retention, expression, copy number and methylation and the set M of layers with at least one shared feature, $\text{MFI} = e^{-0.018p \frac{\sum_{\ell \in M} w_{\ell} s_{\ell}}{\sum_{\ell \in M} w_{\ell}}}$. Scoring an unmeasured layer as $s_{\ell} = 0$ caps an identical mutation+expression PDX at $\text{MFI} = (.35 + .3)/1 = 0.65$ and flags two drifts it never measured (BUG 98);

renormalising gives 1.0. Measured components must not be multiplied by passage decay a second time: $0.98^{10} = 0.817$ and $0.95^{10} = 0.599$ were applied to observed retention and correlation of 1.0 (BUG 99).

(F144) Exact cocktail score. For a cocktail C of lytic phages with isolate susceptibilities s_i , escape probability $e = \prod_{i \in C} (1 - s_i)$, receptor diversity $\rho = |R(C)|/|C|$ (R the distinct primary receptors) and size penalty, the score is $f(C) = 1 - 1.4e + 0.25\rho - 0.03|C|$, maximized by enumerating all $\sum_{k=1}^3 \binom{n}{k}$ subsets. Independent brute-force re-derivation matches the module exactly (E. coli: {T4, λ vir}, $f = 1.148$). The simulation must use the selected cocktail latent period $\bar{\tau} = \frac{1}{60|C|} \sum_i \tau_i^{\min}$ hours; using the library-blind default 0.5 h for a T4+ λ vir cocktail ($\bar{\tau} = 0.5417$ h) mistimes lysis (BUG 103).

(F145) Active-site placement as a nearest-neighbour assignment. For coil positions $C = \{i : ss_i = C\}$ and catalytic residues of count n , the site must be the n coils minimizing $|c - \lfloor L/2 \rfloor|$ (ties by index), sorted ascending; residue k then occupies the k -th selected coil. Restricting the search to $c \geq c^*$ (the single nearest coil) fails whenever fewer than n coils follow c^* : a left-heavy run $C = \{0, \dots, 5\}$, $L = 32$ yields only $\{5\}$, truncating the triad HDE to H while the report still claims a designed active site (BUG 104). The placement is now reported with a completeness flag. Molecular weight follows the average-mass convention $M = 18.0153 + \sum_a n_a m_a$, verified: $M_{GGG} = 3(57.0519) + 18.0153 = 189.171$ Da.

(F146) Fidelity of a rendering pass. A renderer is lossless on scalar data iff every leaf ℓ of the input document (nested maps included) appears in the output; capping a table at $r_{\max}$ rows must be reported as $(n, r_{\max})$ when $n > r_{\max}$, never as $r_{\max}$ alone. Measured on a live RB1 splice assessment: 17 scalar leaves, 0 missing after un-escaping; before the fix, nested leaves such as `branchpoint.drop_bits` were dropped and a 120-row table was headed 50 (BUGs 107-108).

(F147) List scheduling on shared instruments. Task t starts at $s_t = \max(\max_{d \in \text{after}(t)} e_d, f_{\text{inst}(t)})$ and ends at $e_t = s_t + \delta_t$, where f_i is the time instrument i becomes free; utilization is $u_i = \sum_{t: \text{inst}(t)=i} \delta_t / \max_t e_t$. Two 5-min pipette tasks plus a dependent 10-min read give makespan 15 and $u_{\text{pip}} = 10/15 = 0.667$. The preconditions - every dependency names an existing task and ids are unique - were unchecked: a missing dependency was reported as a cycle and duplicates scheduled twice (BUG 111). Reagent activity follows $a = 2^{-a_{\text{days}}/t_{1/2}} e^{-0.03h}$, exactly 0.5 at one half-life.

(F148) Two-stage gene expression and the Gillespie algorithm. Constitutive expression with transcription rate k_{tx} , translation k_{tl} , mRNA decay d_m and protein decay d_p has steady-state means $\langle m \rangle = k_{tx}/d_m$ and $\langle p \rangle = \langle m \rangle k_{tl}/d_p$. A Gillespie simulation ($k_{tx} = 2$, $k_{tl} = 5$, $d_m = 0.3$, $d_p = 0.1$, 500 h, seed 1) converges to these: late-trace mean protein 330.7 vs 333.3 (0.8% off) and mRNA 7.08 vs 6.67. The repressilator ODE $\dot{p}_i = \alpha/(1 + p_{i-1}^n) - \beta p_i$ at the module defaults is damped - first-quarter amplitude 1.36 vs last-quarter 0.40 - so transient peaks are not evidence of sustained oscillation; the report now states per-quarter amplitudes and a sustained flag (BUG 114). Motif anchors live-verified via NCBI: PMID 10659857 (Gardner toggle switch) and PMID 10659856 (Elowitz-Leibler repressilator).

(F149) Hill-regulated gene circuits. A gate repressed by x produces $g_r = 1/(1 + (x/K)^n)$, activated $g_a = (x/K)^n/(1 + (x/K)^n)$; AND combines by product, OR by $1 - \prod_i (1 - t_i)$, and the steady state is $y^* = (b + v_{\max} g)/d$. The module AND gate gives $y^* = (0.01 + (36/37)^2)/0.2 = 4.783$, matching simulation. For the symmetric three-gene repressilator, sustained oscillation requires loop cooperativity $n > 2$ with sufficient gain: the shipped $n = 2.1$ converged to the symmetric fixed point (last-quarter amplitude

0.002 from a broken-symmetry start) while the recalibrated $n = 3.0$ sustains the limit cycle (amplitude 2.77), as in the Elowitz-Leibler construction (PMID 10659856). Symmetric initial conditions stay symmetric by construction, so the limit cycle is observed only from asymmetric starts.

(F150) Competitive inhibition and the serial-pathway linear program. With a competitive inhibitor, $v = v_{\max}S/(K_m(1 + I/K_i) + S)$; occupancy under competition is $(L/K_d)/(1 + L/K_d + C/K_c)$. The pathway LP maximizes terminal flux x_n subject to serial constraints $x_{i+1} \leq x_i$, uptake $x_1 \leq 10$, ATP $\sum_i x_i \leq 20$ and redox $0.5 \sum_i (i+1)x_i \leq 12$; the solved fluxes satisfy all four (committed in benchmarks/sweep_synbio_wizard.json). The Gillespie SSA carries a hard event cap; when it binds, the trajectory stops early - at transcription 10, translation 1000 the cap binds at $t = 10.69$ of a requested 24 h - so the result now reports `truncated` and `simulated_hours` instead of silently labeling the partial trajectory with the full horizon (BUG 119).

(F151) Minimal-genome design as mixed-integer linear programming. Minimize $\sum_g c_g x_g$, $x_g \in \{0, 1\}$, subject to process coverage $\sum_{g:\text{proc}(g)=p} x_g \geq 1$ for each required process p , essentiality $x_e = 1$, and ATP-synthase coupling $x_{\text{atpA}} = x_{\text{atpD}}$; rich medium rescues glyA, so on minimal medium $x_{\text{glyA}} = 1$ is added, growing the optimum from 12 to 13 genes (HiGHS, committed in benchmarks/sweep_synbio_wizard.json). The coupled cell ODE allocates ATP as $8\text{met} - 1.8\text{tx} - 3\text{tl} - 2\text{rep} - \text{burden} - 0.15\text{ATP}$ with Hill saturables $\text{ATP}/(1 + \text{ATP})$ gating transcription and translation. The catalog follows the JCVI-syn3.0 minimal cell (PMID 27013737, live-verified). The SSA behind expression fired 49,761 events in a 50-gene 240-minute run - a count previously misreported as 1000 by the storage cap (BUG 122).

(F152) Burden-selection escape threshold for engineered pathways. With per-gene burden b , yield selection s and n pathway genes, fitness is $W = 1 - bE + s \min(E/n, 1.5)$ for total expression E ; losing one gene changes fitness by $\Delta W = b - s/n$, so loss-of-function sweeps when $b > b^* = s/n$. The threshold predicts the module outcomes exactly: $b = 0.25 > 0.5/3$ silences the pathway (knockout fraction 0.864 by generation 150, seed 1), $b = 0.08 < 0.167$ remains stable, and the balanced design $b = 0.02$, $s = 0.6$ survives. Endpoint honesty decides the verdict: the same run judged at its last sampled generation (100) read knockout fraction 0.084 and was reported as genetically stable (BUG 124). Uncovered environment generations no longer inherit the last-listed phase silently, and censored failure diagnostics report null rather than the final sampled generation (BUGs 127-128). Live anchors: PMIDs 37260076 and 36357387 on evolutionary stabilization of engineered circuits (NCBI).

(F153) Multi-assay essentiality posterior and the deletion-risk keep rule. Gene essentiality pools four assays with weights summing to one, $E = 0.30C + 0.25T + 0.20S + 0.25M$, missing assays defaulting to the category prior; the optimizer keeps genes with $E \geq 1 - r_{\max}$ and adds the top candidates per functional category by $(-E, \text{size})$. On the seven-gene fixture the posteriors compute exactly (rep 0.88, tx 0.825, tl 0.9125, stress 0.17, stress_backup 0.155, mot 0.05, mot_backup 0.038), keeping {rep, tx, tl, stress, mot} and dropping the two backups (committed in benchmarks/sweep_synthetic_life.json). The LP behind flux optimization is infeasible under contradictory demands (maximize uptake with growth floor 10^9) and now reports `objective=None` rather than a feasible-looking 0 (BUG 132); limits name their reactions instead of being silently ignored (BUG 133). Method anchor: Orth, Thiele and Palsson, PMID 20212490 (live-verified); genome-reduced chassis anchor PMID 28893188.

(F154) Diffusion-limited vascular spacing in printed tissue. With channels spaced L apart, the midpoint oxygen fraction is $\max(0, 1 - (L/400\mu\text{m})^2)$, so the largest non-hypoxic spacing satisfies $L \leq 400\sqrt{0.95} =$

389.9 um; the module now recommends the floored 389.8 um, at which its own model gives 5.03% mid-point oxygen, instead of the previous 400 um at which the same model computes exactly 0% (BUG 136, committed in benchmarks/sweep_tissue_eng.json). Cyclic fatigue retention follows $R = \exp(-N\epsilon/50000)$ with integrity requiring $R \geq 0.8$ and strain below the failure estimate; print fidelity is clamped to $[0, 0.99]$ after large nozzles produced negative values (BUG 138). Anchors: perfusable skin-on-chip for drug testing, PMID 40600176, and oxygen-generating scaffolds for thick tissues, PMID 38282554 (both live-verified).

(F155) Whole-cell coupling: Boolean regulation, Hill expression and parsimonious flux. The virtual cell fixes a regulatory steady state synchronously, transcribes through Hill terms $x^n/(1+x^n)$ with $x = P/K$, integrates $\dot{m} = t_x - \delta_m m$, $\dot{p} = k_{tl}m - \delta_p p$ (ODE identity $p^* = k_{tl}(b+v)/(\delta_m\delta_p) = 105$ verified to 0.1%), caps enzyme reactions at $k_{cat}P$, and solves pFBA $\min \sum |v_i|$ s.t. $Sv = 0$ with the objective pinned at its optimum. Honesty repairs: solver-infeasible conditions now report infeasible rather than the biological call no growth (BUG 140), the pFBA fraction is validated in $(0, 1]$ with an honestly labeled fallback (BUG 142), and Boolean initial states are checked (BUG 143). Anchors: whole-cell modeling principles, PMIDs 26115539 and 29275251 (live-verified).

17 Adding evolutionary conservation to the splice-region model

The best splice-region model so far is a logistic regression on PWM and MaxEntScan features. Neither feature sees conservation. We fetched phyloP100way at each variant position from the UCSC Genome Browser REST API for a stratified subsample of 1,501 of the 9,235 modelled variants (379 pathogenic, 1,054 genes, 0 missing). The subsample keeps the class-by-site proportions of the full set; the API rate limit (one request per second) is why we did not fetch all 9,235. Median phyloP is 3.214 at pathogenic and 0.191 at benign positions. On its own, phyloP reaches AUROC 0.8377 (0.9108 at donors, 0.6966 at acceptors). Added to logit+MaxEnt under gene-grouped 5-fold cross-validation repeated over 20 seeds, AUROC rises from 0.963 to 0.9678. A paired bootstrap on the averaged out-of-fold predictions gives a 95% interval for the gain of $[+0.0008, +0.0091]$, which excludes zero. The gain is concentrated at donor-region positions

site	<i>n</i>	phyloP alone	logit+MaxEnt	+ phyloP
donor	876	0.9108	0.9557	0.9634
acceptor	625	0.6966	0.9702	0.9723
all	1501	0.8377	0.9630	0.9678

Table 20: AUROC with and without phyloP (discovery/splice_region_vus/phylop_lift.json). Per-site values use the seed-averaged predictions.

(+3 to +6), where the intronic consensus is short and conservation carries information the PWM misses. At acceptors (the polypyrimidine tract, -3 to -14), MaxEntScan already scores well and conservation adds little.

Caveat. ClinVar submitters may cite conservation as supporting evidence (ACMG criteria PP3/BP4). Some of the gain may therefore be label circularity rather than new biology. We report the gain as modest and conditional on that caveat.

(F47) Model with conservation. $\text{logit}P(\text{path} \mid x) = \beta_0 + \beta^\top \mathbf{z}(x) + \gamma\phi(x) + \delta\phi(x)\mathbb{I}[\text{donor}]$, with $\mathbf{z}$ the standardised PWM/MaxEnt/position features and ϕ the phyloP score; the interaction lets conservation matter differently at donors.

(F48) Paired bootstrap. For resamples $b = 1..B$ of variant indices I_b , $\Delta_b = \text{AUC}(y_{I_b}, \hat{p}_{I_b}^+) - \text{AUC}(y_{I_b}, \hat{p}_{I_b}^0)$; the interval is the 2.5th and 97.5th percentile of $\{\Delta_b\}$. Pairing on the same resample removes the shared sampling noise of the two models.

18 Discussion

Validating each module against an independent reference exposed three kinds of result: implementations that match their reference exactly (CAI vs Biopython, CYP2D6 translation vs CPIC after the fix), implementations whose defaults were stale or weak (CPIC activity values, genome-wide CAI reference, Nussinov folding), and places where a strong published tool beats our model (SpliceAI on splice-region variants). Each fix is committed with the result file that motivated it. The CNN is useful as a calibrated, cheap second opinion at high thresholds, not as a replacement for SpliceAI.

19 Conclusion

The sweep so far corrected two clinically relevant defaults (CPIC activity values, clopidogrel handling of “Likely” phenotypes), improved CAI correlation with measured abundance, and replaced base-pair maximisation with thermodynamic folding. The accession-level dataset target (120) and the formula target (10) are met; the external-tool target (40) is met (manifests/tools.tsv). The manifests in the appendix list exactly what was used.

20 Module inventory and validation coverage

sugarcode-ai contains 95 application modules and 32 bio/ libraries. 83 of these 127 have been checked against an independent reference on real data in this work (51 modules, 32 libraries). The rest are not yet validated. That does not mean they are wrong, only that no claim about them is made here. Validation status is a curated list tied to committed result files (manifests/module_inventory.tsv, built by sweep/module_inventory.py), not a keyword match.

kind	name	py lines	status
bio	align	164	validated
bio	bed	180	validated
bio	chembl	184	validated
bio	codon	170	validated
bio	de	218	validated
bio	entrez	196	validated
bio	fasta	55	validated
bio	fastq	183	validated
bio	genbank	103	validated
bio	gff	237	validated
bio	gnomad	122	validated
bio	gstats	194	validated
bio	kmer	261	validated
bio	motif	198	validated
bio	newick	248	validated
bio	orf	189	validated
bio	pdb	346	validated
bio	phylo	232	validated
bio	pileup	148	validated
bio	primer	278	validated
bio	proteinprops	261	validated

kind	name	py lines	status
bio	pwm	63	validated
bio	reactome	63	validated
bio	restriction	150	validated
bio	rnaseq	201	validated
bio	sam	221	validated
bio	sequence	174	validated
bio	splice	682	validated
bio	stockholm	260	validated
bio	structures	267	validated
bio	uniprot	106	validated
bio	vcf	263	validated
module	acmg_bayesian	469	validated
module	alpha_fold_ui	442	validated
module	bio_material	116	validated
module	bio_switch	511	validated
module	biofactory_1_a	105	validated
module	biogpt_lit	258	validated
module	car_t_designer	104	validated
module	cell_free_opt	109	validated
module	cellfatenet	207	validated
module	cf_d_offtarget	157	validated
module	chem_descriptors	923	validated
module	chem_similarity	234	validated
module	chemgpt_engine	305	validated
module	codon_opt	289	validated
module	crispr_opt	648	validated
module	crisprater	97	validated
module	crisprscan_score	178	validated
module	dark_genome	207	validated
module	deepsplice	751	validated
module	dna_to_code	132	validated
module	docking_studio	535	validated
module	dti_bench	201	validated
module	evidence_mining	418	validated
module	evofold_4d	239	validated
module	gene_analysis	278	validated
module	liquid_biopsy	411	validated
module	metabodesigner	190	validated
module	microbiome_exp	120	validated
module	mit_offtarget	154	validated
module	molecule_eval	149	validated
module	mutdock	337	validated
module	neohunter	358	validated
module	neuroplan_ai	387	validated
module	omega_stats	161	validated
module	oncocircuit	124	validated
module	openclinvar	514	validated
module	pgx_guidelines	287	validated
module	phage_designer	82	validated
module	prime_design	263	validated
module	profile_hmm	747	validated
module	promoter_lib	143	validated
module	qsar_bench	389	validated
module	rarenet_ai	366	validated
module	riboswitch	142	validated
module	rna_decoder	144	validated
module	rna_nussinov	387	validated
module	stability_ai	199	validated
module	str_scope	340	validated
module	structural_biophysics	528	validated
module	syn_stab_ai	77	validated
module	vector_opt	100	validated
module	bio_copilot	245	not yet
module	bioimage_ai	146	not yet
module	bioplayground	98	not yet
module	bioprint_pro	143	not yet
module	biosimvr	200	not yet

kind	name	py lines	status
module	cell_twin	170	not yet
module	cellpainter	289	not yet
module	cellpainter_4d	164	not yet
module	crispr_cargo	156	not yet
module	crispr_muse	278	not yet
module	ecosystem	175	not yet
module	enterprise_bio	260	not yet
module	epi_edit	178	not yet
module	fate_predictor	131	not yet
module	gene_explorer	140	not yet
module	gene_tx_opt	149	not yet
module	genomegpt	279	not yet
module	infinite_diagnosis	130	not yet
module	living_computer	857	not yet
module	living_tx	184	not yet
module	micro_tx	218	not yet
module	microaiverse	138	not yet
module	microbiome_rx	172	not yet
module	neodti_engine	257	not yet
module	neuro_hub	44	not yet
module	neuro_hub_dashboard	25	not yet
module	neuro_pipeline	326	not yet
module	nexus_support	299	not yet
module	organoid_ai	216	not yet
module	organoid_screen	275	not yet
module	pdx_insight	138	not yet
module	phage_tx	161	not yet
module	phageforge	87	not yet
module	proteinPainter	161	not yet
module	report_studio	213	not yet
module	robotic_flow	128	not yet
module	syn_bio_studio	43	not yet
module	synbio_studio	113	not yet
module	synbio_wizard	224	not yet
module	syndroid	231	not yet
module	synlife_evo	278	not yet
module	synthetic_life	197	not yet
module	tissue_eng	193	not yet
module	virtual_cell	504	not yet

A Tools Table

Tool	Version	Status/used in
pyhmmer	0.12.3	installed
ViennaRNA	2.7.2	installed
RDKit	2026.3.6	installed
SpliceAI	1.3.1	installed
TensorFlow	2.15.1	installed
Keras	2.15.0	installed
scikit-learn	1.7.2	installed
SciPy	1.15.3	installed
NumPy	1.26.4	installed
Biopython	1.88	installed
twobitreader	4.0.2	installed
AutoDock Vina (python)	1.2.7	installed
Open Babel	3.1.1.23	installed
pytest	9.1.1	installed
pandas	2.3.3	installed
MHCflurry	2.2.1	sweep/neohunter_mhcflurry
CRISPOR reference scorers (vendored)	web service/vendored	sweep/_crispor_*_ref.py
EBI Clustal Omega REST	web service/vendored	data/README.md
CPIC API	web service/vendored	sweep/pgx_cpic_bench.py
NCBI E-utilities	web service/vendored	src/sc_validate
LOVD3 API	web service/vendored	discovery/splice_region_vus
MaxEntScan (maxentpy)	0.0.4	discovery/splice_region_vus
codon-bias	0.5.0	sweep/codon_tai_enc.py
statsmodels	0.15.0	benchmarks/sweep_codon_ (BH-FDR)
PyTorch	2.14.0+cpu	discovery/splice_region_vus (splice CNN)
Matplotlib	3.10.9	paper/make_figs.py (all figs)
FreeSASA	2.x (python)	sweep/sasa_vs_freesasa.py
primer3-py	2.3.1	marks/sweep_sasa_freesasa
DendroPy	5.1.0	sweep/primer_vs_primer3.p
orfipy	0.0.4	marks/sweep_primer_prime
pysam (htslib)	0.24.1	bio.primer.hairpin_dg
PyDESeq2	0.5.2	sweep/phylo_vs_dendropy.p
parasail	2.6.1	marks/sweep_phylo_dendro
g:Profiler g:GOST API	e114_eg62_p19_374p083	sweep/orf_vs_orfipy.py
STRING API	12.0	marks/sweep_orf_orfipy.js
Reactome AnalysisService	live 2026-09-24	sweep/vcf_vs_pysam.py
networkx	local 3.4.2	marks/sweep_vcf_pysam.js
gseapy	local 1.3.1	of ClinVar VCF)

B Dataset Manifest

#	Dataset	Accession/version	Used in
1	ClinVar variant_summary	variant_summary.txt.gz (GRCh38)	data/clinvar_sample_500.tsv; discovery/splice_region_vus/
2	UCSC hg38 reference genome	hg38.2bit	discovery/splice_region_vus/extract.py
3	LOVD3 shared variant records	top-20 VUS query	discovery/splice_region_vus/lovd_top20.json
4	SKEMPI 2.0	skempi_v2.csv (PMID 30598530)	data/skempi_v2.csv
5	BiGG e.coli_core	e.coli_core (PMID 21988831)	data/e.coli_core.json
6	NCBI RefSeqGene BRCA1	NG_005905.2:60000-70000	data/brca1_10kb.fa; data/crispr_fresh_pairs.tsv
7	CRISPOR reference CFD/MIT matrices	crisporWebsite (PMID 30341956)	data/crispr_cfd_oracle.tsv; sweep/_crispor_mit_ref.py
8	CRISPOR Rule Set 1 reference	crisporEffScores.py	data/doench_rs1_oracle.tsv
9	UniProtKB reviewed globins	25 sequences	data/globins25.fa
10	UniProtKB reviewed non-globins	25 sequences	data/nonglobins25.fa
11	Pfam Globin HMM	PF00042.29	data/pfam_PF00042_globin.hmm
12	Rfam tRNA seed	RF00005	data/rfam/RF00005.sto
13	Rfam 5S rRNA seed	RF00001	data/rfam/RF00001.sto
14	Rfam RNase P RNA seed	RF00010	data/rfam/RF00010.sto
15	ChEMBL approved drugs	1000-compound slice	data/chembl_approved_1000.tsv
16	Kazusa CUTG E. coli K12	taxid 83333	data/README.md (sanity only)
17	NCBI RefSeq E. coli K-12 MG1655 CDS	GCF_000005845.2	sweep/codon_cai_bench.py
18	PaxDb E. coli integrated abundance	511145-WHOLE_ORGANISM-integrated	sweep/codon_cai_bench.py
19	CPIC CYP2C19 diplotype table	666 diplotypes	sweep/pgx_cpic_bench.py
20	CPIC CYP2D6 diplotype table	16836 diplotypes	sweep/pgx_cpic_bench.py
21	CPIC CYP2D6 allele activity table	208 alleles	sugarcode-ai pgx_guidelines/data
22	Edinburgh Genome Foundry codon tables	e.coli_316407	sweep/codon_cai_bench.py (via sugarcode-ai)
23	IEDB MHC-I binding data 2013 (Kim et al. 2014)	bdata.20130222.mhci.txt	sweep/neohunter_iedb_bench.py
24	LP-PDBBind (Li et al. leak-proof split)	LP.PDBBind.csv	sweep/docking_bench.py
25	NCBI RefSeq transcripts (TP53, BRCA1, BRCA2, BRAF, EGFR)	NM_000546.6, NM_007294.4, NM_000059.4, NM_004333.6, NM_005228.5	sweep/primer_vs_primer3.py
26	ClinVar GRCh38 VCF (tabix regions TP53, BRCA1, BRCA2, BRAF, EGFR)	clinvar.vcf.gz (46,603 records)	sweep/vcf_vs_pysam.py
27	GEO GSE52778 airway smooth muscle RNA-seq (Himes et al. 2014), NCBI raw counts	GSE52778_raw_counts_GRCh38.pls	NGM12swgpydeseq2.py
	Platinum flat file (experimental ligand-binding ddG)	1008 mutation records	data/platinum/platinum_flat_file.csv

C Accession-Level Dataset Manifest

Each row is one identifier-backed record fetched individually and used in a committed analysis (counting rule in manifests/README.md).

#	Kind	Study	Accession	Used in
1	study	1	variant_summary.txt.gz (GRCh38)	data/clinvar_sample_500.tsv; discovery/splice_region_vus/
2	study	2	hg38.2bit	discovery/splice_region_vus/extract.py
3	study	3	top-20 VUS query	discovery/splice_region_vus/lovd_top20.json
4	study	4	skempi_v2.csv (PMID 30598530)	data/skempi_v2.csv
5	study	5	e.coli_core (PMID 21988831)	data/e.coli_core.json
6	study	6	NG_005905.2:60000-70000	data/brca1_10kb.fa; data/crispr_fresh_pairs.tsv

#	Kind	Study	Accession	Used in
7	study	7	crisporWebsite (PMID 30341956)	data/crisp
8	study	8	crisporEffScores.py	sweep/.d
9	study	9	25 sequences	data/doi
10	study	10	25 sequences	data/glob
11	study	11	PF00042.29	data/non
12	study	12	RF00005	data/pfar
13	study	13	RF00001	data/rfar
14	study	14	RF00010	data/rfar
15	study	15	1000-compound slice	data/che
16	study	16	taxid 83333	data/RE
17	study	17	GCF_000005845.2	sweep/c
18	study	18	511145- WHOLE_ORGANISM- integrated	sweep/c
19	study	19	666 diplotypes	sweep/p
20	study	20	16836 diplotypes	sweep/p
21	study	21	208 alleles	sugarco
22	study	22	e_coli_316407	sweep/c
23	study	23	bdata.20130222.mhci.txt	sugarco
24	study	24	LP_PDBBind.csv	sweep/n
25	pdb_entry		1CNW	sweep/d
26	pdb_entry		1EJN	sweep/d
27	pdb_entry		1F5L	sweep/d
28	pdb_entry		1G54	sweep/d
29	pdb_entry		1GI1	sweep/d
30	pdb_entry		1K1J	sweep/d
31	pdb_entry		1O3L	sweep/d
32	pdb_entry		1P1O	sweep/d
33	pdb_entry		1Y3X	sweep/d
34	pdb_entry		1ZGI	sweep/d
35	pdb_entry		2ERZ	sweep/d
36	pdb_entry		2J4I	sweep/d
37	pdb_entry		2P4J	sweep/d
38	pdb_entry		2X7T	sweep/d
39	pdb_entry		2XAB	sweep/d
40	pdb_entry		2XJG	sweep/d
41	pdb_entry		2ZDN	sweep/d
42	pdb_entry		3B3C	sweep/d
43	pdb_entry		3C84	sweep/d
44	pdb_entry		3D8W	sweep/d
45	pdb_entry		3HKQ	sweep/d
46	pdb_entry		3HLL	sweep/d
47	pdb_entry		3IBI	sweep/d
48	pdb_entry		3M35	sweep/d
49	pdb_entry		3PO6	sweep/d
50	pdb_entry		3R17	sweep/d
51	pdb_entry		3R4N	sweep/d
52	pdb_entry		4B2I	sweep/d
53	pdb_entry		4DB7	sweep/d
54	pdb_entry		4F5Y	sweep/d
55	pdb_entry		4FL1	sweep/d
56	pdb_entry		4J93	sweep/d
57	pdb_entry		4JFM	sweep/d
58	pdb_entry		4KNJ	sweep/d
59	pdb_entry		4LXZ	sweep/d
60	pdb_entry		4MO8	sweep/d
61	pdb_entry		4NH7	sweep/d
62	pdb_entry		4O04	sweep/d
63	pdb_entry		4OAG	sweep/d
64	pdb_entry		4R76	sweep/d
65	pdb_entry		4RFC	sweep/d
66	pdb_entry		4TIM	sweep/d
67	pdb_entry		4XIT	sweep/d
68	pdb_entry		4YTC	sweep/d
69	pdb_entry		4YXI	sweep/d

#	Kind	Study	Accession	Used in
70	pdb_entry		4ZME	sweep/d
71	pdb_entry		5AML	sweep/d
72	pdb_entry		5CAP	sweep/d
73	pdb_entry		5CAU	sweep/d
74	pdb_entry		5D3L	sweep/d
75	pdb_entry		5EWK	sweep/d
76	pdb_entry		5EWY	sweep/d
77	pdb_entry		5F60	sweep/d
78	pdb_entry		5F62	sweep/d
79	pdb_entry		5G5V	sweep/d
80	pdb_entry		5HTZ	sweep/d
81	pdb_entry		5IX0	sweep/d
82	pdb_entry		5L8C	sweep/d
83	pdb_entry		5MJN	sweep/d
84	pdb_entry		5MPN	sweep/d
85	pdb_entry		5N24	sweep/d
86	pdb_entry		5O87	sweep/d
87	pdb_entry		5SZ4	sweep/d
88	pdb_entry		5SZ5	sweep/d
89	pdb_entry		5Y8Y	sweep/d
90	pdb_entry		6EQ5	sweep/d
91	pdb_entry		6G38	sweep/d
92	pdb_entry		6G6T	sweep/d
93	pdb_entry		6HQY	sweep/d
94	pdb_entry		6HRQ	sweep/d
95	pdb_entry		6OE0	sweep/d
96	pdb_entry		6RFN	sweep/d
97	pdb_entry		6U6W	sweep/d
98	species		10090-ASTROCYTE- integrated.txt	sweep/c
99	species		GCF_000001635.27	sweep/c
100	species		10116- CARDIAC_VENTRICLE- integrated.txt	sweep/c
101	species		GCF_036323735.1	sweep/c
102	species		121845- WHOLE_ORGANISM- integrated.txt	sweep/c
103	species		GCF_000475195.1	sweep/c
104	species		1295- WHOLE_ORGANISM- integrated.txt	sweep/c
105	species		GCF_025558325.1	sweep/c
106	species		1314- WHOLE_ORGANISM- integrated.txt	sweep/c
107	species		GCF_900475035.1	sweep/c
108	species		185431- WHOLE_ORGANISM- integrated.txt	sweep/c
109	species		GCF_000002445.2	sweep/c
110	species		192222- WHOLE_ORGANISM- integrated.txt	sweep/c
111	species		GCF_000009085.1	sweep/c
112	species		196627- WHOLE_ORGANISM- integrated.txt	sweep/c
113	species		GCF_000011325.1	sweep/c
114	species		198214- WHOLE_ORGANISM- integrated.txt	sweep/c
115	species		GCF_000006925.2	sweep/c
116	species		208964- WHOLE_ORGANISM- integrated.txt	sweep/c
117	species		GCF_000006765.1	sweep/c

#	Kind	Study	Accession	Used in
118	species		224308- WHOLE_ORGANISM- integrated.txt	sweep/c
119	species		GCF_000009045.1	sweep/c
120	species		224911- WHOLE_ORGANISM- integrated.txt	sweep/c
121	species		GCF_001642675.1	sweep/c
122	species		243159- WHOLE_ORGANISM- integrated.txt	sweep/c
123	species		GCF_000021485.1	sweep/c
124	species		243230- WHOLE_ORGANISM- integrated.txt	sweep/c
125	species		GCF_020546685.1	sweep/c
126	species		246200- WHOLE_ORGANISM- integrated.txt	sweep/c
127	species		GCF_000011965.2	sweep/c
128	species		2903- WHOLE_ORGANISM- integrated.txt	sweep/c
129	species		GCF_000372725.1	sweep/c
130	species		343509- WHOLE_ORGANISM- integrated.txt	sweep/c
131	species		GCF_000010085.1	sweep/c
132	species		36329- WHOLE_ORGANISM- integrated.txt	sweep/c
133	species		GCF_000002765.6	sweep/c
134	species		403677- WHOLE_ORGANISM- integrated.txt	sweep/c
135	species		GCF_000142945.1	sweep/c
136	species		4081-PERICARP- integrated.txt	sweep/c
137	species		GCF_036512215.1	sweep/c
138	species		4097-LEAF-integrated.txt	sweep/c
139	species		GCF_000715075.1	sweep/c
140	species		418985- WHOLE_ORGANISM- integrated.txt	sweep/c
141	species		GCA_002081605.1	sweep/c
142	species		44689- WHOLE_ORGANISM- integrated.txt	sweep/c
143	species		GCF_000004695.1	sweep/c
144	species		4932- WHOLE_ORGANISM- integrated.txt	sweep/c
145	species		GCF_000146045.2	sweep/c
146	species		54291- WHOLE_ORGANISM- integrated.txt	sweep/c
147	species		GCF_030505655.1	sweep/c
148	species		6239- WHOLE_ORGANISM- integrated.txt	sweep/c
149	species		GCF_000002985.6	sweep/c
150	species		7227-SPERM-integrated.txt	sweep/c
151	species		GCF_000001215.4	sweep/c
152	species		7460-ABDOMEN- integrated.txt	sweep/c
153	species		GCF_003254395.2	sweep/c
154	species		8030-LIVER-integrated.txt	sweep/c

#	Kind	Study	Accession	Used in
155	species		GCF_965601325.2	sweep/c
156	species		83332- WHOLE_ORGANISM- integrated.txt	sweep/c
157	species		GCF_000195955.2	sweep/c
158	species		9031-HEART-integrated.txt	sweep/c
159	species		GCF_016699485.2	sweep/c
160	species		93061- WHOLE_ORGANISM- integrated.txt	sweep/c
161	species		GCF_000013425.1	sweep/c
162	species		9544-BILE_DUCT- integrated.txt	sweep/c
163	species		GCF_049350105.2	sweep/c
164	species		9606-ADRENAL_GLAND- integrated.txt	sweep/c
165	species		GCF_000001405.40	sweep/c
166	species		9615-PLASMA- integrated.txt	sweep/c
167	species		GCF_011100685.1	sweep/c
168	species		9823-LIVER-integrated.txt	sweep/c
169	species		GCF_054392235.1	sweep/c
170	species		9913- LENS_OF_CAMERA_TYPE_EYE- integrated.txt	sweep/c
171	species		GCF_002263795.3	sweep/c
172	species		99287- WHOLE_ORGANISM- integrated.txt	sweep/c
173	species		GCF_000006945.2	sweep/c
174	species		9986- PERIPHERAL_REGION_OF_RETINA- integrated.txt	sweep/c
175	species		GCF_964237555.1	sweep/c
176	gtrnadb		Esch.coli_K_12.MG1655	sweep/c
177	gtrnadb		Baci.subt_subtilis_168	sweep/c
178	gtrnadb		Scere3	sweep/c
179	refseq_transcript	25	NM_000546.6	sweep/p
180	refseq_transcript	25	NM_007294.4	sweep/p
181	refseq_transcript	25	NM_000059.4	sweep/p
182	refseq_transcript	25	NM_004333.6	sweep/p
183	refseq_transcript	25	NM_005228.5	sweep/p
184	study	26	clinvar.vcf.gz	sweep/v
185	study	27	GSE52778	sweep/d
186	rfam_family	RF00004	RF00004	sweep/st
187	rfam_family	RF00015	RF00015	sweep/st
188	rfam_family	RF00017	RF00017	sweep/st
189	rfam_family	RF00023	RF00023	sweep/st
190	rfam_family	RF00050	RF00050	sweep/st
191	rfam_family	RF00059	RF00059	sweep/st
192	rfam_family	RF00162	RF00162	sweep/st
193	rfam_family	RF00167	RF00167	sweep/st
194	rfam_family	RF00168	RF00168	sweep/st
195	rfam_family	RF00174	RF00174	sweep/st
196	rfam_family	RF00234	RF00234	sweep/st
197	rfam_family	RF00504	RF00504	sweep/st
198	refseq_genome	J01749.1	J01749.1	sweep/p
199	refseq_genome	NC_001416.1	NC_001416.1	sweep/p
200	opentree_node	ott1062253	ott1062253	sweep/n
201	opentree_node	ott392223	ott392223	sweep/n
202	opentree_node	ott44565	ott44565	sweep/n
203	opentree_node	ott691846	ott691846	sweep/n
204	opentree_node	ott81461	ott81461	sweep/n
205	opentree_node	ott913935	ott913935	sweep/n
206	study	NA12878.chrom20_lowcov	NA12878.chrom20.ILLUMINA	sweep/f
207	study	SRR622461	SRR622461	sweep/f
208	jaspar_matrix	MA0003.4	MA0003.4	sweep/n

#	Kind	Study	Accession	Used in
209	jaspar_matrix	MA0079.3	MA0079.3	sweep/m
210	jaspar_matrix	MA0080.4	MA0080.4	sweep/m
211	jaspar_matrix	MA0099.3	MA0099.3	sweep/m
212	jaspar_matrix	MA0105.4	MA0105.4	sweep/m
213	jaspar_matrix	MA0106.3	MA0106.3	sweep/m
214	jaspar_matrix	MA0108.2	MA0108.2	sweep/m
215	jaspar_matrix	MA0112.3	MA0112.3	sweep/m
216	jaspar_matrix	MA0113.3	MA0113.3	sweep/m
217	jaspar_matrix	MA0139.1	MA0139.1	sweep/m
218	nucore_record	NC.000913.3	NC.000913.3	sweep/g
219	nucore_record	NC.001422.1	NC.001422.1	sweep/g
220	nucore_record	NC.001477.1	NC.001477.1	sweep/g
221	nucore_record	NC.001498.1	NC.001498.1	sweep/g
222	nucore_record	NC.001802.1	NC.001802.1	sweep/g
223	nucore_record	NC.001806.2	NC.001806.2	sweep/g
224	nucore_record	NC.002549.1	NC.002549.1	sweep/g
225	nucore_record	NC.003977.2	NC.003977.2	sweep/g
226	nucore_record	NC.012920.1	NC.012920.1	sweep/g
227	nucore_record	NC.045512.2	NC.045512.2	sweep/g
228	nucore_record	NG.005905.2	NG.005905.2	sweep/g
229	nucore_record	NM.000038.6	NM.000038.6	sweep/g
230	nucore_record	NM.000249.4	NM.000249.4	sweep/g
231	nucore_record	NM.000251.3	NM.000251.3	sweep/g
232	nucore_record	NM.000314.8	NM.000314.8	sweep/g
233	nucore_record	NM.001126112.3	NM.001126112.3	sweep/g
234	nucore_record	NM.001904.4	NM.001904.4	sweep/g
235	nucore_record	NM.002524.5	NM.002524.5	sweep/g
236	nucore_record	NM.004985.5	NM.004985.5	sweep/g
237	study	29	hg38.cpgIslandExt.chr20	sweep/b
238	study	30	hg38.rmsk.chr20:0-5000000	sweep/b
239	nucore_record	MG772933.1	MG772933.1	sweep/k
240	nucore_record	MN996532.2	MN996532.2	sweep/k
241	nucore_record	MT040333.1	MT040333.1	sweep/k
242	nucore_record	NC.004718.3	NC.004718.3	sweep/k
243	nucore_record	NC.019843.3	NC.019843.3	sweep/k
244	dbnp_variant	rs1042713	rs1042713	sweep/g
245	dbnp_variant	rs10427255	rs10427255	sweep/g
246	dbnp_variant	rs1229984	rs1229984	sweep/g
247	dbnp_variant	rs12913832	rs12913832	sweep/g
248	dbnp_variant	rs1426654	rs1426654	sweep/g
249	dbnp_variant	rs16891982	rs16891982	sweep/g
250	dbnp_variant	rs17822931	rs17822931	sweep/g
251	dbnp_variant	rs1799945	rs1799945	sweep/g
252	dbnp_variant	rs1800562	rs1800562	sweep/g
253	dbnp_variant	rs1801133	rs1801133	sweep/g
254	dbnp_variant	rs1815739	rs1815739	sweep/g
255	dbnp_variant	rs2231142	rs2231142	sweep/g
256	dbnp_variant	rs334	rs334	sweep/g
257	dbnp_variant	rs3827760	rs3827760	sweep/g
258	dbnp_variant	rs429358	rs429358	sweep/g
259	dbnp_variant	rs4680	rs4680	sweep/g
260	dbnp_variant	rs4988235	rs4988235	sweep/g
261	dbnp_variant	rs53576	rs53576	sweep/g
262	dbnp_variant	rs6025	rs6025	sweep/g
263	dbnp_variant	rs671	rs671	sweep/g
264	dbnp_variant	rs7412	rs7412	sweep/g
265	dbnp_variant	rs762551	rs762551	sweep/g
266	hgnc_record	HGNC:ABCC11	HGNC:14639	sweep/u
267	hgnc_record	HGNC:ABL1	HGNC:76	sweep/u
268	hgnc_record	HGNC:ACTN3	HGNC:165	sweep/u
269	hgnc_record	HGNC:ADH1B	HGNC:250	sweep/u
270	hgnc_record	HGNC:ALDH2	HGNC:404	sweep/u
271	hgnc_record	HGNC:APC	HGNC:583	sweep/u
272	hgnc_record	HGNC:APOE	HGNC:613	sweep/u
273	hgnc_record	HGNC:ASXL1	HGNC:18318	sweep/u
274	hgnc_record	HGNC:ATM	HGNC:795	sweep/u

#	Kind	Study	Accession	Used in
275	hgnc_record	HGNC:BRAF	HGNC:1097	sweep/u
276	hgnc_record	HGNC:BRCA1	HGNC:1100	sweep/u
277	hgnc_record	HGNC:BRCA2	HGNC:1101	sweep/u
278	hgnc_record	HGNC:CDKN2A	HGNC:1787	sweep/u
279	hgnc_record	HGNC:CFTR	HGNC:1884	sweep/u
280	hgnc_record	HGNC:CHEK2	HGNC:16627	sweep/u
281	hgnc_record	HGNC:COMT	HGNC:2228	sweep/u
282	hgnc_record	HGNC:DMD	HGNC:2928	sweep/u
283	hgnc_record	HGNC:DNMT3A	HGNC:2978	sweep/u
284	hgnc_record	HGNC:EDAR	HGNC:2895	sweep/u
285	hgnc_record	HGNC:EGFR	HGNC:3236	sweep/u
286	hgnc_record	HGNC:ERBB2	HGNC:3430	sweep/u
287	hgnc_record	HGNC:F5	HGNC:3542	sweep/u
288	hgnc_record	HGNC:FLT3	HGNC:3765	sweep/u
289	hgnc_record	HGNC:FMR1	HGNC:3775	sweep/u
290	hgnc_record	HGNC:GBA1	HGNC:4177	sweep/u
291	hgnc_record	HGNC:HBB	HGNC:4827	sweep/u
292	hgnc_record	HGNC:HERC2	HGNC:4868	sweep/u
293	hgnc_record	HGNC:HFE	HGNC:4886	sweep/u
294	hgnc_record	HGNC:HTT	HGNC:4851	sweep/u
295	hgnc_record	HGNC:IDH1	HGNC:5382	sweep/u
296	hgnc_record	HGNC:IDH2	HGNC:5383	sweep/u
297	hgnc_record	HGNC:JAK2	HGNC:6192	sweep/u
298	hgnc_record	HGNC:KCNQ1	HGNC:6294	sweep/u
299	hgnc_record	HGNC:KRAS	HGNC:6407	sweep/u
300	hgnc_record	HGNC:LCT	HGNC:6530	sweep/u
301	hgnc_record	HGNC:LMNA	HGNC:6636	sweep/u
302	hgnc_record	HGNC:MLH1	HGNC:7127	sweep/u
303	hgnc_record	HGNC:MSH2	HGNC:7325	sweep/u
304	hgnc_record	HGNC:MTHFR	HGNC:7436	sweep/u
305	hgnc_record	HGNC:MYC	HGNC:7553	sweep/u
306	hgnc_record	HGNC:MYH7	HGNC:7577	sweep/u
307	hgnc_record	HGNC:NF1	HGNC:7765	sweep/u
308	hgnc_record	HGNC:NF2	HGNC:7773	sweep/u
309	hgnc_record	HGNC:NPM1	HGNC:7910	sweep/u
310	hgnc_record	HGNC:NRAS	HGNC:7989	sweep/u
311	hgnc_record	HGNC:OCA2	HGNC:8101	sweep/u
312	hgnc_record	HGNC:PALB2	HGNC:26144	sweep/u
313	hgnc_record	HGNC:PIK3CA	HGNC:8975	sweep/u
314	hgnc_record	HGNC:PTEN	HGNC:9588	sweep/u
315	hgnc_record	HGNC:RB1	HGNC:9884	sweep/u
316	hgnc_record	HGNC:RUNX1	HGNC:10471	sweep/u
317	hgnc_record	HGNC:SCN5A	HGNC:10593	sweep/u
318	hgnc_record	HGNC:SLC24A5	HGNC:20611	sweep/u
319	hgnc_record	HGNC:SMAD4	HGNC:6770	sweep/u
320	hgnc_record	HGNC:SOD1	HGNC:11179	sweep/u
321	hgnc_record	HGNC:STK11	HGNC:11389	sweep/u
322	hgnc_record	HGNC:TET2	HGNC:25941	sweep/u
323	hgnc_record	HGNC:TP53	HGNC:11998	sweep/u
324	hgnc_record	HGNC:TTN	HGNC:12403	sweep/u
325	hgnc_record	HGNC:VHL	HGNC:12687	sweep/u
326	study	31	UniProt2Reactome.txt (current release)	sweep/r
327	study	32	gnomad_r4	gnomad.constraint sweep/g
328	study	33	sramp_mature_test.csv (133,826 labels)	sweep/r
329	study	34	GSE63753	sweep/r
330	ensembl_transcript	ENSMUST00000056677	ENSMUST00000056677	sweep/r
331	ensembl_transcript	ENSMUST00000106089	ENSMUST00000106089	sweep/r
332	ensembl_transcript	ENSMUST00000125127	ENSMUST00000125127	sweep/r
333	ensembl_transcript	ENST00000234590	ENST00000234590	sweep/r
334	ensembl_transcript	ENST00000481115	ENST00000481115	sweep/r
335	ensembl_transcript	ENST00000525069	ENST00000525069	sweep/r
336	ensembl_transcript	ENST00000602390	ENST00000602390	sweep/r
337	ensembl_transcript	ENSMUST00000145952	ENSMUST00000145952	sweep/r
338	ensembl_transcript	ENST00000227520	ENST00000227520	sweep/r

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339	ensembl_transcript	ENST00000255641	ENST00000255641	sweep/r
340	ensembl_transcript	ENSMUST00000015889	ENSMUST00000015889	sweep/r
341	ensembl_transcript	ENSMUST00000038791	ENSMUST00000038791	sweep/r
342	ensembl_transcript	ENSMUST00000022709	ENSMUST00000022709	sweep/r
343	ensembl_transcript	ENST00000223368	ENST00000223368	sweep/r
344	ensembl_transcript	ENST00000284551	ENST00000284551	sweep/r
345	ensembl_transcript	ENST00000305885	ENST00000305885	sweep/r
346	ensembl_transcript	ENST00000357186	ENST00000357186	sweep/r
347	ensembl_transcript	ENST00000418388	ENST00000418388	sweep/r
348	ensembl_transcript	ENST00000622639	ENST00000622639	sweep/r
349	ensembl_transcript	ENSMUST00000179669	ENSMUST00000179669	sweep/r
350	ensembl_transcript	ENST00000356872	ENST00000356872	sweep/r
351	ensembl_transcript	ENST00000359867	ENST00000359867	sweep/r
352	ensembl_transcript	ENST00000361952	ENST00000361952	sweep/r
353	ensembl_transcript	ENST00000404824	ENST00000404824	sweep/r
354	ensembl_transcript	ENST00000428417	ENST00000428417	sweep/r
355	ensembl_transcript	ENSMUST00000035295	ENSMUST00000035295	sweep/r
356	ensembl_transcript	ENSMUST00000058667	ENSMUST00000058667	sweep/r
357	ensembl_transcript	ENST00000200453	ENST00000200453	sweep/r
358	ensembl_transcript	ENST00000316836	ENST00000316836	sweep/r
359	ensembl_transcript	ENST00000575785	ENST00000575785	sweep/r
360	ensembl_transcript	ENSMUST00000018716	ENSMUST00000018716	sweep/r
361	ensembl_transcript	ENSMUST00000061425	ENSMUST00000061425	sweep/r
362	ensembl_transcript	ENSMUST00000081747	ENSMUST00000081747	sweep/r
363	ensembl_transcript	ENSMUST00000084743	ENSMUST00000084743	sweep/r
364	ensembl_transcript	ENST00000618250	ENST00000618250	sweep/r
365	ensembl_transcript	ENSMUST00000022272	ENSMUST00000022272	sweep/r
366	ensembl_transcript	ENSMUST00000037235	ENSMUST00000037235	sweep/r
367	ensembl_transcript	ENSMUST00000040967	ENSMUST00000040967	sweep/r
368	ensembl_transcript	ENSMUST00000042203	ENSMUST00000042203	sweep/r
369	ensembl_transcript	ENSMUST00000042227	ENSMUST00000042227	sweep/r
370	pmid	37278811	PMID 37278811	sweep/e
371	pmid	40163448	PMID 40163448	sweep/e
372	pmid	32154747	PMID 32154747	sweep/e
373	pmid	31626570	PMID 31626570	sweep/e
374	pmid	27394157	PMID 27394157	sweep/e
375	pmid	30334692	PMID 30334692	sweep/e
376	pmid	40418577	PMID 40418577	sweep/e
377	pmid	39235967	PMID 39235967	sweep/e
378	pmid	31999662	PMID 31999662	sweep/e
379	pmid	27252987	PMID 27252987	sweep/e
380	nct	NCT04235140	NCT04235140	sweep/e
381	nct	NCT03029455	NCT03029455	sweep/e
382	nct	NCT01614457	NCT01614457	sweep/e
383	nct	NCT01937325	NCT01937325	sweep/e
384	nct	NCT03525574	NCT03525574	sweep/e
385	nct	NCT02797132	NCT02797132	sweep/e
386	nct	NCT05699148	NCT05699148	sweep/e
387	nct	NCT03085485	NCT03085485	sweep/e
388	nct	NCT02965326	NCT02965326	sweep/e
389	nct	NCT07363304	NCT07363304	sweep/e
390	study	35	genes_to_phenotype.txt	sweep/r
391	omim	OMIM:219700	OMIM:219700	sweep/r
392	omim	OMIM:143100	OMIM:143100	sweep/r
393	omim	OMIM:154700	OMIM:154700	sweep/r
394	omim	OMIM:261600	OMIM:261600	sweep/r
395	omim	OMIM:230800	OMIM:230800	sweep/r
396	omim	OMIM:310200	OMIM:310200	sweep/r
397	study	36	pathology.tsv	sweep/o
398	ensembl_gene	ENSG00000164362	ENSG00000164362	sweep/o
399	ensembl_gene	ENSG00000089685	ENSG00000089685	sweep/o
400	ensembl_gene	ENSG00000081051	ENSG00000081051	sweep/o
401	ensembl_gene	ENSG00000142515	ENSG00000142515	sweep/o
402	ensembl_gene	ENSG00000141736	ENSG00000141736	sweep/o
403	chembl_target	CHEMBL203	CHEMBL203	sweep/d
404	chembl_target	CHEMBL279	CHEMBL279	sweep/d
405	chembl_target	CHEMBL1862	CHEMBL1862	sweep/d

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406	chembl_target	CHEMBL2971	CHEMBL2971	sweep/d
407	chembl_target	CHEMBL4282	CHEMBL4282	sweep/d
408	chembl_target	CHEMBL2842	CHEMBL2842	sweep/d
409	uniprot	P00533	P00533	sweep/d
410	uniprot	P35968	P35968	sweep/d
411	uniprot	P00519	P00519	sweep/d
412	uniprot	O60674	O60674	sweep/d
413	uniprot	P31749	P31749	sweep/d
414	uniprot	P42345	P42345	sweep/d
415	pdb	1BWB	1BWB	sweep/m
416	pdb	1JPZ	1JPZ	sweep/m
417	pdb	1MUI	1MUI	sweep/m
418	pdb	1TSL	1TSL	sweep/m
419	pdb	1VKJ	1VKJ	sweep/m
420	pdb	2AOD	2AOD	sweep/m
421	pdb	2IEN	2IEN	sweep/m
422	pdb	2IKI	2IKI	sweep/m
423	pdb	2Q88	2Q88	sweep/m
424	pdb	2R5Q	2R5Q	sweep/m
425	pdb	2Z4O	2Z4O	sweep/m
426	pdb	3FQL	3FQL	sweep/m
427	pdb	3IJW	3IJW	sweep/m
428	pdb	3KYF	3KYF	sweep/m
429	pdb	3NTY	3NTY	sweep/m
430	pdb	3NU3	3NU3	sweep/m
431	pdb	3TKW	3TKW	sweep/m
432	pdb	3UM8	3UM8	sweep/m
433	pdb_ligand	017	017	sweep/m
434	pdb_ligand	065	065	sweep/m
435	pdb_ligand	140	140	sweep/m
436	pdb_ligand	146	146	sweep/m
437	pdb_ligand	1CY	1CY	sweep/m
438	pdb_ligand	1UN	1UN	sweep/m
439	pdb_ligand	2NC	2NC	sweep/m
440	pdb_ligand	388	388	sweep/m
441	pdb_ligand	478	478	sweep/m
442	pdb_ligand	4CS	4CS	sweep/m
443	pdb_ligand	5GP	5GP	sweep/m
444	pdb_ligand	5P3	5P3	sweep/m
445	pdb_ligand	79Z	79Z	sweep/m
446	pdb_ligand	A15	A15	sweep/m
447	pdb_ligand	A3P	A3P	sweep/m
448	pdb_ligand	AB1	AB1	sweep/m
449	pdb_ligand	ACO	ACO	sweep/m
450	kegg_compound	C00010	cpd:C00010	sweep/m
451	kegg_compound	C00011	cpd:C00011	sweep/m
452	kegg_compound	C00022	cpd:C00022	sweep/m
453	kegg_compound	C00024	cpd:C00024	sweep/m
454	kegg_compound	C00031	cpd:C00031	sweep/m
455	kegg_compound	C00042	cpd:C00042	sweep/m
456	kegg_compound	C00048	cpd:C00048	sweep/m
457	kegg_compound	C00074	cpd:C00074	sweep/m
458	kegg_compound	C00085	cpd:C00085	sweep/m
459	kegg_compound	C00092	cpd:C00092	sweep/m
460	kegg_compound	C00111	cpd:C00111	sweep/m
461	kegg_compound	C00118	cpd:C00118	sweep/m
462	kegg_compound	C00149	cpd:C00149	sweep/m
463	kegg_compound	C00158	cpd:C00158	sweep/m
464	kegg_compound	C00186	cpd:C00186	sweep/m
465	kegg_compound	C00197	cpd:C00197	sweep/m
466	kegg_compound	C00236	cpd:C00236	sweep/m
467	kegg_compound	C00354	cpd:C00354	sweep/m
468	kegg_compound	C00469	cpd:C00469	sweep/m
469	kegg_compound	C01089	cpd:C01089	sweep/m
470	kegg_enzyme	EC 1.1.1.1	ec:1.1.1.1	sweep/m
471	kegg_enzyme	EC 1.1.1.27	ec:1.1.1.27	sweep/m
472	kegg_enzyme	EC 1.2.1.12	ec:1.2.1.12	sweep/m

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473	kegg_enzyme	EC 1.2.4.1	ec:1.2.4.1	sweep/m
474	kegg_enzyme	EC 2.3.1.9	ec:2.3.1.9	sweep/m
475	kegg_enzyme	EC 2.3.3.1	ec:2.3.3.1	sweep/m
476	kegg_enzyme	EC 2.3.3.9	ec:2.3.3.9	sweep/m
477	kegg_enzyme	EC 2.7.1.1	ec:2.7.1.1	sweep/m
478	kegg_enzyme	EC 2.7.1.11	ec:2.7.1.11	sweep/m
479	kegg_enzyme	EC 2.7.1.40	ec:2.7.1.40	sweep/m
480	kegg_enzyme	EC 2.7.2.3	ec:2.7.2.3	sweep/m
481	kegg_enzyme	EC 4.1.2.13	ec:4.1.2.13	sweep/m
482	kegg_enzyme	EC 4.1.3.1	ec:4.1.3.1	sweep/m
483	kegg_enzyme	EC 4.2.1.11	ec:4.2.1.11	sweep/m
484	kegg_enzyme	EC 5.3.1.1	ec:5.3.1.1	sweep/m
485	kegg_enzyme	EC 5.3.1.9	ec:5.3.1.9	sweep/m
486	pubchem_cid	54675776	CID 54675776	sweep/b
487	pubchem_cid	6134	CID 6134	sweep/b
488	pubchem_cid	439195	CID 439195	sweep/b
489	pubchem_cid	2153	CID 2153	sweep/b
490	pubchem_cid	241	CID 241	sweep/b
491	pubmed	7510417	PMID 7510417	sweep/b
492	pubmed	1812784	PMID 1812784	sweep/b
493	jaspar	MA0099.3	MA0099.3	sweep/d
494	jaspar	MA0107.1	MA0107.1	sweep/d
495	jaspar	MA0106.3	MA0106.3	sweep/d
496	jaspar	MA0108.2	MA0108.2	sweep/d
497	jaspar	MA0093.3	MA0093.3	sweep/d
498	jaspar	MA0079.4	MA0079.4	sweep/d
499	jaspar	MA0060.2	MA0060.2	sweep/d
500	jaspar	MA0114.4	MA0114.4	sweep/d
501	ucsc_hg38	chr3:36990000-37000000	hg38 chr3:36,990,000-37,000,000	sweep/d
502	ucsc_hg38	chr9:21990000-22000000	hg38 chr9:21,990,000-22,000,000	sweep/d
503	ucsc_hg38	chr9:21976000-21989000	hg38 chr9:21,976,000-21,989,000	sweep/d
504	ucsc_cpg	cpgIslandExt chr3:36900000-37100000	hg38 cpgIslandExt chr3:36,900,000-37,100,000	sweep/d
505	ucsc_cpg	cpgIslandExt chr9:21940000-22000000	hg38 cpgIslandExt chr9:21,940,000-22,000,000	sweep/d
506	pubmed	8242752	PMID 8242752	sweep/d
507	hgnc	OCT4	OCT4	sweep/c
508	hgnc	SOX2	SOX2	sweep/c
509	hgnc	NANOG	NANOG	sweep/c
510	hgnc	GATA4	GATA4	sweep/c
511	hgnc	MEF2C	MEF2C	sweep/c
512	hgnc	TBX5	TBX5	sweep/c
513	hgnc	ASCL1	ASCL1	sweep/c
514	hgnc	NEUROD1	NEUROD1	sweep/c
515	hgnc	MYT1L	MYT1L	sweep/c
516	hgnc	MYOD1	MYOD1	sweep/c
517	hgnc	TNNT2	TNNT2	sweep/c
518	hgnc	MYH6	MYH6	sweep/c
519	hgnc	NKX2-5	NKX2-5	sweep/c
520	hgnc	TUBB3	TUBB3	sweep/c
521	hgnc	MAP2	MAP2	sweep/c
522	hgnc	SCN1A	SCN1A	sweep/c
523	hgnc	MYOG	MYOG	sweep/c
524	hgnc	MYH1	MYH1	sweep/c
525	hgnc	COL1A1	COL1A1	sweep/c
526	hgnc	VIM	VIM	sweep/c
527	hgnc_alias	OCT4- ζ POU5F1	POU5F1	sweep/c
528	pubmed	16904174	PMID 16904174	sweep/c
529	pubmed	20691899	PMID 20691899	sweep/c
530	pubmed	24243019	PMID 24243019	sweep/c
531	pubmed	42775788	PMID 42775788	sweep/b
532	pubmed	42759458	PMID 42759458	sweep/b

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533	pubmed	42746974	PMID 42746974	sweep/b
534	pubmed	42731586	PMID 42731586	sweep/b
535	pubmed	42727368	PMID 42727368	sweep/b
536	pubmed	42701997	PMID 42701997	sweep/b
537	pubmed	42693013	PMID 42693013	sweep/b
538	pubmed	42641908	PMID 42641908	sweep/b
539	icite	pmids_42775788_42759458_42746974_42731586_42727368_42701997_42693013_42641908	citation_count map	sweep/b
540	refseq	NM_000518.5	RefSeq FASTA	sweep/g
541	refseq	NM_000546.6	RefSeq FASTA	sweep/g
542	refseq	NM_000492.4	RefSeq FASTA	sweep/g
543	ncbi_gene	7157	esummary	sweep/g
544	ncbi_gene	672	esummary	sweep/g
545	ncbi_gene	1080	esummary	sweep/g
546	ncbi_gene	3043	esummary	sweep/g
547	ncbi_gene	1956	esummary	sweep/g
548	uniprot	P04637	UniProt REST	sweep/g
549	uniprot	P38398	UniProt REST	sweep/g
550	uniprot	P13569	UniProt REST	sweep/g
551	uniprot	P68871	UniProt REST	sweep/g
552	cbioportal	coadread_tcga_pan_can_atlas_2018	cBioPortal mutations/fetch	sweep/li
553	cbioportal	luad_tcga_pan_can_atlas_2018	cBioPortal mutations/fetch	sweep/li
554	cbioportal	brca_tcga_pan_can_atlas_2018	cBioPortal mutations/fetch	sweep/li
555	cbioportal	paad_tcga_pan_can_atlas_2018	cBioPortal mutations/fetch	sweep/li
556	cbioportal	prad_tcga_pan_can_atlas_2018	cBioPortal mutations/fetch	sweep/li
557	cbioportal	skcm_tcga_pan_can_atlas_2018	cBioPortal mutations/fetch	sweep/li
558	pubmed	25079552	PMID 25079552	sweep/li
559	pubmed	28810144	PMID 28810144	sweep/li
560	pubmed	27428049	PMID 27428049	sweep/li
561	pubmed	26771485	PMID 26771485	sweep/li
562	uniprot	P03135	UniProt REST	sweep/v
563	uniprot	Q6JC40	UniProt REST	sweep/v
564	pubmed	9445046	PMID 9445046	sweep/v
565	pubmed	16378984	PMID 16378984	sweep/v
566	rcsb	1UBQ	RCSB PDB	sweep/a
567	pubchem	243	PubChem PUG-REST	sweep/c
568	pubchem	996	PubChem PUG-REST	sweep/c
569	pubchem	1001	PubChem PUG-REST	sweep/c
570	pubchem	6115	PubChem PUG-REST	sweep/c
571	pubchem	10008	PubChem PUG-REST	sweep/c
572	pubchem	10009	PubChem PUG-REST	sweep/c
573	pubchem	13195	PubChem PUG-REST	sweep/c
574	chembl	CHEMBL2107497	ChEMBL similarity API	sweep/c

References

- [1] Doench JG et al. Optimized sgRNA design to maximize activity and minimize off-target effects of CRISPR-Cas9. *Nat Biotechnol* 34:184-191, 2016.
- [2] Concordet JP, Haeussler M. CRISPOR: intuitive guide selection for CRISPR/Cas9 genome editing experiments and screens. *Nucleic Acids Res* 46:W242-W245, 2018.
- [3] Jankauskaitė J et al. SKEMPI 2.0: an updated benchmark of changes in protein-protein binding energy, kinetics and thermodynamics upon mutation. *Bioinformatics* 35:462-469, 2019.
- [4] King ZA et al. BiGG Models: a platform for integrating, standardizing and sharing genome-scale models. *Nucleic Acids Res* 44:D515-522, 2016.
- [5] Sievers F et al. Fast, scalable generation of high-quality protein multiple sequence alignments using Clustal Omega. *Mol Syst Biol* 7:539, 2011.

- [6] Tavtigian SV et al. Modeling the ACMG/AMP variant classification guidelines as a Bayesian classification framework. *Genet Med* 20:1054-1060, 2018.
- [7] Yeo G, Burge CB. Maximum entropy modeling of short sequence motifs with applications to RNA splicing signals. *J Comput Biol* 11:377-394 (2004).
- [8] Sharp PM, Li WH. The codon adaptation index: a measure of directional synonymous codon usage bias. *Nucleic Acids Res* 15:1281-1295 (1987).
- [9] dos Reis M, Savva R, Wernisch L. Solving the riddle of codon usage preferences: a test for translational selection. *Nucleic Acids Res* 32:5036-5044 (2004).
- [10] Eberhardt J, Santos-Martins D, Tillack AF, Forli S. AutoDock Vina 1.2.0. *J Chem Inf Model* 61:3891-3898 (2021).
- [11] Wang M et al. PaxDb, a database of protein abundance averages across all three domains of life. *Mol Cell Proteomics* 11:492-500 (2012).